Policy Information and Local Operations for Restricted Parity Games

Abstract

Local operations on parity games require different amounts of strategy information. We study this dependence in several explicit finite settings. For a forward two-track interval grammar, we construct an exact polynomial representation, in the complete binary input size, of the joint outcome profiles realizable by one positional Even policy. The representation supports strict-profile queries and legal one-choice updates with original-policy witnesses. A feedback example shows that strict local improvement can produce an exact global tie. A marked feedback alteration also shows that forward joint minimum profiles alone need not determine feedback profiles. For a separate one-feedback syntax, we give an ordered normalization and characterize strictness of a fixed-policy update in an acyclic context. A boundary-forget example identifies information lost by minimal outcome summaries; full positive first-return spectra determine one forget operation for a fixed policy in the stated bounded domain. For two-priority bidirected paths, we prove a linear bound on completed single-edge pivots of universal-tree strategy iteration and give an exact candidate-set evaluator above reachable floors. Its linear rank work per call yields quadratic rank work for two explicit pivot rules; the canonical-tree quadratic baseline is already known. An appendix gives a complete proof of the fixed universal-tree leaf minimum $M(7,3) = 31$, using the published upper construction.

1 Introduction

Parity games combine a finite arena with an infinitary winning condition. For compositional questions, a winning region alone is often too coarse: a play may leave a component through one of several named exits, carrying the maximum priority encountered before exit. A fixed strategy must also make consistent choices when the component is entered at different vertices. Thus two distinct questions arise. Which tuples of local outcomes are realizable by one strategy? Which comparisons between such tuples survive a change of context or boundary?

We examine these questions through explicit representations, counterexamples and local algorithms. The game classes are stated separately: forward interval arenas, a fixed feedback architecture, a literal open-game syntax, bounded boundary observations and two-priority paths. The common object is the information carried by a fixed policy under a specified operation. Each theorem identifies both that information and the hypotheses under which it suffices.

1.1 Results and proof mechanisms

The first result is an exact compiler for a succinct forward interval grammar (Theorem 3.8). Its output describes all tuples of minimal outcomes attainable by a *single* admitted positional Even policy. The proof combines local all-history replacement, a finite capacity partition, transport of a common original policy, and integral difference-bound projection. The representation is polynomial in the complete binary description, with a conservative exponent stated in the theorem. Corollaries 3.23

and 3.25 give, respectively, a strict-profile query and an exhaustive legal one-choice-neighbor query. The latter fixes all other original choices and compares each resulting profile directly with the supplied baseline.

Feedback changes the strictness question. Theorem 4.4 gives a literal example in which a strict local change leaves the entire global raw outcome profile unchanged. The associated vertex minimum (Theorem 4.10) concerns the specified architecture. Proposition 4.11 gives a different obstruction: two complete policies have equal joint minimum profiles in the forward interval grammar, but different minima after a fixed marked feedback connection. The forward and feedback arenas are distinct; the example rules out transfer from the forward minimum profile alone, not a representation using richer policy or spectral information. For a separate finite syntax, Theorem 5.8 gives a sound ordered normalization. The fixed-policy context theorem (Theorem 5.10) then identifies the row transformer that decides whether one legal local update remains strict. Its hypotheses include disjoint syntactic occurrences and an acyclic exterior.

Boundary forgetting reveals a different loss of information. Theorem 6.2 exhibits two policies on the same arena whose boundary keys agree before a forget and differ afterwards. The discrepancy concerns a raw no-return possibility, although the closed winning minima agree. For one fixed original policy, retaining full positive first-return spectra supports an explicit forget formula and reconstruction map (Theorems 6.6 and 6.7). These statements are restricted to the bounded domain declared in Section 6.

The algorithmic results concern Koh–Loho strategy iteration on two-priority bidirected paths. A finite classification of reachable completed labels gives a linear completed-pivot bound (Theorem 7.6). The same structural information yields constant-size candidate sets for an exact evaluator above a reachable carried floor (Theorem 7.15). The proof first places the original least answer in those sets and then uses it to dominate rounded updates. Representation and output costs are stated separately from rank work. Finally, Appendix A establishes the fixed leaf minimum $M(7,3) = 31$ by an all-target lower argument and a specialization of the published upper construction. This tree result is independent of the path evaluator.

Figure 1 records the dependency structure used in the paper.

1.2 Relation to published questions

Watanabe asks how strategy improvement can be lifted to compositional parity-game diagrams while retaining useful intermediate strategies, and proposes an ordered Pareto extension of the diagram language [8, Section 6, p. 14:15]. We use the outcome domain, subpriority order and result-order semantics of that work. The joint compiler retains correlations between starts; the feedback example and context theorem isolate what is required for strict local improvement to remain strict globally. Their restricted domains leave the general compositional question unresolved.

Piedeleu asks for tractable diagram subcategories and a precise relation between local equations and strategy improvement [7, Section VI]. Section 5 supplies an explicit syntax and an ordered normalizer, together with a separate fixed-policy context calculation. We do not identify this syntax with a subcategory of the full calculus or with its discrete-priority denotation. The nineteen equations operate on optimized fronts; the context calculation follows one original policy.

Globally uniform strategy information also appears in the mean-payoff semantics of Watanabe, Eberhart, Asada and Hasuo [9, Definitions 26, 30, 34; Appendix D, Definitions 45, 65, 70]. A fixed pair of memoryless strategies there defines an outcome function over all entrances, and the meager semantics compares collections of these functions. The relation in Section 3 instead minimizes parity outcomes over all Odd histories and retains every admitted Even-policy profile. The objective, strategy quantifiers and orders therefore need to be matched before any theorem can be transferred

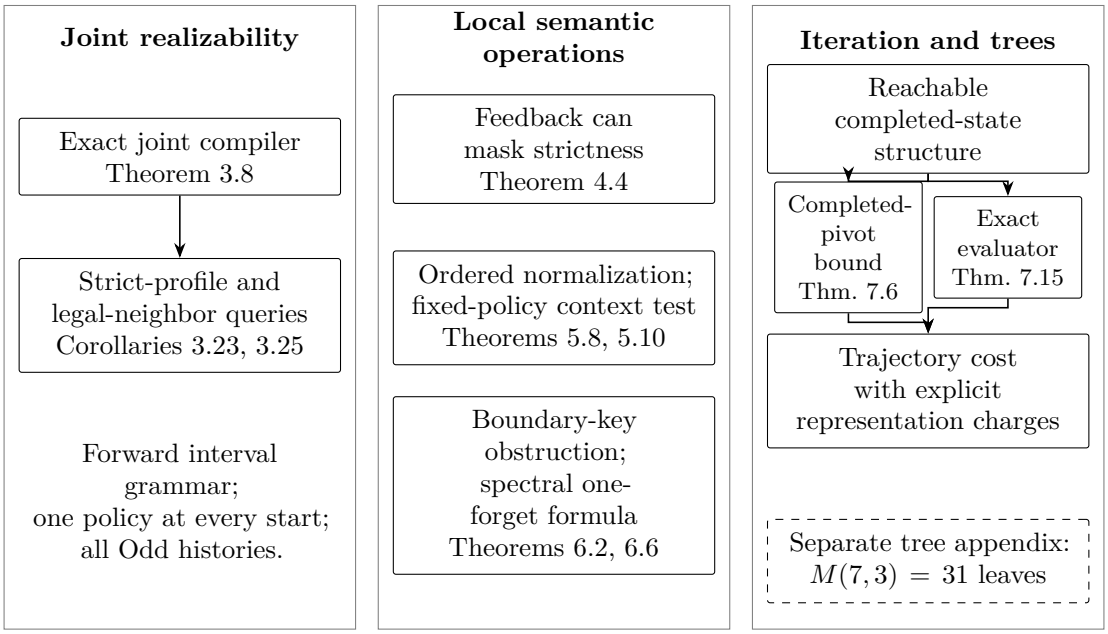


Figure 1: The organization of the results. Arrows indicate proof dependencies. The middle panel groups distinct local operations without an implication between them. The tree appendix uses a separate embedding domain and is not a premise of the path evaluator.

between the two settings.

Fearnley and Schewe ask for a parity-game algorithm of running time $f(k)n^{O(1)}$, with exponent independent of treewidth k [1, Section 9, p. 30]. Our boundary example and spectral formula concern one exact summary operation under a fixed policy. They provide neither a lower bound for winner computation nor an algorithm for a full tree decomposition.

Koh and Loho propose pivot rules improving the total running time of universal-tree strategy iteration, possibly to polynomial time [4, Section 1, p. 63:3], [5, Section 1, p. 24:3]. We use their version-5 algorithm with priorities $1, 2$, a bidirected path and a height-one tree. Their monotone-lift analysis already gives the canonical-tree quadratic total bound, and their framework includes exact evaluation algorithms. Our path argument supplies explicit candidate sets and a rounding proof for the original completed state; it does not improve that known quadratic scale.

Fijalkow's author-hosted question asks for the minimum leaf count of an (n, h)-universal ordered tree [2, "Between lower and upper bounds?"]. The appendix treats the fixed case $(7, 3)$. Its upper construction is Fijalkow's specialization of the Jurdziński–Lazić construction, and its threshold method uses his lower-bound argument [3, Theorems 4–5]. We give the fixed-case proof in full without a priority claim for that equality. These references locate the questions in the cited versions; they do not assert that every question remains open in its original form.

2 Preliminaries

2.1 Arenas and strategies

A finite arena is a directed graph $G = (V, E)$ with a partition $V = V_E \dot\cup V_O$ and a nonempty successor set $E(v) = \{w : (v, w) \in E\}$ at every vertex. Ownership and priority are separate data. A vertex-priority arena has a map $\pi_V : V \to P$; an edge-priority arena has a map $\pi_E : E \to P$, where

P is a finite set of nonnegative integers. Each section fixes the carrier and priority range it uses. An infinite play is won by Even when its largest infinitely recurring priority is even. The finite range ensures that this priority exists.

A positional Even policy is a map

$$\sigma \in \Sigma_E^{\mathrm{pos}}(G) := \prod_{v \in V_E} E(v).$$

It specifies the move at every original Even vertex, including vertices unreachable from a chosen start. A history strategy specifies a legal successor after every finite legal history ending at a vertex of its owner. We write $\Sigma_E^{\mathrm{hist}}(G)$ and $\mathcal{T}_O^{\mathrm{hist}}(G)$ for the two history-strategy spaces. A fixed positional policy gives a selected graph G_σ: Even has its chosen successor and Odd retains every successor. Every path of G_σ is realized by some Odd history strategy, since the prescribed choices along that path can be extended legally to the other histories. This observation applies to finite and infinite paths.

Some constructions restrict Even's positional policies further. We write $\Sigma_E^{\mathrm{adm}}(G) \subseteq \Sigma_E^{\mathrm{pos}}(G)$ when the local grammar imposes such conditions. In particular, the interval compiler requires decreasing menu choices. Positional Odd policies used in strategy iteration will instead be denoted by τ.

2.2 Stopped outcomes and their orders

An open arena has declared entrances and named exits O_{exit}. Its local evaluation stops at the first visit to a declared exit. For vertex priorities, a finite stopped outcome records the maximum priority among the visited vertices; for edge priorities, it records the maximum along the traversed edges. The empty edge prefix has maximum zero. A section's construction specifies the boundary priorities and completion loops. In particular, wiring an exit internally uses the declared continuation in place of its local completion.

For stopped outcomes we include zero in the ambient priority range. The outcome domain is

$$D_O = \{\bot_{\mathrm{play}}, \top_{\mathrm{play}}\} \cup (O_{\mathrm{exit}} \times P).$$

Here (o, p) records an exit and a finite numerical maximum; $\bot_{\mathrm{play}}$ and $\top_{\mathrm{play}}$ record Odd-winning and Even-winning nonexiting plays. Transient priorities affect finite outcomes, but do not determine the winner of a nonexiting play. These conventions follow [8, Definitions 1, 7–9].

We use the subpriority order $\leq_P$: odd priorities are ordered in reverse numerical order, every odd priority precedes every even priority, and even priorities have numerical order. The outcome order $\leq_D$ has least element $\bot_{\mathrm{play}}$, greatest element $\top_{\mathrm{play}}$, and

$$(o, p) \leq_D (o', p') \quad \Longleftrightarrow \quad o = o' \text{ and } p \leq_P p'.$$

Thus finite outcomes at distinct exits are incomparable. Numerical maximum, written max or $\max_{\mathrm{num}}$, retains its ordinary meaning. It is monotone in the subpriority order but is not the maximum operation of that order [8, Definitions 13, 16; Lemma 15].

For a finite poset D, let $\mathcal{A}^+(D)$ denote its nonempty antichains. For a nonempty subset $X \subseteq D$, $\min_D X$ is its antichain of minimal elements. A *row* is an element of $\mathcal{A}^+(D_O)$. Its result order is

$$R \leq_R S \quad \Longleftrightarrow \quad (\forall s \in S)(\exists r \in R)\ r \leq_D s. \tag{1}$$

Equivalently, $\uparrow S \subseteq \uparrow R$, where $\uparrow R = \{d \in D_O : \exists r \in R,\ r \leq_D d\}$. Minimal elements recover an antichain from its generated upset, so $\leq_R$ is a partial order. For rows indexed by declared entrances,

$\leq_J$ denotes its coordinatewise product. Strict comparison means $R \leq_R S$ and $S \not\leq_R R$, with the analogous convention for profiles. These are the result-order conventions of [8, Definitions 17–19, 22–24].

2.3 Joint profiles and optimized fronts

Fix an arena, a start i, and an Even strategy σ. Let $\mathrm{Raw}_i^G(\sigma)$ be the set of outcomes of all plays from i compatible with σ and all legal Odd history strategies. Put

$$R_i^G(\sigma) = \min_D \mathrm{Raw}_i^G(\sigma), \qquad V_G(\sigma) = \big(R_i^G(\sigma)\big)_i.$$

Totality and the stopping convention make the raw set nonempty. For an admitted positional-policy class, the *joint attainable relation* is

$$J_G = \{V_G(\sigma) : \sigma \in \Sigma_E^{\mathrm{adm}}(G)\}. \tag{2}$$

One policy is chosen before the entrance; hence its choices correlate all coordinates. The relation retains every attainable profile, including profiles dominated by another policy's profile. It may be empty if no policy is admitted.

By contrast, the ordered syntax of Section 5 uses the entrance-wise optimized front

$$Q_G(i) = \max_R\{R_i^G(\sigma) : \sigma \in \Sigma_E^{\mathrm{hist}}(G)\}. \tag{3}$$

Here $\max_R$ retains maximal rows in the result order, so a front belongs to $\mathcal{A}^+(\mathcal{A}^+(D_O))$. Different entrances are optimized separately. The nested antichain type and the quantifier order distinguish Q_G from J_G. The fixed-policy context result in that section returns to $V_G(\sigma)$ for a supplied original positional policy.

2.4 Local conventions and resource measures

The boundary sets of Section 6 specify *positive first-return* observations in the original graph. Forgetting a boundary vertex changes the observation set while retaining the vertex and the fixed policy. The spectral record H_C and its attained-record domain are defined there. The labels in Section 7 instead belong to an ordered tree domain with a greatest element $\mathsf{TOP}_{\mathrm{KL}}$. At a correct terminal output this label marks vertices outside Even's winning region; it is distinct from the Even-winning play outcome $\top_{\mathrm{play}}$. No edge/vertex-priority conversion is used without the construction and correspondence stated for the relevant domain.

We distinguish complete binary input length, explicit arena size, literal syntax size, tree leaves and requested output length. The compiler's N includes its succinct interval encoding; it is not the number of generated menu vertices. Path evaluation separates rank operations from bit-cost random-access computation, including input preparation and output. The universal-tree appendix measures leaves under its stated all-node, immediate-edge ordered embedding convention. Each theorem declares the parameters and representation whose cost it bounds.

3 Joint policy profiles in forward interval arenas

The object to be represented is the set of entire outcome profiles attainable by one original policy. Its rows are correlated: choosing a policy separately at each entrance would produce a different relation. The finite grammar below makes this joint relation representable by polynomially many integer zones, even when interval menus are given succinctly in binary.

3.1 Arenas, policies and ordered profiles

Definition 3.1 (Forward interval arena)**.** A *forward two-track interval arena* A has a positive number $\ell \geq 1$ of layers. Its vertices and edges are specified as follows.

1. Each layer contains exactly two distinct original routers of priority zero. Each router is owned by Even or Odd and has one or two outgoing alternatives. A forced or unreachable router retains its identity.

2. Each of the two tracks contains at most one interval-chain gadget in that layer. The gadgets are disjoint. A gadget g has Even hubs $x_0, \ldots, x_{h_g}$, where $h_g \geq 1$, and nonempty integer menus $[L_i, U_i] \subseteq [1, D_{\mathrm{cap}}]$. The hub x_i has one successor $m_{i,q}$ for every $q \in [L_i, U_i]$. This menu vertex is owned by Odd and has two alternatives: a right arm $e_{i,q}$ of priority $4q$ and a left arm $o_{i,q}$ of priority $4q+1$. Each arm is forced. The right arm leads to x_{i+1}, or to O_R if $i = h_g$; the left arm leads to x_{i-1}, or to O_L if $i = 0$. Hubs and menus have priority zero.

3. The four distinct ports I_L, I_R, O_L, O_R are Even vertices of priority zero. The entrances force respectively to x_0, x_{h_g}. An isolated exit has a zero self-loop. Wiring that exit to a continuation replaces the loop by the specified unique move.

4. Exterior wiring uses only gadget ports and declared routers; no exterior edge enters an internal hub, menu or arm. Connections between different layers go from layer i to layer $i+1$. The wiring within a layer is supplied literally. There are no backward interlayer edges or inter-gadget trace operations. In particular, an interface never duplicates a physical Even choice.

5. There are at most two declared original global starts. Each is either a port or router in the first layer, or a separate boundary vertex whose sole successor is such a port or router. No global start is an internal hub, menu or arm. At most two distinct typed global exits terminate the last layer. All these boundary vertices have priority zero. A global exit is a zero sink, and its first visit ends evaluation. Every other move not already described is the supplied unique continuation. The arena is total before stopping at a global exit.

The two-track condition bounds this grammar, rather than graph width. A physical occurrence includes its ownership, layer, ports and complete wiring. The input does not allow an unspecified routing skeleton or arbitrary succinct graph generators.

The *input length* N is the complete binary length of those data, all interval descriptions, and D_{cap}. Menus and their arms are generated from the interval by their declared occurrence identities; the integer menu coordinate is encoded in binary. Thus the layer count and the total hub count r are $O(N)$, and $\log(D_{\mathrm{cap}} + r + 2) = O(N)$. Explicitly listing all generated forced moves is an optional output operation charged by its actual encoded length.

Definition 3.2 (Admitted common policy)**.** An admitted policy σ_E is a positional Even map on every original Even vertex of A. At hub x_i it chooses an existing menu m_{i,q_i}, with

$$L_i \leq q_i \leq U_i, \qquad q_0 > q_1 > \cdots > q_{h_g}$$

for each gadget g. At an Even router it chooses a legal successor; at a forced vertex it takes the unique move. These conditions apply also to unreachable hubs and routers. The entire map is fixed before a global start is selected and is reused at every visit.

Against σ_E, Odd may use an arbitrary legal history-dependent strategy. Equivalently, one may take all paths in the original graph after retaining only the selected successor at each Even

choice. Every such path extends to an Odd history strategy by filling the choices at other histories. Admission therefore restricts Even's policy set, while the opponent quantifier remains unrestricted.

We use the first-exit outcomes and ordered semantics of [8, Definitions 1, 7–9, 13, 16–19, 22–24]. The following local definitions specify their carriers and policy quantifiers for the present arena.

Definition 3.3 (Outcomes in the forward arena)**.** For a fixed admitted σ_E, a play that first reaches a global exit o has outcome (o, p), where p is the numerical maximum priority on the finite prefix through that visit. The empty zero prefix has maximum zero. A nonexiting play has outcome $\top_{\text{play}}$ or $\bot_{\text{play}}$ according as the largest priority recurring infinitely often is even or odd. The latter maximum exists because the arena has a finite priority alphabet. In particular, a transient maximum does not determine the nonexit outcome.

Definition 3.4 (Priority and outcome orders)**.** The subpriority order $\leq_P$ reverses numerical order on odd priorities, places every odd priority below every even priority, and uses numerical order on evens, including zero. On the outcome domain

$$D_{\mathcal{O}} = \{\bot_{\text{play}}, \top_{\text{play}}\} \dot{\cup} \{(o, p) : o \in \mathcal{O},\ p \text{ a priority}\},$$

let $\bot_{\text{play}}$ be least and $\top_{\text{play}}$ greatest, and set

$$(o, p) \leq_D (o', p') \quad \Longleftrightarrow \quad o = o' \text{ and } p \leq_P p'.$$

Here $\mathcal{O}$ is the set of typed global exits. Distinct exits are incomparable. All occurrences of max on priorities below denote numerical maximum, not maximum in $\leq_P$.

Definition 3.5 (Joint attainable relation)**.** For each global start i, define

$$\text{Raw}_i(\sigma_E) = \{\text{outcome}(\rho) : \rho \text{ is a play from } i \text{ under } \sigma_E\}, \qquad R_i(\sigma_E) = \min_{\leq_D} \text{Raw}_i(\sigma_E).$$

Totality and the finite outcome domain make this minimum a nonempty antichain. If bottom occurs, the row is $\{\bot_{\text{play}}\}$. Otherwise it contains the $\leq_P$-least finite priority at each reachable exit, if there is any finite outcome. If no finite outcome occurs, the row is $\{\top_{\text{play}}\}$. Thus a row is either one extremum or at most two finite outcomes at distinct exits.

Write

$$V_A(\sigma_E) = (R_i(\sigma_E))_i, \qquad J_A = \{V_A(\sigma_E) : \sigma_E \text{ is an admitted original policy}\}.$$

The same policy realizes the entire tuple, and every row contains the outcomes of all Odd histories against that policy. No comparison or pruning between policies is performed. In particular, J_A need not be the product of its entrance projections. Different policies may have the same profile, and J_A is empty if no policy is admitted.

Definition 3.6 (Row and profile comparison)**.** For nonempty source-minimal antichains $C, D \subseteq D_{\mathcal{O}}$, set

$$C \leq_R D \quad \Longleftrightarrow \quad (\forall d \in D)(\exists c \in C)\ c \leq_D d.$$

For profiles with the same original starts, define $b \leq_J v$ by $b_i \leq_R v_i$ at every start i. These are the upper/result lift and its product.

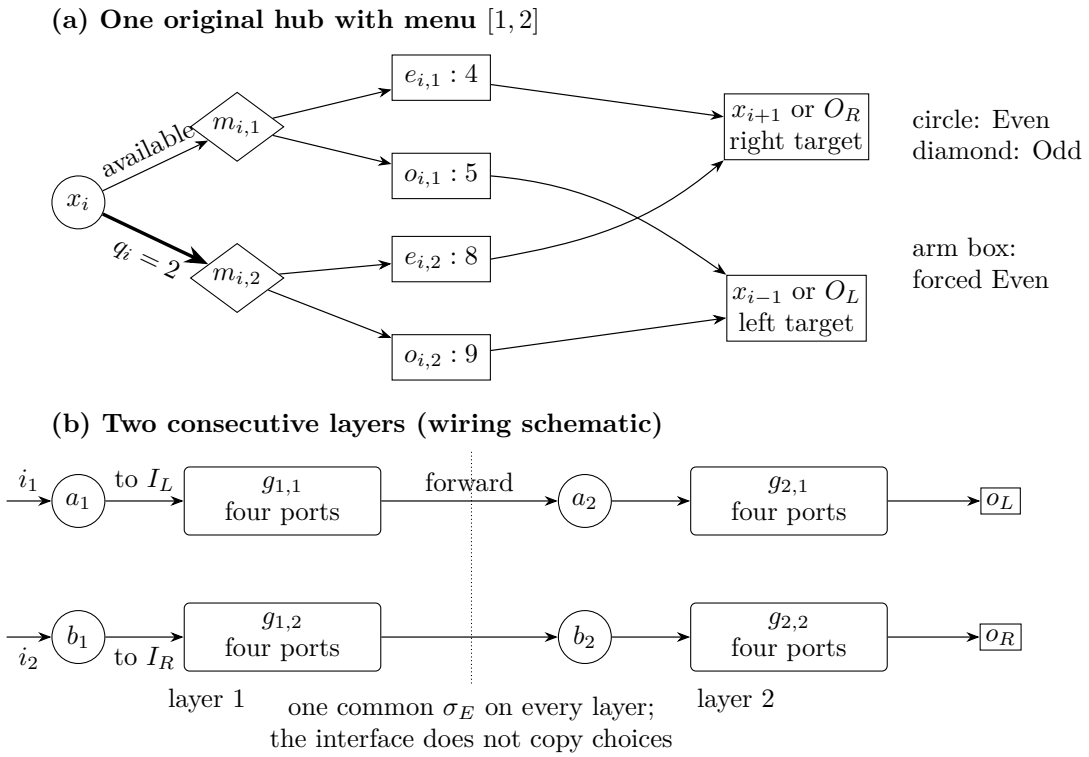


Figure 2: Original choices and shared observations in a forward interval arena. Panel (a) gives the exact incidence at a hub whose two menus are $1, 2$; all unselected menus and arms remain. The bold edge selects $q_i = 2$. Each arm box shows its name and vertex priority; all other displayed vertices have priority zero. At an endpoint, the indicated adjacent hub is replaced by the corresponding exit port. This is a local incidence inset, not a complete gadget. Panel (b) compresses each chain and shows representative wiring only, rather than specifying another complete input. There are two declared routers per layer, drawn Even here. Arrows meet chains at ports, and interlayer arrows point forward. The same original policy determines both rows of the joint profile.

The row order is a partial order. Reflexivity and transitivity follow from $\leq_D$. If $C \leq_R D \leq_R C$, choose for each $c \in C$ elements $d \in D, c' \in C$ with $c' \leq_D d \leq_D c$. The antichain property forces $c' = c$, and antisymmetry then gives $d = c$. Hence $C \subseteq D$, and the reverse inclusion is symmetric. The product order is therefore partial as well.

The preceding row order is the source's result order [8, Definitions 19, 22, Proposition 23]. Its further optimization over Even strategies at an individual entrance [8, Definition 24] is not part of J_A. Figure 2 separates physical choices from the interface through which the same policy is observed.

Definition 3.7 (Tagged integer zones)**.** A tag records, for each start, either its singleton extremum or its present finite exits in ascending original exit-identity order. For each finite entry it also records $\epsilon \in \{0, 1\}$; the corresponding visible integer coordinate u represents

$$p = 4u + \epsilon.$$

Only priorities $0, 4q, 4q + 1$ occur. Zero is represented by $(u, \epsilon) = (0, 0)$; an odd finite entry cannot have $u = 0$. Invalid parity or antichain tags represent no tuple. There are at most four visible coordinates in a global profile.

A *zone* is the set of integral visible tuples satisfying a finite conjunction of difference bounds

$$u_j - u_k \le c, \qquad c \in \mathbb{Z},$$

with a distinguished anchor $u_* = 0$ for unary bounds. Coefficients are binary integers. The representation E_A is an explicit finite union of zones, separately for each tag; $\mathrm{decode}(E_A)$ denotes its set of decoded profiles. No policy, menu or construction variable remains in the output zones. We count zones, zone/tag/coefficient bits, and witness-template bits separately. A witness query asks for one original realizing policy.

3.2 The representation theorem

Theorem 3.8 (Exact joint representation and policy recovery)**.** *Let A be a forward interval arena of binary length N, with admitted policies, outcomes and encoding as in Definitions 3.1–3.7. There is a uniform algorithm that constructs an explicit tagged union E_A of projected integer zones such that*

$$\mathrm{decode}(E_A) = J_A.$$

For each output zone it constructs a common-policy template. Given any encoded profile $v \in \mathrm{decode}(E_A)$, that template returns one admitted original positional policy σ_E with

$$V_A(\sigma_E) = v.$$

The equality holds at all original global starts, with all Odd histories included in each row. The recovered map assigns unreachable choices and has the unique extension to all forced Even moves.

The output has $O((N+1)^{12})$ zones, $O((N+1)^{13})$ zone/tag/coefficient bits, and $O((N+1)^{14})$ template bits. Preprocessing uses $O((N+1)^{125})$ bit operations and $O((N+1)^{64})$ scratch bits. A represented-profile query recovers its profile and nonforced choices with polynomial bit work. If a flat forced map of actual encoded length Q is requested, its output costs an additional $O(Q)$. The constants depend only on the fixed grammar. The relation may be empty.

The construction has three ingredients. A decreasing chain admits an exact four-port reduction (Section 3.3). A finite partition of the possible profile coordinates then records enough integer spacing to transport one policy between any two points of a cell (Section 3.4). Finally, a backward induction uses exact integer projection to decide which cells are attained (Section 3.5). This last step operates on complete suffix tuples, so the correlations between starts survive every layer. The witness templates are the same rank transports used in the constancy proof. Section 3.6 establishes policy recovery and resource bounds and completes the proof of Theorem 3.8.

3.3 Reduction of an interval chain

Lemma 3.9 (Boundary-preserving priority encodings)**.** *The original vertex-priority arena and its source edge-priority encoding have identical all-history outcomes at these zero boundary starts and exits, with original policy identities retained.*

Proof. Put the departing vertex priority $\pi(v)$ on each edge (v, w). The departed priorities before first exit are exactly those in the original prefix, except for the zero exit; the initial vertex is also zero. Finite numerical maxima therefore agree, including an arm that leads immediately to an exit. Every vertex visited on an infinite nonexit play is departed, so recurrent priority sets agree. Conversely, replace a source edge (v, w) with label p by $v \to a_{v,w} \to w$, where $a_{v,w}$ is a distinct forced vertex of priority p and original source vertices have priority zero. Keep owners and choices through

their corresponding arms. Extra zeros affect neither finite maxima nor recurring parity. Stop at the first global exit, whose source self-loop remains zero. Each original Odd history expands, and each expanded history contracts uniquely through forced vertices. Recover the same original Even successor through its edge arm and unique continuation. No priority is assigned merely because an edge enters a nonzero target. This correspondence holds simultaneously at all starts for arbitrary Odd histories. □

Lemma 3.10 (Numerical maximum is subpriority-monotone)**.** *If $a \leq_P b$, then for every priority c,*

$$\max(a, c) \leq_P \max(b, c).$$

This is the known source property [8, Remark 14, Lemma 15], which also credits its predecessor. We include its direct verification.

Proof. For even a, b with $a \leq b$ numerically, $c \leq a$ leaves the ordered evens; $a < c < b$ gives c on the left, below even b whether c is odd or even; $c \geq b$ gives equality. For odd a, b with $a \geq b$ numerically, $c \leq b$ leaves their reverse odd order. If $b < c < a$, the results are a, c; odd a is below c by reverse numerical order when c is odd, and by odd-below-even when c is even. For $c \geq a$, they coincide. The remaining source-order case has odd a and even b. If c is even, the right maximum is even, and the left is either odd a or even $c \leq \max(b, c)$. If c is odd, the left is odd $\max(a, c)$; the right is either even b, or odd c, which is above it in reverse numerical order. These cases cover the order. Commutativity gives monotonicity in either argument. Thus an attained finite excursion may be replaced by its same-exit source minimum before concatenation; numerical maximum has not been identified with maximum in the subpriority order. □

Lemma 3.11 (Feasible chain endpoints)**.** *For a gadget g, define the integer lower envelope and upper endpoint slack by*

$$\begin{aligned} &g_{h_g} = L_{h_g}, && g_i = \max(L_i, g_{i+1} + 1) \quad (i = h_g - 1, \dots, 0), \\ &F_g \iff (\forall i)\ g_i \leq U_i, && c_g = g_0, \quad a_g = U_0, \quad b_g = L_{h_g}, \\ &H_g = \min_{0 \leq i \leq h_g} (U_i - (h_g - i)). \end{aligned}$$

If F_g fails, no endpoint pair is feasible. Otherwise its exact integer relation is

$$c_g \leq s_g \leq a_g, \qquad b_g \leq t_g \leq H_g, \qquad s_g - t_g \geq h_g. \tag{4}$$

For every feasible pair, a decreasing legal choice at every original hub can be recovered using $O(h_g + 1)$ binary arithmetic operations.

Proof. The lower envelope gives necessity; taking its maximum with the prescribed right-endpoint staircase gives sufficiency. Every legal decreasing q majorizes g by backward induction. Also $q_0 \geq L_i + i$, $q_{h_g} \leq U_i - (h_g - i)$ for every i, and $q_0 \geq q_{h_g} + h_g$. These imply F_g and all endpoint bounds. Conversely, for a pair satisfying (4), put

$$q_{h_g} = t_g, \qquad q_i = \max(g_i, t_g + h_g - i)\ (1 \leq i < h_g), \qquad q_0 = s_g. \tag{5}$$

Both sequences in the maximum decrease by at least one toward the next index. The first is bounded by U_i using F_g; the second is bounded by U_i using $t_g \leq H_g$. Their maximum is at least L_i. At the first step, $q_1 + 1 \leq \max(c_g, t_g + h_g) \leq s_g$. For $h_g = 1$, this is directly $t_g + 1 \leq s_g$. The endpoints obey their legal menu bounds. Hence (5) gives a decreasing legal q, selecting the existing m_{i,q_i} at every hub. The same vector is fixed before every entrance and remains assigned when unreachable. This calculation uses every interior interval through F_g. It requires no enumeration of the numerical cap or of products of menus. □

Lemma 3.12 (The four-port reduction)**.** *For fixed decreasing q with $s_g = q_0, t_g = q_{h_g}$, the exact source-minimal finite excursion weights are*

$$\begin{array}{c|cc} & O_L & O_R \\ \hline I_L & 4s_g + 1 & 4s_g \\ I_R & 4s_g + 1 & 4t_g \end{array}. \tag{6}$$

Each is attained by an original finite excursion. Every infinite history confined to the chain wins for Even. Replacing the chain by its four ports and the four weighted Odd excursion alternatives preserves typed finite source minima and bottom availability at every boundary start for the fixed original q and router choices. It therefore preserves the complete minimal-outcome antichains in the forward ladder; it need not preserve all raw finite-priority spectra.

Proof. We first compute finite excursions, then confined infinite plays, and finally verify replacement in the surrounding arena.

Finite excursions. An I_L-to-O_L excursion takes hub 0's left arm and has maximum $4s_g + 1$, as all other labels are no larger. An I_L-to-O_R excursion takes hub 0's right arm, cannot use its outward left arm, and higher-index left arms are smaller than $4s_g$; its maximum is $4s_g$. An I_R-to-O_L excursion eventually takes hub 0's left arm, giving $4s_g + 1$. For I_R-to-O_R, let a be the least visited hub. If $a = h_g$, immediate outward right gives $4t_g$. If $a < h_g$, the excursion eventually crosses right from a; a left move there would contradict minimality of a. Its maximum is $4q_a$, since all visited higher-index arm labels are smaller. It is even and at least $4t_g$. Immediate outward left, straight all-right, straight all-left, and immediate outward right respectively attain the four displayed weights. Their source minima are therefore exactly (6).

Confined infinite plays. For an infinite confined history, let a be the least hub visited infinitely often. Lower hubs are transient. The left arm from a is transient because it visits a lower hub or exits; its right arm therefore recurs. A confined history cannot have h_g alone as its least recurrent hub, since its right arm exits. Every recurrent arm at a larger index is numerically below $4q_a$. Thus the recurrent maximum is even $4q_a$. No eventual periodicity or positional restriction on Odd is used.

Replacement in the arena. Each finite macro walk expands under the same original q using an attaining excursion at every edge, with its numerical maximum unchanged. Conversely, port-only entry lets any original finite first-exit play be partitioned into complete excursions and zero routing moves. Shorten each excursion according to (6); only an I_R-to-O_R even value can decrease in source order. Lemma 3.10 makes the shortened prefix maximum no larger in that order, without changing its global exit. Expansion makes macro outcomes available original outcomes; shortening prevents an original same-exit minimum from being below the macro minimum. Thus the typed minima agree. Macro walks may revisit local vertices: deleting every cycle without preserving a maximum would be invalid.

A losing original nonexit history cannot eventually stay in a decreasing chain. If it completes only finitely many excursions, its exterior tail remains under shortening. Otherwise partition it into finite completed excursion blocks and individual exterior edges. In a finite alphabet the largest recurrent priority equals the largest block maximum recurring infinitely often, even for unbounded finite block lengths: a larger priority occurs in only finitely many finite blocks, while a recurrent block maximum is attained in infinitely many. Shortening retains every odd block maximum and only lowers some even ones. A largest recurrent odd maximum stays largest and odd; no even maximum at least that odd value is introduced. The shortened macro history loses. Conversely, expanding a losing macro history by attaining excursions retains all block maxima and thus its odd recurrent maximum. The exterior moves remain legal under the one fixed policy and all Odd

histories. Zero exterior tails are handled in the same way. If the literal local wiring has no zero routing cycle, this part is unnecessary there; it adds no interlayer feedback to the grammar.

Available $\bot_{\text{play}}$ is the sole minimum. Without it, retain the same typed finite minima; a confined winning history gives $\top_{\text{play}}$, dominated by any finite outcome. If no finite outcome exists either, totality supplies a nonexit play, which wins, so the row is $\{\top_{\text{play}}\}$. This determines every minimal row, even when dominated winning histories or raw finite values are not retained. The endpoint guards and macro parameters are assigned once per physical chain, rather than once per start. □

3.4 Integer zones and transport of a common policy

The backward construction observes all possible entry points into a suffix. Its dimension remains bounded because the interface is bounded by the grammar.

Definition 3.13 (All-port suffix relation)**.** For the suffix S_i of layers $i, \dots, \ell$, let $\mathcal{I}_i$ be the set of all gadget ports and routers in layer i. These are physical zero-priority vertices, with

$$|\mathcal{I}_i| \le B = 2 + 2 \cdot 4 = 10.$$

Its joint relation evaluates every start in $\mathcal{I}_i$ against one common suffix policy, using the original global exits. It consequently has at most $m = 2B \le 20$ visible finite coordinates. A start at a gadget exit follows that exit's actual continuation.

This interface adds no vertex, edge or policy choice. It includes all ports because a preceding layer may enter a next-layer entrance port, exit port or router. The final representation will retain only the original global starts.

Lemma 3.14 (Integer difference constraints)**.** *A finite integral DBM conjunction is feasible over integers exactly when its constraint graph has no negative directed cycle. Intersection is exact conjunction. Eliminating hidden integer variables gives exact integral DBM projections, and every projected integer point can be extended by recovering those variables.*

The zero-anchor representation and the standard operations are described in [6, Sections 2–4, Theorems 1, 2, 5, 10]. We include the integer extension argument, which will also construct policy witnesses after projection.

Proof. Feasibility. Represent $x_j - x_k \le c$ by an edge $k \to j$ of weight c. Summing around a directed cycle excludes negative cycles in any satisfying tuple. Conversely, add a feasibility-only super-source with zero edges to all variables. Without negative cycles, shortest path distances are finite integral potentials satisfying every edge bound. Translate all potentials by minus the zero-anchor potential. Differences are unchanged and $x_* = 0$, giving an integral witness. The super-source is not an output coordinate. Here the graph has constant dimension, so a shortest-path and negative-cycle calculation suffices.

Projection and recovery. To eliminate z, retain bounds not involving it, pair each $x_j - z \le a$ with each $z - x_k \le b$, and add $x_j - x_k \le a + b$; reject a negative self-bound. These bounds are necessary. For a remaining integral tuple, z must be at least each $x_j - a$ and at most each $x_k + b$. The pair bounds say that every integral lower endpoint is at most every integral upper endpoint. Choose an integer between the largest lower and smallest upper. If one side is missing, use a finite endpoint on the other; if both are missing, use zero. Handle $z - z \le c$ separately by nonnegativity of c. Repeat elimination and recover variables in reverse order to extend every projected integer point. Shortest-path closure and restriction to retained coordinates compute the same exact projection. There is no real relaxation or integrality gap, integer-programming oracle, or enumeration of D_{cap}. Intersection retains both constraint lists and uses the same feasibility and recovery argument. □

A zone may contain many integral profiles. To recover a policy at every point from one representative, the partition must remember how many distinct selected menu levels can fit between visible levels and fixed interval endpoints. The following capped integer spacing data are sufficient.

Definition 3.15 (Input-computable capacity cells)**.** For a suffix S with r_S original hubs, let C_S be the sorted distinct interval endpoints L_i, U_i of all its hubs, together with $0, D_{\mathrm{cap}}$. Put $K_S = |C_S| \le 2r_S + 2$. Fix a valid tag with $0 \le m \le 20$ visible integer coordinates. A capacity cell specifies each coordinate's equality to a fixed constant or its position in a consecutive open constant gap; the weak order, including equalities, in every occupied gap; and, after inserting the distinct visible levels, the capped free-slot count for each adjacent anchor pair $a < b$:

$$\mathrm{cap}(a, b) = \min(b - a - 1, r_S).$$

All coordinates lie in $[0, D_{\mathrm{cap}}]^m$ and obey fixed-tag legality. Unoccupied fixed gaps retain their endpoints and need no additional cap.

Positions and weak orders are difference constraints: $a < b$ is $a - b \le -1$, and equality uses both reverse bounds. Fixed constants use $u_* = 0$. A cap $c < r_S$ means $b - a = c + 1$; a saturated cap r_S means $b - a \ge r_S + 1$. These are DBM bounds for visible or fixed anchors. The $+1$ is essential because the free-slot number is $b - a - 1$. If $r_S = 0$, saturation at zero just requires $b > a$.

Lemma 3.16 (The capacity partition)**.** *Discarding empty integer cells gives a partition of every valid tagged tuple in* $[0, D_{\mathrm{cap}}]^m$*. It is computable from the input alone and has*

$$O((K_S + 1)^m (r_S + 1)^{2m}) = O((N + 1)^{3m})$$

cells. Each has a constant number of bounds for fixed m *and* $O(N)$ *tag, identity, and coefficient bits. Consequently there are* $O((N + 1)^{60})$ *auxiliary cells,* $O((N + 1)^{12})$ *final cells, and* $O((N + 1)^{13})$ *total final zone, tag, and coefficient bits, including extrema tags with* $m = 0$*.*

Proof. Each coordinate has at most $2K_S + 1$ endpoint or gap positions. For fixed m, the weak orders are a constant-size family. If the s occupied fixed gaps contain $t_1, \ldots, t_s$ distinct visible levels, they have $\sum_j (t_j + 1) \le m + s \le 2m$ adjacent-anchor gaps. Each cap has $r_S + 1$ choices, giving the bound. Sort $O(N)$ endpoints, enumerate positions, orders and caps, form their DBMs, and discard infeasible integer cells by Lemma 3.14. Each actual tuple uniquely determines those data, proving coverage and disjointness within a tag. Discard empty tag cases; the grammar has constantly many tags.

Every coefficient is an original endpoint, endpoint difference, or one of these shifted by at most $r_S + 1$. Its bit length is $O(\log(D_{\mathrm{cap}} + r_S + 2)) = O(N)$. Constant dimension bounds the number of constraints. Multiplying the final $O((N + 1)^{12})$ cells by $O(N)$ bits gives the stated final size. No loop enumerates the numerical cap D_{cap}. □

Lemma 3.17 (Transport within a capacity cell)**.** *For every all-port suffix interface and valid fixed tag, membership in its full joint attainable relation is constant on each capacity cell. If* u, u' *are in the same cell and one original suffix policy realizes* u*, a rank transport produces one original suffix policy realizing* u' *at all interface starts simultaneously, including unreachable choices. The reverse implication holds as well.*

Proof. We construct one increasing relabelling of all selected menu levels. Because it fixes interval endpoints and preserves parity, it transports every history and all rows simultaneously.

Fix a realizing policy and its original router map. Prune each Even hub to its selected menu and each Even router to its selected successor. Unselected menus and arms have no reachable incoming

edge from any permitted start: external edges enter only ports, and a menu's sole predecessor is its hub. For comparison, normalize selected names $m_{i,q_i}, e_{i,q_i}, o_{i,q_i}$ to m_i, e_i, o_i. With the router map fixed, the selected topology, physical hubs, ports, routers, and all Odd branches are independent of q. Only the labels $4q_i, 4q_i + 1$ depend on values.

Sort the fixed constants, visible levels, and selected-q equality classes. Fix every original interval endpoint and zero, and send each visible anchor to its corresponding u' anchor. The cell fixes visible equalities and equalities with constants, so these assignments are consistent and preserve the complete anchor order.

Suppose an adjacent-anchor gap contains k hidden selected-q classes. They occupy distinct integer slots, with $k \le r_S$. If its cap is $c < r_S$, both gaps have exactly c slots, and the old witness gives $k \le c$. A saturated target gap has at least r_S slots. Send the classes in rank order to $a' + 1, \dots, a' + k$ in the target (a', b'). Use the same injection in unoccupied fixed gaps, whose endpoints and slots are unchanged. Classes equal to anchors follow the anchors. This produces an integral map f, strictly increasing on distinct classes, fixing endpoints and zero, preserving every equality, and mapping every visible level to its prescribed new value.

Replace q_i by $f(q_i)$, retaining the original router map. Since L_i, U_i are fixed, legality $L_i \le q_i \le U_i$ implies $L_i \le f(q_i) \le U_i$; strict decreases are preserved. This includes all unreachable hubs and routers in the same policy. The new choice uses an existing original $m_{i,f(q_i)}$, and every forced continuation is filled uniquely.

The normalized graphs are isomorphic at every suffix start under this one transported policy. Labels map by $4q + \epsilon \mapsto 4f(q) + \epsilon$, with zero fixed. For $q < q'$, $4q + 1 < 4q'$; for equal q, ϵ determines the order. Thus the numerical map is increasing and parity preserving. Every finite maximum maps to the maximum along the mapped history. Recurrence and parity on infinite histories are preserved, fixing $\top_{\text{play}}, \bot_{\text{play}}$; exit identities stay fixed. Source order is preserved too: equalities, reverse odd order, ordinary even order, and odd-below-even all agree. Raw histories, and hence their minimum antichains, transport simultaneously at every start.

Each visible finite coordinate is zero or a selected q level, since a finite maximum is an attained label. Its image is therefore the prescribed u', not merely a dominating profile. Starting with a witness for u' proves the reverse implication. Thus the transport is defined from an individual realizing policy, independently of any procedure for recognizing attainable cells. □

3.5 An exact backward construction

After fixing local router choices and replacing each chain by its four ports, only a constant-size graph remains in one layer. The next lemma bounds the finite walks needed to evaluate it and identifies all possible losing local infinite histories.

Lemma 3.18 (Finite maxima and recurrent cycles)**.** *In a local macro graph with at most $B \le 10$ current-layer ports and routers and absorbing next-layer or final boundary targets, each finite local walk maximum before first boundary is attained by a walk of length at most $2B + 1$ with the same start and target. A losing infinite local history exists exactly when a reachable simple directed local cycle has odd numerical maximum.*

Proof. Finite walks. For a walk of maximum λ, mark an edge attaining it. In the subgraph of edges with numerical labels at most λ, choose a simple prefix to its source and a simple suffix from its target to the same boundary. The original walk supplies both reachabilities. Boundary targets are terminal, so the prefix does not cross one and the suffix stops at first boundary. The two paths and marked edge have length at most $2B + 1$ and maximum exactly λ. A marked boundary edge has empty suffix. Permitted empty boundary-evaluation paths have maximum zero. Constant B and

bounded local outdegree give constantly many walk words. Simple paths alone would lose a finite odd maximum on a detour; retaining the marked edge matters.

Infinite histories. For a losing infinite history let its largest recurrent label M_{rec} be odd. After a finite prefix no larger edge occurs. The recurring part lies in a strongly connected component of the graph of labels at most M_{rec}, containing a recurring edge of label M_{rec}. A simple return path from its target to its source produces a simple cycle of that maximum. A finite route to it does not affect recurrence. Conversely, repeat any reachable odd-maximum simple cycle. This concerns all path histories, not a selected Odd policy. Only cycles in the supplied local graph are tested; the lemma adds no backward interlayer edge. □

Lemma 3.19 (The suffix transition)**.** *Given the normalized exact relation of suffix S_{i+1}, an explicit constant-variable DBM branch construction has union equal to the joint relation of S_i at all its interface starts. Each integral branch point has a recoverable original local-plus-suffix common policy.*

Proof. The construction retains one whole suffix tuple while enumerating the constant amount of local order information. We describe its constraints, verify the induced rows, and prove the two inclusions.

Branch constraints. Choose one accepted suffix cell Y and its tag. Its $m_y \le 20$ variables give all next-layer starts under one suffix policy. Choose one legal vector ρ of the current physical Even router moves, at most four possibilities, keeping all Odd alternatives. For each of at most two chains add its one pair (s_g, t_g) with (4). If any original chain fails F_g, reject the full input for lack of an admitted policy, even when that chain is unreachable. Assume all such guards have passed.

Build the actual macro graph from the original ports, routers, wiring, and (6). It has at most $B \le 10$ local vertices; its forward targets lie in $\mathcal{I}_{i+1}$, or are global exits in the last layer. No suffix edge returns to this layer. Read both next-layer entrance and exit ports from the all-port interface instead of collapsing it to two routers.

There are at most $m_y + 4 \le 24$ variable level atoms, plus zero. Enumerate their complete weak orders, a finite constant independent of N, D_{cap}, and conjoin their DBM equalities and strict integer bounds with Y and the endpoint guards. The order and fixed ϵ tags determine all numerical and source comparisons of $4u + \epsilon$: $u < u'$ gives $4u + 1 < 4u'$; equality is resolved by ϵ; odds reverse numerical order, evens retain it, and all odds precede all evens.

For every local start compute its entire minimum row with this same ρ, endpoints, and whole suffix tuple y. It is $\{\bot_{\mathrm{play}}\}$ if a reachable suffix row is bottom or a reachable local cycle has odd numerical maximum. Otherwise enumerate the bounded walks of Lemma 3.18 to boundaries with finite rows. Combine each local prefix maximum with its same-exit suffix minimum by numerical maximum, and take the source minimum for each typed global exit. Direct final exits have boundary value zero. If neither a bottom nor a finite candidate exists, the sole row is $\{\top_{\mathrm{play}}\}$. For a fixed weak order, reachability, cycles and walks are constant-size computations. Infinite confined chain tails win by Lemma 3.12 and are dominated unless the last case applies.

Each resulting finite priority is a selected atom. Introduce output coordinate x equal to its u level by two reverse bounds; store its ϵ in the tag. Record absence or extrema only in the tag. Conjoin these bounds with the earlier constraints. There are at most $1 + 20 + 4 + 20 = 45$ variables including the zero anchor, with constantly many bounds. Project the old and local variables by Lemma 3.14; retain the original conjunction, elimination data, ρ, and the whole next-stage cell identity for recovery. Each Y gives constantly many branches, hence $O((N+1)^{60})$ projected branches at a stage before normalization. No menu or router variable remains, and no factor four remains in a projected bound.

Evaluation of a branch. To verify the semantic calculation, a nonexit history stays in the current layer or eventually enters the forward suffix. A confined chain tail wins. Other losing local histories

correspond to odd-maximum cycles by Lemmas 3.12 and 3.18. After suffix entry its finite prefix is transient, so bottom availability is exactly a reachable bottom suffix row. For a finite global exit, the prefix reaches a boundary and the suffix supplies a finite outcome there. Without bottom in that row its stored same-exit minimum is attained, and any larger source reward can be replaced by it under the same suffix policy. Lemma 3.10 makes the concatenated maximum no worse for Odd. Conversely, that minimum is attainable after the chosen prefix by a legal history-dependent continuation. Lemma 3.18 retains every possible local maximum, and Lemma 3.12 supplies attaining original excursions. Bottom rows already dominate their finite outcomes, and top-only rows have none. Totality and Definition 3.5 verify the last case. This calculation is simultaneous for the whole y tuple and one local policy.

Soundness. An integral preprojection point gives legal current endpoints and a whole tuple $y \in Y$. Induction supplies one original suffix policy for y. Use (5) once per current physical chain and the chosen ρ. Distinct layers have disjoint choice vertices, so the choices combine into one admitted policy assigning all unreachable choices. The calculation proves its exact profile is x. Every projected point extends to this conjunction by Lemma 3.14, so no spurious profile is introduced.

Completeness. Every admitted policy has a current router vector, legal endpoints, and one suffix profile y in an accepted Y. Its levels have an enumerated weak order. The calculation determines its full current x, and its tuple satisfies that branch conjunction. Thus x lies in the projected union. Recovery need not reproduce that same original q: Lemma 3.12 proves that any recovered decreasing q with those endpoints has the same minima and bottom availability. The union is exactly the full joint attainable relation. □

Lemma 3.20 (Exact backward construction)**.** *A backward induction over the layers maintains the following invariants at every stage: the accepted normalized cells equal the full joint suffix relation; the explicit projected branch union is exact and retains its preprojection conjunctions and whole next-stage cell pointers; each accepted cell has a feasible integer intersection with a matching branch and an integral representative; and every relation point extends through a containing branch to one original suffix policy, including all unreachable choices. It terminates with exactly J_A.*

Proof. The induction represents a relation as a union of whole capacity cells. A nonempty intersection with an exact branch certifies one attainable point, and transport then certifies the entire cell.

Terminal boundary. After the last layer, every global exit has row $\{(o, 0)\}$. There is no remaining policy choice: use the singleton tagged zero DBM and the empty suffix policy. If fewer than two exits are declared, use only the existing rows. Totality is inherited and no artificial finite outcome is added. The base invariants hold.

Inductive step. Form the projected branch list $\mathcal{L}_i$ by Lemma 3.19. Generate the complete input-only capacity partition of S_i by Lemma 3.16. For each nonempty candidate cell C and matching-tag branch $Z \in \mathcal{L}_i$, test integer feasibility of $C \cap Z$ by Lemma 3.14. Accept C exactly when a test succeeds. Such a point is attainable by branch soundness; Lemma 3.17 then makes every point of C attainable. Conversely, a true point lies in one partition cell and one projected branch, so that cell is accepted. This proves exact normalization and representatives. The transition supplies the retained exact branches. Pointwise extension, the next-stage invariant, and (5) prove policy recovery. An empty suffix makes every earlier list and relation empty. A failed chain guard was already rejected globally.

Global starts. Since $\ell \geq 1$, repeating this step from layer ℓ through layer 1 is well defined. Evaluate each actual global start by its stated zero-priority edge to a physical first-layer target, or by its existing first-layer row. Keep shared or duplicate targets rather than assigning separate policies. This constant-size transition retains one whole first-layer tuple, projects unwanted coordinates, and

normalizes using the final partition with $m \leq 4$ and the full ladder's interval anchors. Lemma 3.17 applies simultaneously at these starts, giving exactly J_A.

All acceptance tests use constructed branches, input-only cells, and integral difference constraints. No accepted-cell oracle or nonuniform witness advice is an input. □

3.6 Witness templates and binary resource bounds

The semantic induction already gives a policy for each attained profile. For a uniform query procedure, we first recover a representative policy for each final cell and then use the transport lemma within that cell.

Lemma 3.21 (Policy templates)**.** *The backward construction produces one common original-policy template per accepted final cell. For every represented profile it returns one admitted original σ_E with exactly that profile at all starts and all Odd histories, including unreachable choices and the unique forced extension. A template can have polynomial size even when a requested flat forced map has actual output length Q.*

Proof. First recover a policy at one representative of each cell. Then store exactly the order data needed to transport it to every other point.

Representative recovery. An accepted $C \cap Z$ yields a representative x^* by Lemma 3.14. Recover old and local variables by reverse elimination. At the resulting entire next-stage tuple y, find its matching accepted cell and scan the next-stage projected list for a branch containing y, which exists by the exactness invariants. Extend through it and recurse once on its whole next tuple. Use one router vector and (5) once per gadget at each stage. The base policy is empty. Disjoint physical layers combine these choices into one original policy for exactly x^*, with unreachable choices assigned. A representative's branch need not contain every point of its cell, so recovery does not reuse that pointer for an arbitrary point or recurse separately per entrance.

Template and query. For each final cell, recover a representative policy and sort its selected-q classes relative to all interval endpoints and visible anchors. Store its original router map, each hub's class, equality-to-anchor assignments, and each hidden class's rank in its adjacent-anchor gap. Original intervals and physical identities remain accessible in A. This is an actual produced template, not supplied advice. A conservative size is $O(N^2)$ bits per cell.

For a query profile in J_A, use its tag and bounds to find its accepted cell and apply Lemma 3.17. Fixed-anchor classes stay fixed; visible-anchor classes take query levels; hidden classes are injected by rank into target gaps. Keep the original router map. Transport proves every menu's legality, strict chain descent, all-history correspondence, and the exact queried profile simultaneously. Select existing $x_i \to m_{i,q_i}$ and original router successors, and fill forced moves uniquely. The result is one positional policy fixed before the start, rather than witnesses chosen after starts or Odd choices.

The input explicitly describes nonforced hubs and routers, so their arithmetic and identity work is polynomial in N. If all forced vertices are explicit, the full map has polynomial size too. For succinctly generated unselected menus and arms, a requested flat listing is enumerated only as output at cost its actual encoded length Q in the stated identity format. There is no numerical-D_{cap} scan in preprocessing or nonforced extraction, and no claim that exponentially long flat output is produced in polynomial total time in N. □

Lemma 3.22 (Binary complexity)**.** *For this grammar, preprocessing uses $O((N+1)^{125})$ bit operations*

and $O((N+1)^{64})$ *scratch bits. The output has*

$$O((N+1)^{12}) \text{ zones},$$
$$O((N+1)^{13}) \text{ zone/tag/coefficient bits},$$
$$O((N+1)^{14}) \text{ common-policy template bits}.$$

A represented-profile query uses polynomial bit work for the profile and nonforced original choices, plus $O(Q)$ *for a requested flat forced map. Constants depend on the fixed grammar.*

Proof. We count the cell–branch tests, bound the integers they manipulate, and then charge storage and witness output separately.

Number of tests. Every auxiliary stage has $m \le 20$, $r_S, K_S = O(N)$, and $O((N+1)^{60})$ candidate cells by Lemma 3.16. Lemma 3.19 gives the same projected-branch bound. Tags, constant-dimensional DBMs and recovery records have $O(N)$ bits. Testing every matching cell–branch pair uses $O((N+1)^{120})$ constant-dimensional tests per layer. Each test performs constantly many additions and comparisons on $O(N)$-bit integers, giving $O((N+1)^{121})$ bit work per layer and $O((N+1)^{122})$ over $O(N)$ layers.

Coefficient size. For uniform arithmetic bounds, normalized cells use only original endpoints, endpoint differences, and shifts of size at most $r_S + 1$. The endpoint recurrence has magnitude $O(D_{\mathrm{cap}}+r)$. A branch conjunction adds these guards, source-cell bounds, weak-order constants $0, -1$, and output equalities. Projection uses shortest paths through at most 45 variables, summing only constantly many such coefficients. They retain $O(N)$ bits. Normalization returns to original-anchor cap bounds before the next layer, preventing uncontrolled growth over $O(N)$ stages. Potential witnesses and reverse extensions have polynomial, here $O(N)$, entry bits; visible and endpoint variables are bounded by their guards.

Storage and extraction. Endpoint recurrences, grammar and identity checks, sorting and branch generation fit smaller polynomial costs. Router vectors, weak orders, local walk words and cycle words form constant families because the grammar bounds B and local outdegree. They do not enumerate global policies or complete layer-order sequences. Retaining cells, branches and preprojection records over $O(N)$ layers fits the stated $O((N+1)^{64})$ scratch bound; the direct list and record accounting is conservatively $O((N+1)^{62})$. For each of $O((N+1)^{12})$ final cells, representative recovery scans one $O((N+1)^{60})$ list per stage and extends a bounded-dimensional whole tuple. This remains far within $O((N+1)^{125})$, including template materialization.

Finally $m \le 4$ gives $O((N+1)^{12})$ zones and $O((N+1)^{13})$ zone, tag and coefficient bits. Multiplying $O(N^2)$ template size by the zone count gives $O((N+1)^{14})$. A query scans the zones and transports $O(N)$ classes with polynomial binary arithmetic and identity work, then pays $O(Q)$. The large numerical cap is binary data, not an enumeration bound except in actual requested output. These conservative exponents establish encoded polynomiality, not practical performance. □

Proof of Theorem 3.8. The local endpoint relation and its witness are given by Lemma 3.11. Lemmas 3.9–3.12 justify the four-port evaluation for all histories. The integer feasibility and projection operations are exact by Lemma 3.14. The capacity partition and common-policy transport are supplied by Lemmas 3.16 and 3.17; in particular, each output zone is an accepted final capacity cell.

Lemma 3.19 constructs the exact one-layer branches, and Lemma 3.20 proves the backward induction and the final restriction to the original global starts. Lemma 3.21 constructs a template for each accepted final cell and recovers one original policy at every represented point. Finally, Lemma 3.22 gives the stated binary resource bounds. Every feasibility witness and every hidden variable used in this argument is recovered by the specified finite operations. □

3.7 A strict-profile query

The row and product orders of Definition 3.6 make strict improvement a precise predicate, including rows with different exit supports or extrema. Here the baseline is a profile, rather than the identity of a previously used policy.

Corollary 3.23 (Strict improvement over an attainable profile)**.** *Let A have the exact domain and encoding of Theorem 3.8, with length N. Let b be a fixed, explicitly encoded attainable profile in J_A, of length L_b, and set $M = N + L_b$. A uniform algorithm decides*

$$\exists v \in J_A : \quad b \leq_J v \quad \text{and} \quad \neg(v \leq_J b). \tag{7}$$

If yes, it returns an exact such v and one admitted original σ_E realizing it at all starts and all Odd histories, including unreachable choices and the unique forced extension. Safe bit time is $O((M+1)^{125}) + O(Q)$, with $O((M+1)^{64})$ scratch bits, where Q is the actual requested flat forced-output length. The baseline profile does not identify a particular old literal policy.

Proof. For a fixed tag, strict improvement is a Boolean combination of constantly many unary integer comparisons. We intersect its satisfying cases with the exact zones from Theorem 3.8.

Run the compiler. For a candidate tag T, write $v_i(u)$ for its row and decode the baseline in its own tag. Candidate and baseline may have different supports or extrema. Form

$$\begin{aligned} F_T(u) &= \bigwedge_i \bigwedge_{d \in v_i(u)} \bigvee_{c \in b_i} [c \leq_D d], \\ G_T(u) &= \bigwedge_i \bigwedge_{c \in b_i} \bigvee_{d \in v_i(u)} [d \leq_D c], \\ H_T(u) &= F_T(u) \wedge \neg G_T(u). \end{aligned}$$

The orientations are exactly Definition 3.6. There are at most two starts and two finite entries per row, or one extremum, so there are constantly many primitive comparisons.

Extremum comparisons are constants, with bottom least and top greatest, including equality. Finite outcomes at different exits compare false. At a common exit, let baseline and candidate priorities be $4a + \epsilon$ and $4u + \delta$. For $\epsilon = 1, \delta = 0$, baseline-to-candidate is true and reverse is false; for $\epsilon = 0, \delta = 1$, these reverse. For two evens they mean respectively $u \geq a$ and $u \leq a$; for two odds, respectively $u \leq a$ and $u \geq a$. Zero is covered by the even case. Every nonconstant atom is thus an integer unary bound.

Enumerate truth assignments to these constantly many atoms, keeping those satisfying H_T. Conjoin true bounds and integer complements of false bounds:

$$\neg(u \leq a) : \quad u_* - u \leq -a - 1, \qquad \neg(u \geq a) : \quad u - u_* \leq a - 1.$$

The shifts retain every strict integer boundary. Each case is an integral DBM comparison cell on at most four visible coordinates and the zero anchor. Repeated atoms may be shared or left separate; Lemma 3.14 discards inconsistent assignments. Their union is exactly H_T, including absent exits and extrema, with no nonlinear policy variables or real relaxation.

For each accepted zone C with tag T, and each comparison cell D, test exact integer feasibility of $C \cap D$. Success gives a profile v by the potentials of Lemma 3.14, with the zone's $[0, D_{\text{cap}}]$ guards. Lemma 3.21 recovers one original common policy with exact profile v, including all unreachable choices. The profile satisfies H_T, so it proves (7). It is a whole-profile witness, rather than independent row witnesses.

Conversely, any improving policy's full profile belongs to an accepted zone by Theorem 3.8. Its tag determines the formulas and its primitive truth values give an enumerated satisfying case. That zone–case intersection contains the profile and is feasible. All tests fail exactly when no improving profile exists.

The compiler has $O((N+1)^{12})$ final zones, with constantly many comparison cases per zone in dimension at most five. Added coefficients are baseline integers and shifts by one, with $O(M)$ bits. Intersection and potentials use constantly many $O(M)$-bit operations, so a conservative scan bound is $O((M+1)^{13})$. Entries stay bounded by the arena guards. Template extraction is polynomial in M, followed by $O(Q)$ actual forced output. The compiler's preprocessing and scratch bounds imply the displayed total bounds. Neither numerical priority enumeration nor an iteration bound is used.

Attainability is a hypothesis. Without its promise, Theorem 3.8's exact membership test can check $b \in J_A$ and recover some baseline witness. Several policies can realize b, so this does not recover a specific formerly used policy. The procedure searches arbitrary admitted policies: it prescribes no unit endpoint or router move, fixed exterior, number of changed choices, repeated-pivot process, convergence, or total-runtime bound.

□

3.8 Strict improvement within a legal one-choice neighborhood

A local update is a condition on the original policy map. The same profile can be realized by several maps, with different legal neighbors. We therefore take the old policy itself as input and restrict a candidate arena to each proposed edit. Its joint relation becomes a singleton, whose profile can be compared directly with the baseline.

Definition 3.24 (One-choice neighborhood)**.** Let A have the domain of Definition 3.1. Write $V_A(\sigma) = (R_i(\sigma))_i$. The additional input is an actual admitted original policy σ_0, an encoded profile $b = V_A(\sigma_0)$, and a set F of mutable original hubs and Even routers. Define $\mathcal{N}(A, \sigma_0, F)$ to consist of precisely the admitted policies obtained by changing one hub in F from q_i to $q_i + 1$ or $q_i - 1$, or one Even router in F to its other distinct existing successor. Every other original Even choice is fixed, including unreachable choices and all forced moves. A forced router has no alternative. An endpoint change followed by reconstruction of several hubs is not a member of this neighborhood.

Corollary 3.25 (Exact strict-neighbor decision)**.** *Let r be the original hub count and ℓ the layer count. A uniform procedure decides whether some $\sigma \in \mathcal{N}(A, \sigma_0, F)$ satisfies*

$$b \leq_J V_A(\sigma) \quad \textit{and} \quad V_A(\sigma) \not\leq_J b,$$

and if so returns such an original policy and its exact whole profile. Let L_σ, L_b, L_F be the actual lengths of the supplied old-policy, baseline and mutable-set representations, and put $M = N + L_\sigma + L_b + L_F$. There are at most $K = 2r + 2\ell$ candidates. The total bit cost is

$$O((K+1)(M+1)^{125}) + O(Q),$$

with $O((M+1)^{64})$ scratch bits when candidates are processed in sequence. Here Q is the requested flat listing length of all original forced Even moves of A. The supplied equality $b = V_A(\sigma_0)$ can also be checked within this bound.

Proof. The algorithm enumerates legal original-coordinate edits. For each edit, it pins the corresponding common policy in a restricted copy of the arena, computes its singleton profile relation, and compares that profile with the baseline. The restriction preserves the entire set of histories from every permitted start.

Legal edits. Read the old map and check its original identities, menu legality, strict descent in every chain and router legality. Reject an invalid old map as outside the input specification. For each mutable hub test both signed unit edits against its interval and its immediate predecessor and successor inequalities, where present. The other descent inequalities are unchanged. For each mutable Even router test its distinct other successor. All remaining choices are copied exactly. Odd routers are not coordinates of the Even policy. Redundant encodings of an identical successor do not create a new policy. There are at most two proposals per hub and one per original Even router, hence at most $2r + 2\ell$. Each retained map σ_c is admitted by these checks, and every map in the stated neighborhood occurs. Equal profiles are not used to identify different policy maps. An empty legal list correctly gives no strict neighbor.

A pinned arena. For each retained σ_c, make a separate arena A_c with an explicit identity embedding into A. Retain every original hub, port, router, layer, track, start and typed exit, with the same priorities and ownership. At hub i, replace the interval by the singleton $[q_i^c, q_i^c]$. Retain its originally named menu m_{i,q_i^c}, both arms e_{i,q_i^c}, o_{i,q_i^c}, their priorities and their original forced targets. Omit the other menus and their arms from this restricted copy only. At an Even router retain the selected successor; at an Odd router retain every original alternative. Keep each router's identity even when it becomes forced or unreachable. All other retained forced continuations are unchanged.

The singleton intervals are legal, contiguous and nonempty. Chain lengths, track disjointness, the two declared routers per layer, ownership and forward layer direction are unchanged. Router targets survive because exterior wiring enters only permitted ports and routers, not omitted internal menus or arms. The targets of selected arms are retained adjacent hubs or exits. Thus totality before stopping and all wiring restrictions of Definition 3.1 hold. A singleton menu formerly numbered $q > 1$ retains that number and priority; this operation does not relabel it to 1. There is exactly one admitted complete policy τ_c on A_c, including unreachable choices, because every Even choice is unique and the selected chains are decreasing.

Lift τ_c to A through the original identities and give every omitted original forced Even arm its original unique continuation. Other forced Even moves are completed the same way. Odd menus require no Even-policy choice. All nonforced original coordinates were retained, so this is exactly σ_c pointwise on every original Even vertex. The generator for unique forced moves may represent that extension unless a flat listing is requested. In particular, the policy outside the one edit equals σ_0, whether or not those choices are reachable.

History correspondence. Fix an original start. Under σ_c, an omitted menu cannot be reached: its sole predecessor is its hub, whose selected menu is retained. An omitted arm can be entered only from its own omitted menu. Starts and exterior wiring do not enter internal vertices. Inductively, every finite history from the start remains among retained vertices. At a hub or Even router its selected successor is identical in the two arenas. At a retained Odd menu both arms survive; at an Odd router all alternatives survive; at a forced vertex the successor is identical. The extension sets of every such finite history are therefore identical in A and A_c. The reverse inclusion follows from the same identity embedding.

Finite first-exit histories coincide vertex for vertex, including the first exit and priority sequence. Infinite nonexit histories coincide as well, since every finite prefix is legal on each side. Any such path extends to an Odd history strategy by filling legal choices at off-path histories; at omitted menus these choices never affect a path from the permitted start. The quantifier is consequently over all history-dependent Odd strategies, not only positional strategies. Numerical finite maxima and largest recurrent priorities agree. Taking the same outcome minima simultaneously over all starts gives

$$V_{A_c}(\tau_c) = V_A(\sigma_c), \qquad J_{A_c} = \{V_A(\sigma_c)\}.$$

The same argument works from every retained port or router, so it also fits the compiler's intermediate all-port interface. It does not identify the whole selected graph of A with A_c: the former still has the omitted, unreachable vertices.

The decision and its witness. Run Theorem 3.8 on A_c. Its decoded union is the nonempty singleton just proved. Find any nonempty accepted final cell and an integral point by Lemma 3.14; decode its whole profile as v_c. Every such point gives the same $v_c = V_A(\sigma_c)$, even if the representation has duplicate cells. The template can return only τ_c, whose lift is the already constructed σ_c. No separate reconstruction per row is involved.

Test $b \leq_J v_c$ and $v_c \leq_J b$ directly with Definition 3.6. There are constantly many comparisons because there are at most two starts and two finite entries per row. Extrema compare by bottom/finite/top; finite atoms require the same exit, with reverse numerical order on odds, ordinary order on evens and every odd below every even. Accept exactly when the first product comparison succeeds and the second fails. Equal and incomparable profiles are rejected. Soundness follows from legality, exact histories and this exact predicate. Completeness follows because every legal neighbor was enumerated and its unique exact profile is extracted. Failure means that this neighborhood has no strict neighbor, and says nothing about an arbitrary admitted improving policy elsewhere in J_A.

Optionally apply the same restriction to σ_0, extract its singleton profile and compare it for equality with the supplied b. A mismatch rejects the input; this checks the distinguished old map, rather than merely finding some policy realizing b. Notice that we do not apply Corollary 3.23 to (A_c, b): if v_c is strict above b, then $b \notin J_{A_c}$, violating that corollary's attainable-baseline premise. The direct comparisons require no such premise in the restricted arena.

Bit complexity. All supplied data are charged. A long explicit forced map is read within L_σ; the compact convention checks its declared unique extension. Each restriction has size $n_c = O(M)$: copy the physical skeleton and binary chosen values, retain original identities and delete only unselected options in the declared generator. There is no loop up to D_{cap} or over products of menus. Legality checks and construction fit the stated polynomial cost. By Lemma 3.22, each compiler call costs $O((n_c+1)^{125})$ bit work and $O((n_c+1)^{64})$ scratch. Its final union has $O((n_c+1)^{12})$ zones and $O((n_c+1)^{13})$ coefficient/tag bits. Scanning the zones and producing one integral point uses constantly many $O(M)$-bit DBM operations per zone, within $O((M+1)^{13})$ work. The row comparison also fits this bound. Process candidates successively, reusing scratch, and include at most one old-baseline validation call.

The nonforced map is already known and has polynomial encoded length. If requested, output all original forced moves, including arms omitted from every A_c, at their actual length cost $O(Q)$ in the original identity format. This is the output of A, not merely the smaller pinned arena. An exponentially long flat map is therefore not claimed to be produced in polynomial time in N. The displayed bounds follow; with $K = O(N)$, a coarser time bound is $O((M+1)^{126}) + O(Q)$. □

Example 3.26 (A strict neighbor and a nonvacuous tie)**.** Take one layer with cap 3, one two-hub gadget with menus $[3,3], [1,2]$, and an empty second track. Its exactly two original routers a, b are zero-priority forced Even vertices and are the two starts. First wire $a \to I_L$ and $b \to I_R$; the two distinct global exits are O_L, O_R. Set $\sigma_0 = (3,1)$ and allow the second hub to change. The sole legal unit neighbor is $(3,2)$. All original menus, both arms and ports are present; there is no exterior feedback.

Write $l = (O_L, 13)$ and $r_p = (O_R, p)$. At a, a left exit has maximum 13, a right exit passes through the first right arm of priority 12, and a nonexit play repeats the internal cycle with priorities $12, 4t+1$ for $t = 1, 2$. Hence the raw set is $\{l, r_{12}, \top_{\mathrm{play}}\}$ in both states. At b, immediate right exit gives r_{4t}; visiting the first hub before right exit gives r_{12}; left exit gives l; and the same nonexit cycle gives top. These exhaust all histories, including arbitrary finite repetitions. The old and new

minimum profiles are therefore

$$(\{l, r_{12}\}, \{l, r_4\}) \quad \text{and} \quad (\{l, r_{12}\}, \{l, r_8\}).$$

The forward comparison holds, and the reverse fails at old r_4.

For a separate control, wire both distinct routers to I_L, retaining all other data. Both rows now equal $\{l, r_{12}\}$ in both policies. The second hub is reachable, and the same sole neighbor remains admitted, but the whole profile ties. The exact neighbor decision returns no. This is a control in the forward grammar itself; it does not import the feedback arena below.

Finite strict iteration in a fixed finite policy set cannot repeat a profile, by the strict partial order. Its number of policies may depend on products of numerical menu lengths. Neither a polynomial iteration bound nor global optimality at a state with no strict legal neighbor follows. Thus the corollary supplies an exact local-update test in this grammar. A bound for an iterated improvement procedure would require additional information about the length of strict chains and the chosen pivot rule.

4 Strict local improvement and a feedback tie

Strict comparison of local profiles need not remain strict after feedback. We give two original policies that differ at exactly one hub. Their complete local profiles are strictly ordered, whereas their complete global raw outcome sets are equal. The construction uses a fixed exterior and retains all original menus, including unselected options.

4.1 Literal fragments and the fixed exterior

Definition 4.1 (Original literal interval fragment)**.** A fragment X has $r_X \geq 2$ selectors $x_0, \ldots, x_h$, $h = r_X - 1$, and nonempty positive integer interval menus $[L_i, U_i]$. Its distinct original vertices are the r_X Even selector hubs, four Even ports I_L, I_R, O_L, O_R, one Odd menu $m_{i,q}$ for every $q \in [L_i, U_i]$, and two distinct forced arms $e_{i,q}, o_{i,q}$ for every menu. Port I_L forces to x_0, and I_R forces to x_h. Each hub has all its original menus as successors, and each menu has both arms. The right arm goes to x_{i+1}, or O_R at $i = h$; the left arm goes to x_{i-1}, or O_L at $i = 0$. Isolated O_L, O_R have their own loops. Forced arms may be Even-owned because each has one successor. Their priorities are $4q$ and $4q + 1$, and hubs, menus, and ports have zero priority. Every identity includes the fragment prefix. No policy pruning deletes an original vertex, option, or arm.

A feasible fragment policy chooses legal $q_0 > \cdots > q_h$, with endpoints $(s, t) = (q_0, q_h)$. For a prescribed feasible endpoint pair, use the decreasing witness of Lemma 3.11, equation (5), to fix the full policy before the entrance is chosen. This is a statement about one interval chain; it imposes no forward-layer restriction on its exterior. For a longer chain, changing an endpoint may also change the recovered interior choices. The construction below has exactly two selectors in each fragment, so its endpoint change changes exactly one original hub. For $r_X = 2$, its witness is simply $q_X = (s, t)$, and

$$\mathcal{R}_X = \{(s, t) : L_0 \leq s \leq U_0,\ L_1 \leq t \leq U_1,\ s > t\}.$$

An endpoint neighbor changes one fragment's pair by one of $(s + 1, t), (s - 1, t), (s, t + 1), (s, t - 1)$, keeping only feasible pairs and fixing the other fragment and exterior. No cross-fragment inequality $t_A > s_B$ is required.

For this section, local evaluation restores a fragment's exit loops and stops at either exit. Global evaluation instead uses the specified feedback wiring and stops only at the remaining typed global

exit. The outcome convention and row order are those of [8, Definitions 1–3, 7–9, 13, 16–19, 22–24, Proposition 23]; we spell out the local specialization.

Definition 4.2 (All-history profiles and strict comparison)**.** In a finite total arena, fix one positional Even policy σ_E at every original Even vertex, including unreachable vertices, before choosing an entrance. Use it after every return. Odd may choose a legal successor from the entire finite history.

Typed exits are absorbing boundaries. A finite first-exit play has outcome (o, m), where m is the ordinary numerical maximum of source edge priorities before first arrival at o. An infinite nonexit play is $\top_{\text{play}}$ if its largest recurrent priority is even and $\bot_{\text{play}}$ otherwise. Lemma 4.6 gives the same outcomes in the forced-arm vertex encoding used below.

Choose an even ambient priority bound d_W at least as large as every priority in the arena. For interval fragments this means $d_W \geq \max_{X,i}(4U_i^X + 1)$, with the maximum taken over all original menus, including unselected ones. Let $P_W = \{0, \ldots, d_W\}$ and

$$D_{\mathcal{O}} = \{\bot_{\text{play}}, \top_{\text{play}}\} \cup \{(o, p) : o \in \mathcal{O},\ p \in P_W\}.$$

The subpriority order is

$$d_W - 1 <_P d_W - 3 <_P \cdots <_P 1 <_P 0 <_P 2 <_P \cdots <_P d_W.$$

It reverses numerical order on odds, puts all odds below all evens, and retains numerical order on evens. Outcome order $\leq_D$ has bottom least, top greatest, and compares finite outcomes only at the same exit using $\leq_P$.

For entrance i, define

$$T_i(\sigma_E) = \{\text{outcome}(\rho) : \rho \text{ is a legal play from } i \text{ under } \sigma_E\}, \qquad F_i(\sigma_E) = \min_{\leq_D} T_i(\sigma_E).$$

Every Odd history is included. Each trace extends to an Odd history strategy by assigning legal moves at unvisited histories. Totality gives nonempty raw sets and minimum antichains. For nonempty antichains and an ordered entrance list J, use

$$C \leq_R D \iff (\forall d \in D)(\exists c \in C)\ c \leq_D d, \qquad F \leq_J F' \iff (\forall i \in J)\ F_i \leq_R F'_i.$$

The strict predicate is

$$\text{Strict}_J(F, F') = (F \leq_J F') \wedge \neg(F' \leq_J F).$$

It rejects equality and incomparability. Local and global tests use this same predicate on their explicitly specified arenas.

Definition 4.3 (Two fragments with fixed feedback)**.** Take disjoint original fragments with two selectors each and menus

$$A : [3, 3], [1, 2], \qquad B : [2, 2], [1, 1].$$

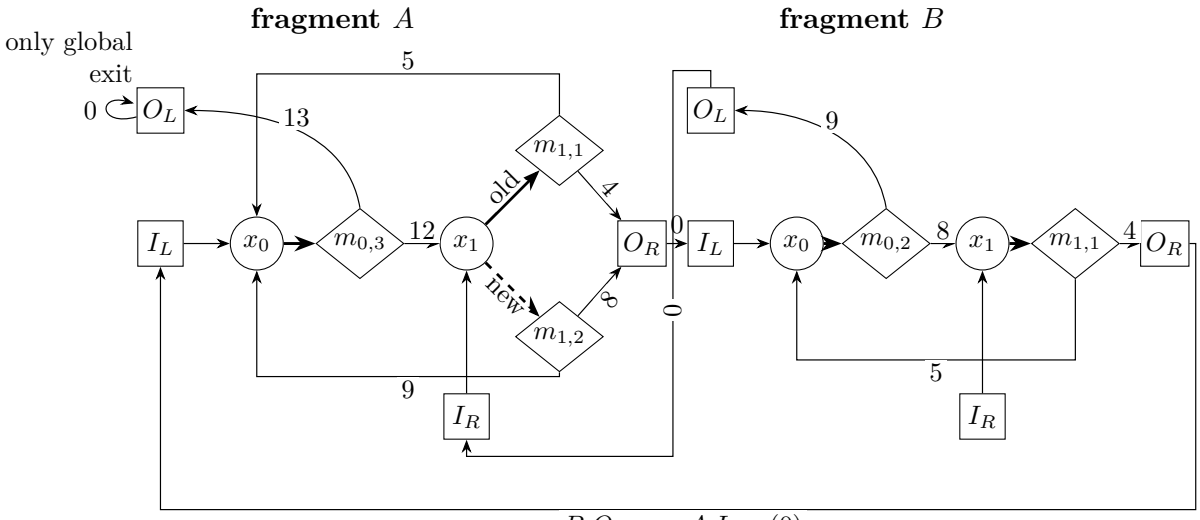
The largest original priority is 13, so we may take $d_W = 14$. The policies select

$$q_A^{\text{old}} = (3, 1), \qquad q_A^{\text{new}} = (3, 2), \qquad q_B = (2, 1).$$

Together with every forced original Even move, these define $\sigma_E^{\text{old}}, \sigma_E^{\text{new}}$. Only $A.x_1$'s successor changes, from $A.m_{1,1}$ to $A.m_{1,2}$; every other selected successor is identical.

Replace the indicated exit loops by forced zero-priority edges

$$A.O_R \to B.I_L, \qquad B.O_L \to A.I_R, \qquad B.O_R \to A.I_L.$$



	selected q_A	selected q_B	local row at $A.I_R$	global row at every start
old	$(3,1)$	$(2,1)$	$\{l, r_4\}$	$\{\bot_{\text{play}}\}$
new	$(3,2)$	$(2,1)$	$\{l, r_8\}$	$\{\bot_{\text{play}}\}$

Figure 3: The two original policies in the fixed feedback arena. Circles and port boxes are Even; diamonds are Odd. All drawn vertices have priority zero. An arrow from a menu labelled p abbreviates the two-edge path through its own distinct forced Even arm of vertex priority p; other arrows are zero. Thus the picture retains all original incidences, including both menus at $A.x_1$. Its labelled solid and dashed bold edges mark the old and new selections; the other bold edges are selected in both policies. Neither option is deleted. Unmarked crossings are not vertices. The literal counts are $A : 2 + 4 + 3 + 6 = 15$ and $B : 2 + 4 + 2 + 4 = 12$ (hubs, ports, menus, arms), not the number of drawn nodes. The table uses $l = (A.O_L, 13)$ and $r_p = (A.O_R, p)$. The local $A.I_L$ row is unchanged at $\{l, r_{12}\}$, so the local profile is strictly improved in the product row order. With all three displayed exterior edges, the global raw set from each of the four entrances is exactly $\{\bot_{\text{play}}, \top_{\text{play}}, l\}$ in both policies.

The first two are crossovers; the third is the sole additional outer return. Preserve the port vertices. Only $A.O_L$ remains a typed global exit, with its zero loop. The exterior policy ξ_{ext} is this fixed collection of forced moves, with no added vertex, identical in both states. Declare $J = (A.I_L, B.I_R)$. We will additionally evaluate the inner entrance starts $A.I_R, B.I_L$ in the same arena.

Figure 3 shows the complete incidence data with forced arms compressed to labelled paths. Both choices at $A.x_1$ remain present; only the selected edge differs between the two policies.

Theorem 4.4 (Local strictness with a global tie)**.** *In the two-disjoint-original-fragment architecture of Definitions 4.1–4.3, requiring decreasing policies within each fragment, the legal unit neighbor* $A : (3,1) \to (3,2)$ *strictly improves the whole local two-entrance profile of A, also after appending the unchanged profile of B. The recomposed global profile on* $J = (A.I_L, B.I_R)$ *is equal. Equality still holds when both inner entrance starts are included. The global strict predicate rejects this sole legal neighbor as a tie. Both original Even policies admit a losing nonexit Odd history from each named start.*

The proof separates the local profile calculation from the recurrence created by the exterior.

First, the interval menus determine exactly which update is legal.

Lemma 4.5 (The unique legal update)**.** *Both policies are feasible, and the only legal unit endpoint neighbor of the old state is* $A : (3,1) \to (3,2)$*.*

Proof. The exact endpoint sets are $\mathcal{R}_A = \{(3,1),(3,2)\}$ and $\mathcal{R}_B = \{(2,1)\}$. The first menu of A fixes $s = 3$; its second permits $t = 1, 2$, both below 3. Fragment B fixes $s = 2, t = 1$. Changing A's first coordinate or either coordinate of B leaves a singleton menu, and lowering A's t from 1 gives illegal zero. Raising it to 2 is the sole allowed change. With two selectors these pairs are the full original witnesses, using existing menus with strict descent. Neither old $t_A = 1$ nor new $t_A = 2$ satisfies $t_A > s_B = 2$. Such a condition would define a stronger composite-descending family outside this hypothesis. □

4.2 The local calculation

The literal construction assigns priorities to forced arm vertices. The following encoding permits the source edge-priority notation while retaining the original choices and every history.

Lemma 4.6 (The edge-priority encoding)**.** *For each of the four entrance starts, every legal literal vertex history has the same typed first-exit or nonexit parity outcome in its source edge-priority encoding.*

Proof. Give a selected menu's right and left alternative edges labels $4q, 4q+1$, and subdivide each by its own forced arm of that priority. Original hubs, menus and boundary vertices have zero priority. Each menu choice is followed by that arm and its sole successor. Subdivision adds no player choice and removes no Odd alternative; it retains one original selector policy at every occurrence and revisit.

Before first exit, each traversed nonzero source edge contributes exactly its arm's priority; every visited arm completes its step before exit arrival. The exit is zero, so including it in the vertex maximum does not change the edge maximum before arrival. Other added values are zero. Equivalently, keep the arm graph and put each arm priority on its menu-to-arm edge, with all other edges zero; complete traversals have the same maxima.

On an infinite nonexit play a source alternative recurs exactly when its own arm recurs. Forced arms cannot retain a play. The recurrent nonzero priority set and its largest value and parity are unchanged, and recurring zeros do not alter that parity. All-history raw sets and minima are preserved without identifying ports or deleting unselected original menus. This is the endpoint-safe use of [8, Definitions 1, 8]. □

For the isolated evaluation of A, write $t = 1$ or 2 for the second selected menu in the old or new policy. Put

$$l = (A.O_L, 13), \qquad r_{12} = (A.O_R, 12), \qquad r_4 = (A.O_R, 4), \qquad r_8 = (A.O_R, 8).$$

Lemma 4.7 (The isolated fragment)**.** *The two-entrance local tuple of* A *has*

$$T^{\text{old}}_{A.I_L} = T^{\text{new}}_{A.I_L} = \{l, r_{12}, \top_{\text{play}}\},$$
$$T^{\text{old}}_{A.I_R} = \{l, r_4, r_{12}, \top_{\text{play}}\}, \qquad T^{\text{new}}_{A.I_R} = \{l, r_8, r_{12}, \top_{\text{play}}\}.$$

Consequently,

$$F^{\text{old}}_{A.I_L} = F^{\text{new}}_{A.I_L} = \{l, r_{12}\}, \qquad F^{\text{old}}_{A.I_R} = \{l, r_4\}, \quad F^{\text{new}}_{A.I_R} = \{l, r_8\}.$$

Proof. We classify first-exit plays by their exit and by whether they visit the first hub. A nonexit play has only one possible internal cycle.

A left-exit play ends its nonzero arms with $A.o_{0,3}$ of priority 13, and all other priorities are at most 13. Its maximum is exactly 13. Left exit is reachable directly from $A.x_0$, or from $A.x_1$ through its left arm and then $A.x_0$.

Every right-exit play from $A.I_L$ passes through $A.x_0$. Avoiding left exit there requires arm $A.e_{0,3}$ of priority 12; no remaining selected arm is larger. Thus its maximum is 12, attained on a straight rightward play. From $A.I_R$, immediate outward right at $A.x_1$ attains $4t = 4$ or 8. If it first visits $A.x_0$, it later uses arm 12 to reach right exit, giving maximum 12. One left step followed by the straight right route attains it. A right-exit history either visits $A.x_0$ or does not, so these are all possibilities.

A nonexit infinite local play cannot use an outward endpoint arm; it repeats the internal walk

$$\begin{aligned} A.x_0 \to A.m_{0,3} \to A.e_{0,3} \to A.x_1 \\ \to A.m_{1,t} \to A.o_{1,t} \to A.x_0. \end{aligned}$$

Its recurrent nonzero priorities are $12, 4t+1$, with $4t+1 = 5$ or 9, so its largest is even 12. Every nonexit local history wins, and top is attained from both entrances; bottom never occurs. This exhausts local histories, including arbitrary finite repetitions.

Top is above each finite outcome. At the right exit, $4 <_P 12$ and $8 <_P 12$. Left and right outcomes are incomparable because their exits differ. Removing exactly the dominated outcomes gives the displayed minima. □

Lemma 4.8 (Strict local comparison)**.** *The old whole local two-entrance profile of A is strictly below the new one in the product upper/result order. Appending B's unchanged whole local profile retains this strict comparison.*

Proof. At $A.I_L$ the antichains agree. At $A.I_R$, the forward comparison uses $l \le_D l$ and $r_4 \le_D r_8$, since $4 <_P 8$. For the reverse test, old r_4 has no new predecessor: l has another exit and $r_8 \not\le_D r_4$. The forward product inequality holds and the reverse fails at $A.I_R$. The isolated arena and selected full policy of B are identical, so all its components agree. Appending them changes neither conclusion. □

4.3 The global calculation

The local change raises a finite right-exit minimum from 4 to 8. In the feedback arena both values lie below the recurring odd priority 9 of an available interface cycle. This cycle forces the same minimum row under both policies. The next lemma proves the stronger equality of all raw outcomes.

Lemma 4.9 (The feedback arena)**.** *Under either policy, each $i \in \{A.I_L, A.I_R, B.I_L, B.I_R\}$ has*

$$T_i = \{\bot_{\text{play}}, \top_{\text{play}}, (A.O_L, 13)\}, \qquad F_i = \{\bot_{\text{play}}\}.$$

Proof. We exhibit each of the three outcomes from every start, then classify all remaining histories. The losing cycle uses the two crossover edges and avoids the first hub of A; the outer return makes the finite exit and the winning cycle reachable from the B starts as well.

A losing history. For $t = 1$ or 2, Odd can repeat

$$\begin{aligned} A.x_1 \to A.m_{1,t} \to A.e_{1,t} \to A.O_R \to B.I_L \to B.x_0 \\ \to B.m_{0,2} \to B.o_{0,2} \to B.O_L \to A.I_R \to A.x_1. \end{aligned}$$

Every menu is the same policy's selected successor on every visit, including B's original left alternative. The recurring nonzero sets are $\{4, 9\}$ or $\{8, 9\}$. Their largest priority is odd 9, and $A.O_L$ is never visited.

This cycle is reachable without $A.O_L$ from all four starts. From $A.I_L$, take arm 12 to $A.x_1$; $A.I_R$ enters there directly. From $B.I_L$, take $B.x_0$'s left arm 9 into $A.I_R$. From $B.I_R$, take left arm 5 to $B.x_0$, then arm 9 to the same entrance. The first route's priority 12 is transient and not in the recurring set. Thus bottom is attained in both policies from each start.

Finite exit and winning histories. Every first-exit play reaches $A.O_L$ through $A.o_{0,3}$, priority 13. All priorities are at most 13, so its maximum is 13. Each start reaches $A.x_0$ and takes that arm: $A.I_L$ enters directly, $A.I_R$ uses its left arm, and either B start can move right through $B.O_R$'s return to $A.I_L$. Thus $(A.O_L, 13)$ is attained from all starts.

Using those routes to $A.x_0$, instead repeat the internal A cycle of Lemma 4.7. Its recurrent maximum is even 12, and it avoids the global exit. Hence top is also attained from each start.

Exhaustion. To exhaust infinite histories, A's left arm 13 never occurs in a nonexit play. If $A.x_0$ recurs, its right arm 12 recurs and dominates all other usable arms, giving top. Otherwise, after its last visit neither $B.O_R$ nor $A.I_L$ can occur. Each later visit to $A.x_1$ takes its right arm $4t$, and each later visit to $B.x_1$ takes its left arm 5. If B's arm 9 recurs, the largest recurrent priority is 9, giving bottom. If it does not, after its last occurrence no return to $A.x_1$ is possible, and the history eventually repeats B's internal cycle with priorities $\{8, 5\}$, giving top. These cases exhaust the forced-port and selector graph.

Every finite exiting history gives the one exit outcome; every infinite nonexit history gives one of the two extrema, and all three are attained. The raw sets therefore agree exactly. Since bottom is below every outcome, every minimum antichain is its singleton. □

Proof of Theorem 4.4. Lemma 4.5 proves legality for exactly the stated menus and per-fragment descent. Lemmas 4.6 through 4.8 give all-history local profiles and their strict comparison. Lemma 4.9 gives identical raw global sets and singleton minima in both states. The global product inequalities hold in both directions, so Strict_J is false. The exhibited interface cycle is a legal losing history for both policies. It is an Odd play cycle; it is not an accepted global pivot or an algorithm state cycle. □

4.4 Size of the literal construction

The vertex count is taken before fixing a policy: every original menu and both of its arms count, as do all four distinct boundary ports.

Theorem 4.10 (Minimum size in the literal architecture)**.** *Let $M_X = \sum_i (U_i - L_i + 1)$ count original menu options in a literal fragment with $r_X \geq 2$. Then*

$$|V_X| = r_X + 4 + 3M_X.$$

A static pair of disjoint fragments has at least 24 *original vertices. A unit endpoint change between two legal original policies in this architecture requires at least* 27 *in the original pair, before extra exterior vertices. Definition 4.3 attains* 27*, with* 15 *in A and* 12 *in B.*

Proof. The disjoint categories are r_X selectors, four ports, M_X Odd menus and $2M_X$ forced arms, giving the formula. Nonempty menus imply $M_X \geq r_X$, hence

$$|V_X| \geq 4r_X + 4 \geq 12.$$

Disjointness and boundary-preserving crossover edges give $12 + 12 = 24$ for a static pair. A unit change uses two different legal original labels at its changed endpoint selector, so that selector has at least two options and all others at least one. The changed fragment has $M_X \geq r_X + 1$, giving

$$|V_X| \geq 4r_X + 7 \geq 15.$$

The other still needs 12. Edges do not identify or delete ports, and extra exterior vertices cannot lower the count. The pair thus needs at least $15 + 12 = 27$.

In the instance, $r_A = 2, M_A = 3, r_B = 2, M_B = 2$, giving $2 + 4 + 9 = 15$ and $2 + 4 + 6 = 12$, with no extra exterior vertex. Lemma 4.5 supplies an actual legal unit change, so the bound is attained. The distinct actual priorities are $\{0, 4, 5, 8, 9, 12, 13\}$. The even ambient bound $d_W = 14$ adds no actual priority or vertex. □

The lower bound is specific to this literal two-fragment architecture. Allowing another encoding or identifying boundary vertices changes the counting problem. The two policies need only decrease within their own fragments; neither cross-fragment descent nor a winning old policy is assumed.

The tie is compatible with numerical-maximum monotonicity and the source's static shortcut and compositionality results [8, Lemma 15, Definition 41, Propositions 42–43]. It shows why an iterated local strategy improvement procedure would need to test its global strictness condition. The displayed cycle is an Odd play, not an accepted global pivot or a cycle of algorithm states. Neither the tie nor the counting argument gives a termination or optimality certificate or a running-time bound. This arena has return edges excluded from Section 3; no correspondence with the carried-floor iteration state of Section 7 is assumed.

4.5 Forward minimum profiles do not determine marked feedback

The forward compiler of Theorem 3.8 represents the joint minimum profiles of complete original policies. Those minima need not retain enough information to evaluate a marked backward connection. The following example uses the all-history outcome and result orders of Definitions 3.3–3.5, adapted from the ordered open-game semantics of [8, Definitions 7–9, 13, 16–19, 22–24]. It compares two arenas: a forward arena A, and an altered arena A^{fb} obtained by replacing one port continuation. It does not identify a forward profile with a pre-forget boundary record in the altered arena.

Take three layers with routers R_{j0}, R_{j1}, $j = 1, 2, 3$, and typed zero sinks s_0, s_1. The three disjoint two-hub gadgets are

gadget	layer, track	x_0 menu	x_1 menu
L_1	1, left	$[2, 3]$	$[1, 1]$
U_1	1, right	$[4, 4]$	$[1, 1]$
L_2	2, left	$[3, 3]$	$[1, 1]$

with the four distinct ports, original menu vertices and forced arms of Definition 3.1; set $D_{\mathrm{cap}} = 4$. There is no layer-2 right gadget or layer-3 gadget. The starts are $i_0 = R_{10}$ and $i_1 = R_{11}$. Router R_{10} is Odd-owned with alternatives $L_1.I_L, U_1.I_L$; every other router has the single continuation specified below. Both exits of L_1 lead to $L_2.I_L$, both exits of U_1 lead to R_{20}, and both exits of L_2 lead to R_{30}. The remaining router moves are

$$R_{11} \to R_{21}, \qquad R_{20} \to R_{30}, \qquad R_{21} \to R_{31}, \qquad R_{30} \to s_0, \qquad R_{31} \to s_1.$$

Every unused entrance port retains its forced move to its designated hub; all forced, unreachable and unchosen original vertices remain present. This is a literal forward arena of Definition 3.1 with

$15 + 12 + 12 + 6 + 2 = 47$ original vertices. Its exactly two admitted complete policies, σ_2, σ_3, choose respectively 2 or 3 at $L_1.x_0$, choose 1 at $L_1.x_1$, and take all other singleton or forced Even choices. In particular, Odd's R_{10} branch is not another Even policy.

Mark the layer-2 local exit $p = L_2.O_R$ and the layer-1 entrance $q = L_1.I_L$. In A^{fb}, replace only the forward continuation $p \to R_{30}$ by $p \to z \to q$, with a fresh forced priority-zero vertex z. This makes 48 original vertices and preserves the two policies and the distinct global starts and sinks. The local exit p is not a stopped global exit. Evaluate both arenas by the same first-global-exit and infinite-play rules, using the same complete original Even policy at every return.

Proposition 4.11 (A marked feedback minimum-profile collision)**.** *For the fixed marked alteration above, the profiles at (i_0, i_1) are*

	$V_A(\sigma)$	$V_{A^{\mathrm{fb}}}(\sigma)$
σ_2	$(\{(s_0, 17)\}, \{(s_1, 0)\})$	$(\{(s_0, 17)\}, \{(s_1, 0)\})$
σ_3	$(\{(s_0, 17)\}, \{(s_1, 0)\})$	$(\{\bot_{\mathrm{play}}\}, \{(s_1, 0)\})$

Thus no function of a policy's forward joint minimum profile alone returns its feedback joint minimum profile, even for this fixed marked arena. Moreover, replacing only the menu $[2, 3]$ at $L_1.x_0$ by $[2, 2]$ leaves the forward attainable relation J_A unchanged but removes the second feedback profile. Consequently no uniform operator whose sole input is the forward attainable minimum-profile relation determines the feedback relation throughout this marked class.

Proof. The only nonforced Even menu is $L_1.x_0$. Its two values exceed the forced $L_1.x_1 = 1$; the other gadgets also descend strictly. Hence σ_2, σ_3 are exactly the admitted complete original policies, including their assignments at unreachable vertices.

From i_1, the forced zero path reaches only s_1, giving $\{(s_1, 0)\}$ under either policy in either arena. Every finite play from i_0 reaches only s_0. Odd can choose $U_1.I_L$, then the left arm at $U_1.x_0$, and reach s_0 with numerical maximum 17. No priority in the original arena exceeds 17, and 17, the greatest odd priority present, is least among its priorities in $\leq_P$. Thus $(s_0, 17)$ is the least finite i_0-outcome in all four evaluations.

In A, exterior connections only move forward. Any infinite nonexiting play must eventually cycle inside one gadget. Such a cycle traverses a first-hub right arm of priority $4q_0$ and a second-hub left arm of priority $4q_1 + 1$; strict descent $q_0 > q_1$ makes $4q_0$ its greatest recurrent priority, which is even. Hence no forward play has outcome $\bot_{\mathrm{play}}$.

In A^{fb}, every external return through p must traverse $L_2.x_0$'s right arm of priority 12 and $L_2.x_1$'s right arm of priority 4. Under σ_2, every other priority on such a return is at most 12; in particular, $L_2.x_0$'s odd arm 13 goes to $L_2.O_L$ and then to s_0, not to p. If returns recur infinitely often, their recurrent maximum is therefore the even priority 12. If only finitely many occur, an infinite play eventually stays in one gadget, where the preceding even-cycle argument applies. Thus σ_2 has no losing nonexit history.

Under σ_3, Odd may repeatedly take the left arm of $L_1.x_0$, of priority 13, then the right arms of $L_2.x_0, L_2.x_1$, of priorities $12, 4$, and return through p, z, q. This legal infinite history reuses the same original Even policy and has greatest recurrent priority 13, so its outcome is $\bot_{\mathrm{play}}$. By Definition 3.5, bottom is the least outcome, making that entire row $\{\bot_{\mathrm{play}}\}$, despite the still-attainable finite $(s_0, 17)$.

The table follows. Its two equal forward inputs and unequal feedback outputs disprove a policy-level minimum-profile-only map. For the relation-level assertion, the singleton menu $[2, 2]$ deletes only σ_3 from the admitted policies; the sole remaining policy still has the same forward profile and the nonbottom feedback profile.

This compares the two physical arenas, not merely two policy sets on one arena. The restriction removes exactly the original menu vertex $L_1.m_{0,3}$ and its two forced arms; the forward and marked original vertex counts become 44 and 45. In the graph selected by σ_2, the only possible incoming edge to this menu is the unselected edge from $L_1.x_0$, and each removed arm has its only incoming edge from that menu. No exterior edge enters an internal menu or arm by Definition 3.1. Thus the removed triple is unreachable from either global start, before or after marking. On the remaining vertices the selected graphs are identical: the same Even moves, all Odd alternatives, priorities, starts, exits and marked continuation remain. The identity therefore matches every finite first-exit history and every infinite nonexit history, preserving its outcome. In particular, the complete raw and minimum profiles of σ_2 agree across the original and singleton-menu arenas.

The two forward attainable relations are therefore equal singletons, while their feedback attainable relations differ. □

There is also a one-vertex boundary forget in an unrestricted-width tree decomposition, from $B = \{s_0, s_1, z\}$ to $B' = \{s_0, s_1\}$ within the *feedback* arena: a sufficiently wide bag containing all 48 vertices covers every edge, and z has no occurrence above its forget node. This observation does not turn V_A into a pre-forget record for A^{fb}, nor does the forget create the feedback edge. The 48-vertex instance is outside the numerical and width caps of Definition 6.1 and its fixed-policy spectral theorem. Registered Theorem 6.2 instead changes a raw no-return flag while its closed minima stay equal; Proposition 4.11 changes a closed minimum after a marked alteration. An algorithm also given the arena, policy, marker or richer spectral data is not excluded. In particular, the example is no lower bound on the size of an all-policy post-feedback zone representation.

5 An ordered calculus and fixed-policy contexts

Two kinds of local reasoning will be distinguished. First, a restricted syntax of open games admits a canonical form for its entrance-local Pareto front. Second, an acyclic context determines an explicit monotone map on the outcome rows of a fixed original policy. The latter map decides whether a particular legal policy change remains strict in its context.

The ordered semantics and the shape of the canonical form come from Watanabe's outcome, result and shortcut constructions [8, Definitions 2, 7–24 and 41, Lemma 15, Theorem 28, Propositions 42–43]. Priority pushing and distributive normalization also occur in Piedeleu's completeness proof [7, Theorems IV.4–IV.6]. His discrete-priority semantics differs from the ordered semantics below; we therefore prove the equations directly for the declared syntax.

5.1 Literal syntax and its semantic types

Definition 5.1 (Literal open games)**.** Fix an ordered exit list O_{exit}, with $1 \le |O_{\mathrm{exit}}| \le 2$, and the edge-priority alphabet $P = \{0, 1, 2, 3, 4\}$. An arena is finite and total. Each entrance is Even-owned and has one outgoing edge. Each exit is Even-owned and has only its priority-zero self-loop; entrances and exits are disjoint. A play stops on its first visit to an exit. When an exit becomes internal wiring, its completion loop is replaced by the prescribed continuation.

The following constructors have one entrance and the same exit list.

1. The atom $a_{o,p}$ is a forced priority-p edge from its entrance to o. The other exits are present and unreachable. The constants T, B have a zero edge from their entrance to a forced terminal loop of priority zero or one, respectively; all declared exits are unreachable.

2. The term $\mu_r X$ adds a new forced entrance edge of priority r into the entrance of X. Here μ_r denotes a finite prefix, not a fixed-point operation.

3. The fork $E(X, Y)$, respectively $O(X, Y)$, adds a forced zero entrance edge to a fresh Even-owned, respectively Odd-owned, binary choice. Its zero arms enter disjoint copies of X and Y. For each exit name, the two former exit stubs lose their completion loops and have zero edges to a fresh forced merger, followed by a zero edge to the new external exit of that name. Each merger has two incoming edges and one outgoing edge.

Every forced port, merger and original choice retains its identity. Repeated syntactic variables denote fresh copies, never shared choices. These fixed templates add a bounded number of vertices and edges per constructor. Forced zero paths may be drawn compactly, but their explicit vertices are retained when graph size or original-policy output is counted.

There is one additional possible leaf $L^o_A(p, q)$, where $A \in \{E, O\}$, $p \in \{1, 2, 3, 4\}$, and $q \in P$. Its body has entrances x, y_{in} and exits o, y_{out}. Both entrances have forced zero paths to one original choice c, owned by A. The exit arm carries priority q to o; the return arm carries priority p to y_{out}. Replace the completion loop at y_{out} by a zero edge to y_{in}. Every lap thus revisits the same choice c. The remaining external interface is x, o; a second unused declared exit, if any, is added after this wiring.

A term contains at most one such leaf. Apart from this leaf and the terminal completion loops, the graph is acyclic: no outer fork is revisited, no child choices are shared, and no other feedback is allowed. A two-entrance forest consists of two terms with disjoint internals and only their stopped external exits in common, with at most one L across the entire forest. Denote this syntax by $\mathcal{G}(4, 2, \text{onefork})$. Every actual interface has at most two entrances and two exits.

The atoms and constants have outcomes in

$$D_O = \{\mathsf{B}, \mathsf{T}\} \cup (O_{\text{exit}} \times P).$$

Here $\mathsf{B} = \bot_{\text{play}}$ and $\mathsf{T} = \top_{\text{play}}$ at the semantic level: they denote losing and winning particular nonexit plays. The same letters also name their literal constant terms. They are unrelated to the algorithmic label TOP_{KL}. The outcome order $\leq_D$ has these least and greatest elements; finite outcomes are comparable only at the same exit, using

$$3 <_P 1 <_P 0 <_P 2 <_P 4. \tag{8}$$

A finite exit outcome (o, p) records the *numerical* maximum of the preceding edge priorities. For a nonexit play, the greatest infinitely recurring edge priority decides between B and T. Write $\max_{\text{num}}$ for numerical maximum.

Recall that $\mathcal{A}^+(X)$ denotes the nonempty antichains of a finite poset X. The semantic types used here are

$$\text{outcomes } D_O, \qquad \text{rows } \mathcal{L} = \mathcal{A}^+(D_O), \qquad \text{fronts } \mathcal{A}^+(\mathcal{L}).$$

The order on rows is

$$R \leq_R S \quad \Longleftrightarrow \quad (\forall s \in S)(\exists r \in R)\ r \leq_D s. \tag{9}$$

For an entrance i and a full-history Even strategy σ, let $\text{Raw}_i^G(\sigma)$ be the outcomes against all full-history Odd strategies. Totality makes this set nonempty. The entrance-local front is

$$Q(G)_i = \max_R \{\min_D \text{Raw}_i^G(\sigma) : \sigma \text{ a full-history Even strategy}\}. \tag{10}$$

Both extrema are taken in finite posets, so the displayed front is nonempty. Even's witnesses are optimized separately for each entrance. Thus $Q(G)_i = Q_G(i)$ in the notation of the preliminaries. For a one-entrance term write $Q(G)$. This construction records a front of rows, rather than a joint relation requiring one original policy to realize all entrance coordinates.

5.2 The antichain algebra

The basic algebra has a simple set representation: rows are represented by generated upsets, and fronts by nonempty generated downsets. These representations also explain why local pruning is sound.

Lemma 5.2 (Rows and fronts)**.** *The row order makes $\mathcal{L}$ a finite poset with least element $\{\mathsf{B}\}$, greatest element $\{\mathsf{T}\}$, and meet*

$$R \wedge S = \min_D(R \cup S).$$

For fronts define

$$F \vee G = \max_R(F \cup G), \qquad F \wedge G = \max_R\{R \wedge S : R \in F,\ S \in G\}.$$

These operations form a distributive lattice.

Proof. For a row R, put $\uparrow R = \{d : \exists r \in R,\ r \leq_D d\}$. Then $R \leq_R S$ is equivalent to $\uparrow S \subseteq \uparrow R$. The minimal elements recover R, so the representation is injective and establishes antisymmetry. For nonempty $U, V \subseteq D_O$, finiteness gives

$$\min_D(U \cup V) = \min_D(\min_D U \cup \min_D V), \tag{11}$$

because each element of either set lies above one of its minima. Consequently $\min_D(R \cup S)$ is below both R and S. If $X \leq_R R, S$, then X supplies a predecessor to every element of their union and hence to each minimum. Thus this row is their greatest lower bound, and row meet is monotone in both arguments.

For a front F, put $\downarrow F = \{R : \exists S \in F,\ R \leq_R S\}$. The maximal elements recover F, and

$$\downarrow (F \vee G) = \downarrow F \cup \downarrow G, \qquad \downarrow (F \wedge G) = \downarrow F \cap \downarrow G.$$

Indeed X belongs to the intersection exactly when it is below some $R \in F$ and $S \in G$, equivalently below $R \wedge S$. All represented downsets are nonempty. Every nonempty downset contains $\{\mathsf{B}\}$, so intersection remains nonempty. Union and intersection give commutativity, associativity, idempotence, absorption and distributivity of the front operations. □

Lemma 5.3 (Literal forks)**.** *For equally typed terms with disjoint internal occurrences,*

$$Q(E(X,Y)) = Q(X) \vee Q(Y), \qquad Q(O(X,Y)) = Q(X) \wedge Q(Y).$$

Proof. At an Even fork, a strategy chooses one child and can then use any full-history strategy of that occurrence. Conversely, any child strategy extends to a whole strategy by legal choices on unused histories. The zero paths and forced mergers add neither a priority contribution nor a new nonexit alternative. Thus the family of attainable raw sets is the union of the child families, giving the front join.

At an Odd fork, a fixed Even strategy restricts to one strategy in each child. Conversely, both restrictions can be chosen independently: the children are disjoint, their histories have fixed branch

prefixes, and no choice is revisited outside a child. Odd may select either arm and then any history within it. The whole raw set is therefore exactly the union of the two child raw sets. Equation (11) makes its minimum their row meet.

Every attainable row in a finite poset lies below an attainable maximal row. By monotonicity of meet, replacing each child family by its maximal rows leaves the maximal pairwise meets unchanged. This proves the second identity for full-history strategies. □

Lemma 5.4 (Finite prefixes)**.** *For $r \in P$, define $f_r : D_O \to D_O$ by*

$$f_r(\mathsf{B}) = \mathsf{B}, \qquad f_r(\mathsf{T}) = \mathsf{T}, \qquad f_r(o,p) = (o, \max_{\text{num}}(r,p)).$$

The row map $\phi_r : \mathcal{L} \to \mathcal{L}$, $\phi_r(R) = \min_D f_r(R)$, is monotone and preserves row meets. Moreover

$$Q(\mu_r X) = \max_R \{\phi_r(R) : R \in Q(X)\},$$

and this induced front map preserves both joins and meets.

Proof. The numerical-maximum map is monotone in the subpriority order [8, Lemma 15]. On our alphabet the full calculation, with columns in subpriority order, is

r	3	1	0	2	4
0	3	1	0	2	4
1	3	1	1	2	4
2	3	2	2	2	4
3	3	3	3	3	4
4	4	4	4	4	

Each row is nondecreasing in $\leq_P$. Since exit names and extrema are preserved, f_r is monotone on D_O. Every element of a nonempty finite U lies above a minimum, whence

$$\min_D f_r(U) = \min_D f_r(\min_D U). \tag{12}$$

To see equality, the image on the right before minimization is a subset of $f_r(U)$ and supplies a predecessor to every element of $f_r(U)$. The two images therefore have the same minima.

If $R \leq_R S$, each predecessor $a \leq_D b$ maps to $f_r(a) \leq_D f_r(b)$, and minimizing the left image retains a predecessor. Thus ϕ_r is monotone. Images commute with unions, so (11) and (12) give $\phi_r(R \wedge S) = \phi_r(R) \wedge \phi_r(S)$.

A forced finite prefix bijectively relates the relevant suffix histories to whole histories. Unused histories extend by totality. Finite outcomes are mapped by f_r, while nonexit parity ignores the prefix. After applying (12), monotonicity permits deletion of nonmaximal child rows, proving the front formula. Union gives preservation of joins; the pairwise-meet formula and row-meet preservation give preservation of front meets. □

5.3 The feedback leaf and the equations

Define $\Omega(p) = \mathsf{T}$ for even p and $\Omega(p) = \mathsf{B}$ for odd p. The restricted feedback leaf can be removed by considering all possible numbers of visits to its single original choice.

Lemma 5.5 (Eliminating the literal feedback leaf)**.** *For the body in Definition 5.1,*

$$Q(L^o_E(p,q)) = Q(E(a_{o,q}, \Omega(p))),$$
$$Q(L^o_O(p,q)) = Q\big(O(a_{o,q}, O(a_{o,\max_{\text{num}}(p,q)}, \Omega(p)))\big).$$

Proof. If c is Even-owned, a full-history strategy exits immediately, exits after a positive finite number of laps, or returns forever. These cases have outcomes

$$(o, q), \qquad (o, \max_{\text{num}}(p, q)), \qquad \Omega(p),$$

respectively, and every case is attainable. If p is even, perpetual return attains T, which dominates all singleton results. If p is odd, then $\max_{\text{num}}(p, q) \leq_P q$: for odd q use the reverse numerical order on odds; for even $q > p$ the values coincide; for even $q < p$ the maximum is odd and precedes q. The perpetual return is then B. Thus immediate exit dominates every other result in the odd case. These are exactly the fronts of the displayed Even fork.

If c is Odd-owned, Even has only forced choices. Odd's full-history decisions produce exactly the raw set

$$\{(o, q), (o, \max_{\text{num}}(p, q)), \Omega(p)\}.$$

The nested Odd fork has the same raw set and hence the same minimum. The exit-arm priority q does not recur on perpetual return, and an outer finite prefix does not change $\Omega(p)$.

Every lap in this argument revisits the same original choice. In the Even case a delayed exit can produce a dominated raw outcome unavailable from either original positional decision. Its removal is justified by front optimization; equality of raw families is not asserted. □

Equational presentation. For a nonempty minimized row R, let $M(R)$ be an Odd fold of its atoms and extrema; a singleton fold is its sole term. All variables below have the same ordered exit type. In nonfeedback schemas they range over feedback-free terms, and repeated variables mean fresh copies. The atom guards allow extrema as well as finite atoms.

$$\begin{array}{lll}
\text{R01} & E(X,Y) = E(Y,X), & \\
\text{R02} & E(E(X,Y),Z) = E(X,E(Y,Z)), & \\
\text{R03} & E(X,X) = X, & \\
\text{R04} & O(X,Y) = O(Y,X), & \\
\text{R05} & O(O(X,Y),Z) = O(X,O(Y,Z)), & \\
\text{R06} & O(X,X) = X, & \\
\text{R07} & E(X,O(X,Y)) = X, & \\
\text{R08} & O(X,E(X,Y)) = X, & \\
\text{R09} & O(X,E(Y,Z)) = E(O(X,Y),O(X,Z)), & \\
\text{R10} & \mu_r(E(X,Y)) = E(\mu_r X, \mu_r Y), & \\
\text{R11} & \mu_r(O(X,Y)) = O(\mu_r X, \mu_r Y), & \\
\text{R12} & \mu_r(a_{o,p}) = a_{o,\max_{\text{num}}(r,p)}, & \\
\text{R13} & \mu_r(\mathsf{T}) = \mathsf{T}, & \\
\text{R14} & \mu_r(\mathsf{B}) = \mathsf{B}, & \\
\text{R15} & O(a,b) = a, & a \leq_D b, \\
\text{R16} & E(a,b) = b, & a \leq_D b, \\
\text{R17} & L^o_E(p,q) = E(a_{o,q}, \Omega(p)), & \\
\text{R18} & L^o_O(p,q) = O(a_{o,q}, O(a_{o,\max_{\text{num}}(p,q)}, \Omega(p))), & \\
\text{R19} & E(M(R),M(S)) = M(S), & R \leq_R S.
\end{array} \tag{13}$$

In R19 the guard is a comparison of two explicit finite minimized lists. It does not call the semantics of a whole game.

Lemma 5.6 (Soundness)**.** *Every schema in* (13) *preserves Q, including in every admitted outer context.*

Proof. Lemmas 5.2 and 5.3 identify the forks with the lattice operations on fronts, giving R01–R09. Lemma 5.4 gives R10–R11. Concatenation of finite paths gives R12, and irrelevance of a finite prefix to recurrent parity gives R13–R14. For $a \leq_D b$, the Odd fork has raw set $\{a, b\}$ and minimum $\{a\}$, while the Even fork has maximal singleton result $\{b\}$. These are R15–R16. Lemma 5.5 gives R17–R18. Finally Odd can select exactly the outcomes listed in $M(R)$, so $Q(M(R)) = \{R\}$. Under $R \leq_R S$, the maximal front of the two choices in R19 is therefore $\{S\}$.

All guarded lists are nonempty and equally typed. In particular a same-exit guard never deletes an incomparable outcome at another exit. The fork and prefix formulas preserve semantic equality in every permitted outer context. Feedback occurs only in the proved literal leaf, not as an arbitrary surrounding operation. □

5.4 Canonical forms and completeness

Fix a total serialization order on typed atoms and finite lists. For a front F, let $\mathrm{NF}(F)$ be the sorted Even fold of its rows, each represented by a sorted Odd fold $M(R)$. A canonical description includes the ordered entrance and exit lists; unreachable declared ports remain present. The resulting Even-choice, Odd-result, typed-outcome form has the shape of Watanabe's shortcut [8, Definition 41], realized here by literal binary forks and mergers rather than by identifying the physical graphs.

Lemma 5.7 (Constructive normalization)**.** *Every term G has a finite derivation to* $\mathrm{NF}(Q(G))$ *using R01–R19. The following four-phase procedure terminates.*

Proof. The phases remove feedback, eliminate prefixes, distribute the forks, and prune the resulting finite lists.

Feedback. Replace the sole optional L by R17 or R18 in its outer context. This finite step occurs before any duplication.

Prefixes. Process child terms before their parents. Once a child is prefix-free, push a prefix through its forks using R10–R11, recursively visiting the smaller children, and apply R12–R14 at atoms. The outer recursion is over the original finite syntax. Each push traverses a finite already prefix-free tree with strictly smaller recursive arguments. Thus nested prefixes are eliminated by numerical maximum in finitely many steps.

Distribution. Convert the prefix-free term into an Even fold of Odd folds. An atom gives one singleton atom list. An Even fork concatenates the two lists of rows; an Odd fork forms their finite Cartesian product, concatenating the atom lists in each pair. R09 distributes Odd over a right Even fold, and R04 gives the analogous distribution on the left. R01, R02, R04 and R05 flatten and reorder the resulting folds. Structural recursion and finiteness of each Cartesian product give a finite equality derivation.

Pruning and sorting. In each Odd fold, R04–R05 sort the atoms, R06 removes duplicates, and R15 removes dominated atoms. Every deletion shortens a finite list and retains a predecessor of the deleted atom, leaving its unique nonempty minimum antichain. Next R01–R02 sort the Even fold, R03 removes duplicate rows, and R19 removes strictly dominated rows. Its guards now apply to explicitly minimized lists. Each deletion retains a dominator and leaves the unique nonempty maximal antichain. R16 can shorten singleton cases; R07–R08 are sound but unnecessary for this procedure.

Sorting is finite and every deletion decreases a list length. Recording each applied equation and its context gives the required derivation. Lemmas 5.3–5.5 identify the final row minima and maximal row list with $Q(G)$.

One may equivalently evaluate recursively, minimizing and maximizing after each constructor: atoms give singleton fronts, Even takes front join, Odd takes pairwise row meet followed by maximization, a prefix applies ϕ_r, and L uses its expansion. The domination and monotonicity arguments above justify this early pruning and give the same canonical description. □

Theorem 5.8 (Restricted ordered completeness)**.** *Every finite well-typed term of $\mathcal{G}(4, 2, \text{onefork})$ has a canonical normal form derived by R01–R19 with exactly its entrance-local ordered front. The specified normalization procedure terminates. For equally typed G, H, the following are equivalent:*

1. *their canonical descriptions are equal;*
2. *$Q(G) = Q(H)$;*
3. *$G = H$ has an equality derivation in the admitted typed contexts.*

The result holds coordinatewise for a two-entrance disjoint forest with at most one feedback leaf across the entire input.

Proof. Soundness is Lemma 5.6. Lemma 5.7 derives $G = \mathrm{NF}(Q(G))$ and $H = \mathrm{NF}(Q(H))$. Nonempty minimum atom lists, their nonempty maximal row list, the fixed serialization and the ordered ports recover the front uniquely. Thus equal fronts have the same canonical description, and concatenating the derivation for G with the reverse derivation for H gives $G = H$. Conversely any equality derivation preserves the front.

The derivation first removes the optional leaf of G; only the reverse derivation for H reintroduces its optional leaf. No intermediate requires two such leaves. For a forest, normalize both components before reversing either target derivation. A play stays within its entrance component until the shared stopped exit, so the argument applies to each ordered coordinate. □

The theorem concerns the given normalization algorithm. Bidirectional rewriting is neither asserted to terminate globally nor to be confluent. Distributive Cartesian products can enlarge the syntax; the proof gives no polynomial normalization, output-size or derivation-length bound. It also supplies no correspondence between original policies of equated graphs. The next result retains those policies explicitly.

5.5 One-choice policy changes in acyclic contexts

Let $R, S \in \mathcal{L}$ be explicit minimized rows with $R <_R S$, and let $K = E(M(R), M(S))$. An acyclic one-hole context is generated by

$$C ::= [\,] \mid \mu_r C \mid E(C, D) \mid E(D, C) \mid O(C, D) \mid O(D, C),$$

where D is a closed feedback-free term. All child internals are disjoint. Acyclicity permits forced terminal and absorbing exit loops, but excludes a return to a choice, any L, shared choice or backward wiring. Internalized outputs have their prescribed continuations. A possible second entrance component is disjoint and unchanged, apart from sharing stopped external exits.

Fix a complete original positional Even policy σ_0 on $G = C[K]$, including every forced and unreachable vertex. At the original root c of K, it selects the R arm. Define σ_1 by changing only that choice to the existing S arm. Both policies are legal; this grammar imposes no further menu or descent condition.

For a fixed such policy define its row profile

$$V_G(\sigma)_i = \min_D \mathrm{Raw}_i^G(\sigma), \qquad V_G : \{\text{complete original positional Even policies}\} \longrightarrow \prod_i \mathcal{L}.$$

The same σ is fixed before the entrance is selected, and Odd ranges over all full-history strategies. We use the coordinatewise row order $\leq_J$ on this product. No optimization over Even occurs in V_G.

Lemma 5.9 (Exact fixed-policy evaluation)**.** *For a closed occurrence D, use the restriction of the original policy and define*

$$\begin{aligned} v_\sigma(a_{o,p}) &= \{(o,p)\}, & v_\sigma(\mathsf{B}) &= \{\mathsf{B}\}, & v_\sigma(\mathsf{T}) &= \{\mathsf{T}\},\\ v_\sigma(\mu_r D) &= \phi_r(v_\sigma(D)),\\ v_\sigma(O(D_1,D_2)) &= v_\sigma(D_1) \wedge v_\sigma(D_2),\\ v_\sigma(E(D_1,D_2)) &= v_\sigma(D_j), \end{aligned}$$

where j is the arm selected by this original Even root. Then $v_\sigma(D)$ is exactly the minimum row of the occurrence's raw outcomes, with its exits treated as stopped boundaries.

Proof. We prove the corresponding raw-set equalities by structural induction. An atom or extremum has the specified unique outcome. A forced prefix maps the child raw set exactly by f_r: each child history extends by that prefix, and each whole history restricts to the child. At an Even fork only the supplied selected arm is available. At an Odd fork, every whole history uses one child, and any child history follows Odd's matching initial choice; the raw set is their union. All unused histories extend legally by totality. No choice is revisited, so there is no additional consistency constraint between the children. Zero connectors and name-preserving forced mergers change neither finite maxima nor the available outcomes. Taking minima and using (11) and (12) gives the recurrence. □

Every closed sibling is evaluated with its restriction of the same exterior policy. Define a row transformer $h^\sigma_C : \mathcal{L} \to \mathcal{L}$ by

$$\begin{aligned} h^\sigma_{[\,]}(U) &= U,\\ h^\sigma_{\mu_r C}(U) &= \phi_r(h^\sigma_C(U)),\\ h^\sigma_{O(C,D)}(U) &= h^\sigma_C(U) \wedge v_\sigma(D),\\ h^\sigma_{E(C,D)}(U) &= \begin{cases} h^\sigma_C(U), & \text{the exterior choice selects } C,\\ v_\sigma(D), & \text{the exterior choice selects } D. \end{cases} \end{aligned}$$

There are symmetric clauses for a hole in the right child. The disjoint second component has a constant transformer. Write h_i for the resulting transformer at entrance i. Each description contains at most one occurrence of U; a bypass makes the value constant without deleting any original policy assignment.

Theorem 5.10 (Fixed-policy context correspondence)**.** *For the declared acyclic context and policies σ_0, σ_1,*

$$V_G(\sigma_0)_i = h_i(R), \qquad V_G(\sigma_1)_i = h_i(S).$$

Each h_i is monotone. In particular the legal one-choice update satisfies $V_G(\sigma_0) \leq_J V_G(\sigma_1)$.

Proof. Odd can navigate $M(R)$ to every listed atom or extremum and to no other outcome, so its raw set is exactly R; likewise for S. The selected root arm gives the corresponding row, even when that root is unreachable from a global entrance. Lemma 5.9, applied up the unique hole path, gives both profile identities. Sharing only stopped exits leaves a disjoint second component unchanged. In particular sibling rows use the supplied exterior policy; they are not chosen anew after the argument U is supplied.

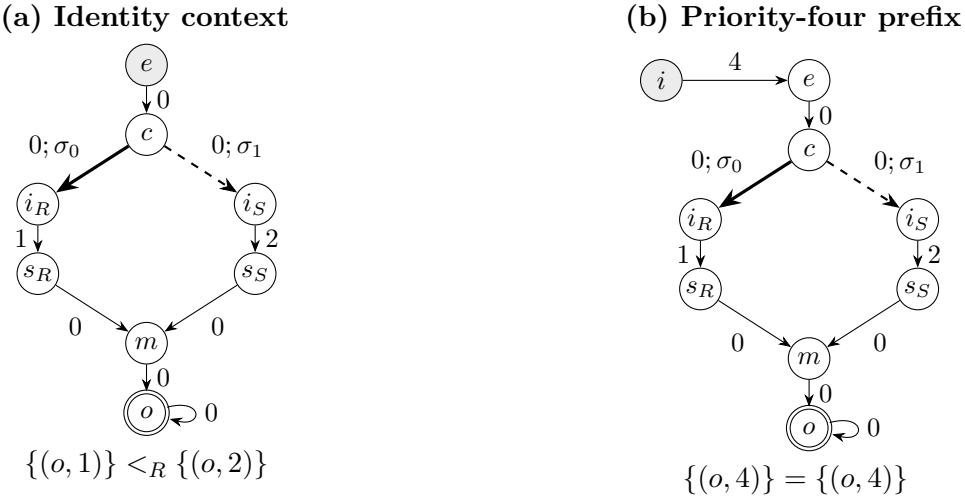


Figure 4: Two exact literal context graphs for $K = E(a_{o,1}, a_{o,2})$. Every vertex is Even-owned; only c has a choice. Thick solid and dashed choice edges are selected by σ_0 and σ_1, respectively; all other selections are forced. The shaded vertex is the external entrance, and the double circle is the stopped exit, with its total completion loop shown. Separate arm entrances i_R, i_S, internalized stubs s_R, s_S, and forced merger m retain their original identities. Panel (a) has eight vertices and nine edges; panel (b) adds only $i \xrightarrow{4} e$. The update changes one original choice in each fixed graph. Numerical maximum preserves the local strictness in (a) and maps both rows to $\{(o,4)\}$ in (b).

Identity, ϕ_r, meet with a constant, and constant maps are all monotone by Lemmas 5.2 and 5.4. Induction along the hole path proves monotonicity of each h_i, and $R \leq_R S$ gives the product comparison. □

Figure 4 gives a literal example in which the identity context preserves strictness and a priority-four prefix removes it. Take $R = \{(o,1)\}$, $S = \{(o,2)\}$. The graph of $E(a_{o,1}, a_{o,2})$ has entrance e, original choice c, separate arm entrances i_R, i_S, internalized exit stubs s_R, s_S, a forced merger m, and exit o. Its edges are

$$e \xrightarrow{0} c, \quad c \xrightarrow{0} i_R \xrightarrow{1} s_R \xrightarrow{0} m, \quad c \xrightarrow{0} i_S \xrightarrow{2} s_S \xrightarrow{0} m, \quad m \xrightarrow{0} o, \quad o \xrightarrow{0} o.$$

All vertices are Even-owned; only c has a choice. The prefix graph adds $i \xrightarrow{4} e$. Changing only $c \to i_R$ to $c \to i_S$ changes the identity-context row from R to S, since $1 <_P 2$. In the prefix context, numerical maxima give the same row $\{(o,4)\}$ under both policies.

Exact comparison. Define $d(a,b)$ to be the Boolean relation $a \leq_D b$: it is true if $a = \mathsf{B}$ or $b = \mathsf{T}$; for finite atoms it tests equality of exits and nondecreasing priority rank in (8); in the remaining extremum cases it holds only for equality. For $A_i = h_i(R)$ and $B_i = h_i(S)$, set

$$\mathrm{FWD} = \bigwedge_i \bigwedge_{b \in B_i} \bigvee_{a \in A_i} d(a,b), \tag{14}$$

$$\mathrm{REV} = \bigwedge_i \bigwedge_{a \in A_i} \bigvee_{b \in B_i} d(b,a). \tag{15}$$

The same finite list comparisons test the local premise $R <_R S$.

Corollary 5.11 (Strictness and equality certificates)**.** *For a legal annotated instance with $R <_R S$, the update is strict if and only if* FWD $\wedge \neg$REV*, and is a tie if and only if* FWD $\wedge$ REV*. A strictness certificate consists of forward predecessor choices and one pair (i, a^*), $a^* \in A_i$, such that $d(b, a^*)$ is false for every $b \in B_i$. A tie certificate consists of predecessor choices in both directions. In either case the rows are bound to their local evaluation equations at every syntax node.*

Proof. Equations (14)–(15) are exactly the forward and reverse product-row comparisons. Theorem 5.10 establishes the forward comparison on every admitted instance; row antisymmetry identifies two-way comparison with equality. Thus the two cases exhaust the domain. A false forward comparison instead indicates a failed input premise or evaluation.

The pair (i, a^*) is precisely a witness to failure of the reverse universal-existential formula. The remaining predecessor choices certify its positive comparisons. Checking every leaf, chosen Even arm, row meet and prefix equation binds these comparisons to the true evaluated rows. Predecessors justify discarded atoms. Attained witness plays alone would not establish completeness of those rows. □

For witness extraction, each retained atom may point to a surviving child atom: the selected child at Even, either child at Odd, or a preimage under a prefix. At $M(R)$ or $M(S)$ it points to the listed atom. Following pointers and restoring zero wiring gives either a finite original first-exit play or a finite prefix followed by the forced terminal loop. Each play follows the specified original Even policy and extends to an Odd history strategy by totality. Prefix pointers record numerical maximum. The updated policy itself is the full supplied σ_0 with the single original choice changed.

Three further controls. Let $O_{\text{exit}} = \{o\}$, $R = \{(o, 0)\}$ and $S = \{(o, 2)\}$, so $R <_R S$. For $C = \mu_0[\,]$, h_C is identity: the two rows are R, S, forward comparison holds, and reverse comparison fails at $(o, 0)$. The original plays use the selected priority-zero or priority-two arm after a zero prefix.

For $C = O([\,], a_{o,0})$, the map is $U \mapsto U \wedge \{(o, 0)\}$, and both minima are $\{(o, 0)\}$. Odd's sibling arm realizes this outcome under either policy. The new raw set also contains $(o, 2)$; equality of minima here does not imply raw equality.

For $C = E([\,], a_{o,0})$ with its exterior policy selecting the sibling, the map is constant $\{(o, 0)\}$. Both raw sets agree. The changed choice at c is unreachable, but remains a legal original assignment. A disjoint unchanged second component appends equal coordinates in every case.

5.6 Cost of a context evaluation

Theorem 5.12 (One annotated context check)**.** *Let n be the node count of the explicit unshared syntax forest, including both M folds, all siblings whether selected or not, and any second component. With indexed constant-arity nodes and direct access to the supplied original-policy choices, the two valuations, comparison guard and row certificate require $O(n)$ word operations and $O(n)$ words. If L is the uncompressed label and auxiliary encoding length in bits, they require*

$$O(L + n \log(n + 2))$$

bit work and storage in the log-cost indexed random-access model specified in the proof.

Proof. At a fixed exit, finite outcomes form a chain. A minimized row therefore contains at most one finite atom per exit; a row containing an extremum is a singleton. Thus every row has at most two atoms, a meet examines at most four, and a prefix at most two. After exit names are replaced by their one or two declared indices, priorities, ranks and row comparisons take constant word work.

A postorder scan evaluates every closed sibling once, including unreachable ones. Along the unique hole path it propagates two rows using these same sibling values; an Even bypass copies a constant row. The second component, if present, is evaluated once. Each syntax node uses constantly many constant-size row operations. Local equations and provenance pointers also use constant space per node, and the final entrance comparisons are of constant size.

More precisely, words have width $w = \lceil \log_2(n+2) \rceil$; each access, arithmetic or index operation on a word costs $O(w)$ bit operations. Thus indexed work and storage cost $O(n \log(n+2))$. At each occurrence, arbitrary exit strings are compared with at most two declared strings, charging their read/comparison length in L; the supplied auxiliary encoding is charged in the same way. This proves the stated bound for well-formed indexed syntax with the corresponding policy access. Recognition of an independently encoded arbitrary graph and recovery of its syntax and original-identity map are separate preprocessing tasks. The model is an indexed log-cost RAM, not a claimed multitape-Turing simulation or compressed-DAG bound. □

The fixed literal templates give an explicit graph with $n_G = O(n)$ vertices and edges. Storing that graph and the complete original policy costs $O(n_G)$ indexed words. Checking, copying or serializing the full policy has this additional word cost, or its actual arbitrary-ID encoding cost; a policy delta records one original vertex and its new existing arm. Across two policies and at most two entrances, there are at most eight retained-atom witness plays. Each uses $O(n_G)$ edges, with at most one loop edge to specify an infinite terminal tail. Their explicit extraction and output cost $O(n_G)$ indexed words plus the actual ID bytes. The comparison witness itself uses only constantly many row indices.

The state of the context construction is the original graph, its complete policy, and the annotation specifying c, its existing arms and the rows R, S. R19 is an equation between front semantics of different terms; the policy update instead keeps G fixed, while $V_G(\sigma)$ changes. The evaluation bound concerns one such annotated update. Arbitrary feedback, shared choices, whole-front normalization, optimum-policy search and sequences of pivots require additional arguments.

Piedeleu's questions about tractable subcategories and the relation of equations to strategy improvement [7, Section VI, PDF p. 12], and Watanabe's compositional strategy-improvement direction [8, Section 6, p. 14:15], concern broader constructions. The present results provide a literal ordered normalizer and a fixed-policy acyclic-context calculation, with the distinct domains and cost measures stated above.

6 Boundary information under one forget operation

Forgetting a boundary vertex can turn repeated excursions into an infinite history that never returns to the remaining boundary. A summary retaining only minimum excursion weights can lose the parity information needed to recognize such histories. We exhibit this loss for two policies on one width-two arena, and then show that full positive first-return spectra determine the exact new summary for a fixed policy.

6.1 The arena and its boundary observations

Definition 6.1 (Fixed-policy boundary observations)**.** Let $G = (V, E, V_{\mathrm{Even}}, V_{\mathrm{Odd}}, \pi)$ be a finite total original arena with vertex priorities $\pi : V \to P = \{0, 1, 2, 3, 4\}$ and $n = |V| \leq 12$. A supplied nice tree decomposition has width at most two. Each original edge has both endpoints in a bag, the occurrences of each vertex form a connected subtree, and the width is one less than the maximum

bag size. Introduce and forget nodes change a bag by one vertex; the two children of a join have the same bag as the join. Fix one legal forget

$$B' = B \setminus \{z\}, \qquad 2 \le |B| \le 3,$$

where z has no exterior occurrence.

Fix a complete original positional Even policy σ, assigning an original outgoing edge at every Even vertex, including forced and unreachable vertices. A σ-walk follows this same edge whenever it visits that vertex; Odd may choose any legal edge using the whole preceding history. A finite walk is *positive* if it has at least one edge. Its weight is the numerical maximum of *all* visited vertex priorities, including both endpoints.

For a boundary $C \subseteq V$ and $u, v \in C$:

1. a positive first return from u to v has no interior vertex in C;
2. a positive repeated-return walk has endpoints u, v, but may visit C internally;
3. a no-return infinite history from u has no visit to C at a positive time.

Outside the boundary, vertices may repeat. The parity of an infinite history is $e \in \{0, 1\}$, the parity of its greatest infinitely recurring priority. Thus $e = 0$ gives $\top_{\text{play}}$, and $e = 1$ gives $\bot_{\text{play}}$.

For the rest of this section the graph, its priorities and σ stay fixed; only the observed boundary changes. Source minima use the common subpriority order

$$3 <_P 1 <_P 0 <_P 2 <_P 4. \tag{16}$$

This order is distinct from the numerical maximum defining a walk weight.

The key $K_C(\sigma)$ is the tuple of the following five observations:

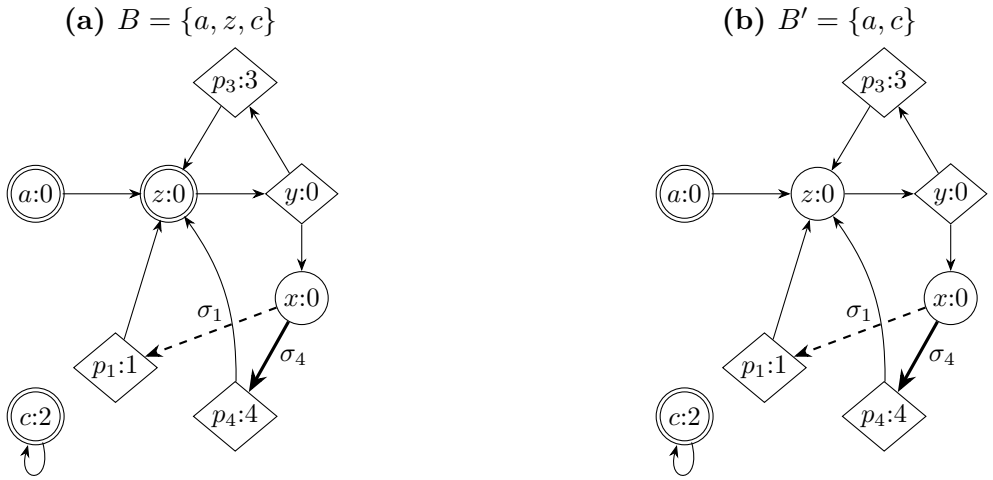
field	value
β_C	the selected original edge ID at each $v \in C \cap V_{\text{Even}}$
first returns	the $\le_P$ -minimum weight for each $(u, v) \in C^2$
no return	two existence flags, one for each parity, at each $u \in C$
repeated returns	the $\le_P$ -minimum weight for each $(u, v) \in C^2$
closed minimum	$\bot_{\text{play}}$ if some infinite history loses, and $\top_{\text{play}}$ otherwise, at each $u \in C$.

An empty finite-return spectrum is recorded as unr (unreachable). No fictitious weight zero is inserted. All fields are measured under the same original policy. Neither the complete interior selector vector nor the raw closed parity spectrum is part of K_C.

6.2 An equal-key pair on eight vertices

Consider the original arena

vertex	owner	π	successors
a	Even	0	z
z	Even	0	y
c	Even	2	c
y	Odd	0	p_3, x
x	Even	0	p_4, p_1
p_3	Odd	3	z
p_4	Odd	4	z
p_1	Odd	1	z



$$K_B(\sigma_4) = K_B(\sigma_1), \qquad N_{B'}(a,0) = \begin{cases} 1 & \text{under } \sigma_4, \\ 0 & \text{under } \sigma_1. \end{cases}$$

Figure 5: The exact eight-vertex arena before and after the boundary forget. Circles are Even-owned, diamonds Odd-owned; each vertex is labelled by its name and vertex priority. Double outlines mark the observed boundary. All ten original edges are shown in each panel. The thick solid edge $x \to p_4$ belongs to σ_4, and the dashed edge $x \to p_1$ to σ_1; the other three Even choices are forced and common. Each policy is held fixed when passing from (a) to (b). Forgetting removes only the boundary status of z; it deletes no vertex or edge, including the unselected arm. The old first-return spectra at (z, z) are $\{3, 4\}$ and $\{3, 1\}$, both with subpriority minimum three. After forgetting, the winning no-return flag at a differs. The closed minima at the surviving starts remain $(\bot_{\text{play}}, \top_{\text{play}})$ under both policies.

It has ten edges and no parallel edges, so an edge ID can be its ordered source-target pair. There are precisely two original Even policies: for $j \in \{4, 1\}$, let

$$\sigma_j = (a \to z,\ z \to y,\ c \to c,\ x \to p_j).$$

The unselected arm is unreachable from a, z, but its original vertex and forced Odd edge remain in the arena. Odd may choose p_3 or x on each visit to y, using full history. Figure 5 displays the graph, both selector choices, and the two boundary observations.

Here is a full nice decomposition, written from parent to child. The root chain is

$$\{a, c\} \longrightarrow \{a, z, c\} \longrightarrow \{z, c\} \longrightarrow \{z\} \longrightarrow \{z, y\}.$$

The last node is a join with two children of bag $\{z, y\}$. One child has the chain

$$\{z, y\} \longrightarrow \{z, y, p_3\} \longrightarrow \{z, p_3\} \longrightarrow \{p_3\} \longrightarrow \varnothing.$$

The other continues as

$$\{z, y\} \longrightarrow \{z, y, x\} \longrightarrow \{z, x\},$$

whose last node is a join with two children of bag $\{z, x\}$. Their chains, for $j = 4, 1$, are

$$\{z, x\} \longrightarrow \{z, x, p_j\} \longrightarrow \{z, p_j\} \longrightarrow \{p_j\} \longrightarrow \varnothing.$$

Each nonjoin step differs by one vertex. With the bottom-up convention, the root forgets z from $B = \{a, z, c\}$, leaving $B' = \{a, c\}$.

For edge coverage, use the first triple for a, z, the pair for z, y, and the indicated triples for y, p_3, y, x, x, p_j and p_j, z. The loop at c is covered by any bag containing c. The occurrences of z stay connected through both joins, those of y through the first join, and those of x through the second. Every other vertex occurs along one connected chain. All bags have size at most three, and the root forget has no exterior occurrence of z. Repeated occurrences in bags are the same original vertices, with the same policy choices.

Theorem 6.2 (The minimum key is not preserved by forget)**.** *For this arena, the two complete policies satisfy*

$$K_B(\sigma_4) = K_B(\sigma_1), \qquad K_{B'}(\sigma_4) \neq K_{B'}(\sigma_1).$$

Their only differing new field is the winning no-return flag at a. Consequently no function of K_B alone computes the exact policy-conditioned $K_{B'}$ for every original policy, even on this fixed arena. Keeping one canonical original policy witness for each old key loses an attained new record after this forget.

Proof. We compute every component of the two old keys, then identify the sole new discrepancy. Their boundary restriction is $\beta_B = (a \to z, z \to y, c \to c)$. The common first-return minimum matrix, with rows and columns a, z, c, is

	a	z	c
a	unr	0	unr
z	unr	3	unr
c	unr	unr	2

. (17)

The edge $a \to z$ has weight zero. The two positive first returns from z are z, y, p_3, z, of weight three, and z, y, x, p_j, z, of weight j. Since three precedes both one and four in $\leq_P$, their minima agree. The loop c, c has weight two. No edge enters a, and the c-component is separate, proving all unreachable entries.

Both old no-return flags are false at every start. From a or c the next vertex is in B; from z, each possible excursion returns in three or four edges. This exhausts all Odd histories.

The common positive repeated-return minimum matrix is

	a	z	c
a	unr	3	unr
z	unr	3	unr
c	unr	unr	2

. (18)

A positive walk from a to z is its initial edge followed by zero or more z-excursions. Its spectrum is $\{0, 3, 4\}$ under σ_4 and $\{0, 1, 3\}$ under σ_1; both minima are three. The repeated z-return spectra are $\{3, 4\}$ and $\{1, 3\}$, again with minimum three. At c all repetitions have weight two. The remaining pairs are unreachable for the same graph-theoretic reasons as before. In particular, the repeated a-to-z minimum is three, whereas its first-return minimum is zero.

The old closed minima are $(\bot_{\text{play}}, \bot_{\text{play}}, \top_{\text{play}})$. Odd can always choose p_3, giving recurrent maximum three from a or z; the only history from c repeats priority two. Thus all five fields of the old keys agree.

After forgetting z, every history from a remains outside B' at positive times. Under σ_4, infinitely many p_4-episodes give recurrent maximum four. If only finitely many occur, eventually every episode

uses p_3, giving recurrent maximum three. Both cases are attainable, so both new no-return flags at a are true. Under σ_1, infinitely many p_3-episodes give maximum three; otherwise eventually every episode uses p_1, giving maximum one. These cases partition all history-dependent Odd choices, and both lose. The new losing flag is therefore true and the winning flag false. For example,

$$a, z, (y, x, p_4, z)^\omega, \qquad a, z, (y, p_3, z)^\omega$$

witness the winning flag for σ_4 and the losing flag for either policy, respectively.

At c both new no-return flags remain false: its next step is in B'. In either new finite-return matrix the only reachable pair is (c, c), of weight two. The new edge restriction is $\beta_{B'} = (a \to z, c \to c)$; forgetting does not change the original edge into z. The closed minima remain $(\bot_{\text{play}}, \top_{\text{play}})$. Thus the winning no-return flag at a is the sole difference.

A function of the shared old key must assign one value to two distinct true new keys. Likewise a canonical-witness rule that retains only σ_4 loses the record with winning flag false, while retaining only σ_1 loses the record with winning flag true. □

This obstruction concerns the specified key. The complete interior selectors already distinguish the policies, as does a representation retaining both witnesses. The equal closed minima also show why the example gives no impossibility result for winner selection.

6.3 Full first-return spectra

The lost information is restored by retaining all attained excursion weights before applying the boundary operation.

Definition 6.3 (Spectral record)**.** For the same original σ, define

$$H_C(\sigma) = (\beta_C, S_C, N_C), \qquad S_C : C^2 \to \mathcal{P}(P), \quad N_C : C \times \{0,1\} \to \{0,1\}.$$

Here $S_C(u, v)$ is the full set of weights of positive first returns, and $N_C(u, e) = 1$ exactly when there is a no-return history of parity e from u. An empty spectrum means unreachable. We consider only *attained* records: the entries come from this same graph and complete policy. The original graph and σ remain available separately for interpreting original witnesses.

The proofs rely on two elementary facts: pieces formed under one policy can be concatenated, and recurrent maxima survive decomposition into positive finite blocks.

Lemma 6.4 (Excursion concatenation and decomposition)**.** *First-return excursions with matching endpoints in C concatenate under the same σ, with weight the numerical maximum of their weights. Every positive finite σ-walk with endpoints in C decomposes into first-return excursions at its successive boundary visits.*

Proof. Concatenate the original edge lists, identifying each shared endpoint once. Every Even edge still follows the same original policy, and every Odd edge is legal. An Odd history strategy realizes any prescribed finite or infinite legal walk by selecting its prescribed edge at each successive finite prefix and arbitrary legal edges on unused histories. Prefixes of different lengths are different histories, so these choices cannot conflict. Totality supplies the unused assignments.

Counting a shared endpoint twice would not change numerical maximum. Thus the concatenated weight is the maximum of the block weights, including the priorities at both endpoints.

Conversely, consecutive boundary-visit times delimit positive pieces with no interior boundary visit. Splitting at all such times retains the entire original walk and all its priorities. Self-loops, immediate boundary edges, repeated endpoints and repeated vertices outside C are included. □

Lemma 6.5 (Infinite excursion blocks)**.** *Let an infinite original σ-history $(v_t)_{t\geq 0}$ start at $v_0 \in C$ and visit C infinitely often. Split it into positive first-return blocks at its successive visit times*

$$0 = t_0 < t_1 < t_2 < \cdots .$$

If m_i is the numerical maximum in block i, then the greatest original priority occurring infinitely often equals the greatest block weight occurring infinitely often. The block lengths need not be bounded. Conversely, every infinite weighted boundary walk using attained spectral edges expands into an original history of the same parity.

Proof. Let r be the greatest original priority recurring infinitely often. After a finite time no priority exceeds r, so sufficiently late block weights are at most r. Infinitely many occurrences of r lie in infinitely many blocks: any finite collection of finite blocks covers only finitely many time positions. Hence infinitely many block weights equal r. Conversely, infinitely many blocks of maximum r supply infinitely many original positions with priority r, since each shared endpoint belongs to at most two consecutive blocks. This proves the equality without a bound on block lengths.

There are finitely many spectral edge types (u, p, v). Choose an original witness for every type used by an infinite boundary walk and substitute it at every occurrence of that edge. Lemma 6.4 gives legality under the same policy. Each witness has positive length, so the expansion is infinite; the preceding argument proves equality of parity. This existence construction may repeat fixed witnesses, while the forward argument covers arbitrary history-dependent Odd play. □

6.4 The one-forget formula

Write H_B and $H_{B'}$ for the attained records of the same graph and σ. The next formulas use numerical maxima of complete spectra. Figure 6 depicts the finite and infinite decompositions.

Theorem 6.6 (One fixed-policy spectral forget)**.** *Assume the arena and single legal forget of Definition 6.1: $n \leq 12$, priorities $0, \ldots, 4$, width at most two, $2 \leq |B| \leq 3$, and no exterior occurrence of z. For a fixed complete original positional Even policy, with all Odd histories retained, the following identities hold for $u, v \in B'$ and $e \in \{0, 1\}$:*

$$\beta_{B'} = \beta_B\big|_{B' \cap V_{\mathrm{Even}}}, \tag{19}$$

$$\begin{aligned} S_{B'}(u,v) = S_B(u,v) \cup \\ \{\max_{\mathrm{num}}(a,c,b) : a \in S_B(u,z),\ b \in S_B(z,v),\ c \in \{0\} \cup S_B(z,z)\}, \end{aligned} \tag{20}$$

$$\begin{aligned} N_{B'}(u,e) = N_B(u,e) \vee \\ \big[S_B(u,z) \neq \varnothing\ \wedge (N_B(z,e)\ \vee\ \exists p \in S_B(z,z),\ p \bmod 2 = e)\big]. \end{aligned} \tag{21}$$

They define an exact map $F_z : H_B(\sigma) \mapsto H_{B'}(\sigma)$ on the attained records in this domain. Only the boundary changes.

Proof. A new first return either avoids z internally or factors through it. A new no-return history either never reaches z, has a last visit to z, or visits z infinitely often. We treat these cases and their converse realizations.

Take a positive B'-first-return walk from u to v. If it has no internal z, it is already a B-first return. Otherwise, split at its internal z-visits. The first and last pieces have weights $a \in S_B(u, z)$, $b \in S_B(z, v)$. The middle consists of $k \geq 0$ positive old z-returns of weights $c_1, \ldots, c_k$. If $k = 0$, set $c = 0$, the neutral element for maximum on P. If $k > 0$, the numerical maximum of the finite nonempty list is one of its members and therefore belongs to $S_B(z, z)$. Lemma 6.4 gives total weight $\max_{\mathrm{num}}(a, c, b)$, proving one inclusion in (20).

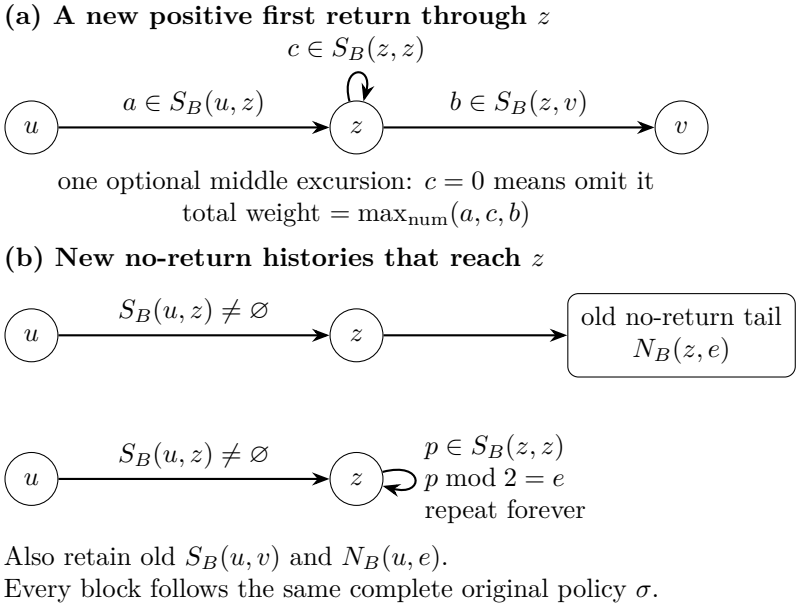


Figure 6: Schematic decompositions for one legal forget $B' = B \setminus \{z\}$ on attained records, under the caps of Theorem 6.6. Here $u, v \in B'$, possibly with $u = v$. Each weighted arrow denotes a positive old first-return excursion, whose interior avoids B and whose weight includes both endpoints; it is not necessarily one original edge. In (a), any nonempty finite sequence of middle loop weights has a maximum attained by one loop, which suffices for (20). The neutral value zero also permits the absence of a middle loop; it does not assert a zero-weight excursion. In (b), an initial excursion is followed either by a tail after the last visit to z, or by infinitely many z-returns. The illustrated repeated loop proves sufficiency in (21). Necessity uses Lemma 6.5 for arbitrary histories with possibly unbounded finite block lengths. The graph and σ remain fixed throughout this single boundary operation.

Conversely, an old u-to-v excursion remains a new one. For a displayed triple, concatenate witnesses for a and b. If $c \in S_B(z,z)$, insert one witness for that loop weight. The extra value $c = 0$ can instead be realized by inserting no loop; an actual zero-weight loop, when present, also works. The only internal old boundary vertex is z. Thus the result is a positive new first return with exactly the claimed maximum, including when $u = v$.

For sufficiency in (21), any old no-return history from u is still new no-return. An old u-to-z excursion may also be followed by an old no-return history from z; the prefix does not affect parity. Alternatively, if $p \in S_B(z,z)$ has parity e, repeat a positive original weight-p loop after that prefix. Its recurrent maximum is p, and the resulting history avoids B'.

For necessity, let a new no-return history from u have parity e. If it never reaches z, it is old no-return. If it visits z a positive finite number of times, its prefix to the first visit is an old first return and its suffix from the last visit is old no-return of the same parity. If it visits z infinitely often, the initial excursion is followed by infinitely many positive old z-returns. Lemma 6.5, now starting at $z \in B$, identifies the greatest infinitely recurring block weight p with the original recurrent maximum. Thus $p \in S_B(z,z)$ and $p \bmod 2 = e$. The alternatives exhaust all Odd histories, including those with unbounded finite block lengths.

Finally, (19) is literal restriction of original selected edge IDs. In particular a remaining vertex's selected edge into z is unchanged. □

6.5 Reconstruction of the minimum key

Full first-return spectra also recover repeated returns and closed outcomes. Let $C = B$ or B', write $q = |C| \le 3$, and form the weighted multigraph D_C on C with

$$(x,p,y) \in E(D_C) \quad \Longleftrightarrow \quad p \in S_C(x,y).$$

It has at most $5q^2$ edges. Let $R_p(x,y)$ be reflexive reachability using numerical edge weights at most p, and $R_*(x,y)$ unrestricted reflexive reachability. Reflexivity is only an auxiliary path relation; it inserts neither zero-length spectral edges nor positive-return witnesses.

Define the repeated-return spectrum by

$$\mathrm{Rep}_C(u,v) = \{p : \exists x,y \in C,\ R_p(u,x),\ p \in S_C(x,y),\ R_p(y,v)\}. \tag{22}$$

The designated edge of weight p enforces positivity even when the auxiliary paths are empty. Define raw closed parity attainability by

$$\begin{aligned} A_C(u,e) = {} & [\exists x \in C,\ R_*(u,x) \wedge N_C(x,e)] \\ & \vee [\exists x,y \in C,\ \exists p \in P,\ R_*(u,x) \wedge p \in S_C(x,y) \\ & \qquad \wedge R_p(y,x) \wedge (p \bmod 2 = e)]. \end{aligned} \tag{23}$$

The second clause describes a reachable positive closed weighted walk of maximum p. Its transient prefix may have larger weights.

Theorem 6.7 (Exact reconstruction)**.** *For an attained H_C, the following map ρ_C recovers the entire key K_C: copy β_C and the two no-return flags; take $\min_P S_C(u,v)$ for each first-return entry and $\min_P \mathrm{Rep}_C(u,v)$ for each repeated-return entry, using* unr *exactly for empty sets; and set the closed minimum at u to $\bot_{\mathrm{play}}$ if $A_C(u,1)$ holds and to $\top_{\mathrm{play}}$ otherwise. Consequently*

$$K_{B'}(\sigma) = \rho_{B'}(F_z(H_B(\sigma)))$$

under the hypotheses of Theorem 6.6.

Proof. The edge restriction, first-return and no-return fields have their defining values. To recover repeated returns, observe that a positive weighted boundary walk of maximum p contains an edge of weight p. Its prefix and suffix satisfy the two threshold-reachability conditions in (22). Conversely, those threshold paths together with the designated edge form a positive walk of maximum p. Lemma 6.4 decomposes and expands original walks in both directions, proving exactness of Rep_C. No positive self-return is created by reflexive reachability alone. Source minimization is applied only after computing the numerical spectra. Each threshold path may be chosen simple, of length at most $q-1$, or empty. Thus any witnessed repeated weight uses at most $2q-1$ boundary edges.

We next prove that $A_C(u,e)$ is exactly raw closed parity attainability. In the first clause of (23), concatenate the boundary prefix with the no-return suffix. The prefix does not change its parity. In the second, reach x and then repeat the weight-p edge followed by a threshold path from y to x. Choose that path simple; the positive closed walk has at most q edges, including the one-edge case $x=y$. Its original expansion follows σ and has recurrent maximum p, by Lemma 6.5 or directly by repeating the finite expanded closed walk.

For completeness take any infinite original history from u. If it visits C only finitely often, let x be its last visit, possibly u at time zero. Decomposition of the finite prefix gives $R_*(u,x)$, and the remaining history gives $N_C(x,e)$. If it visits C infinitely often, Lemma 6.5 produces an infinite weighted boundary walk whose greatest infinitely recurring weight p has parity e. Discard a finite prefix after which no weight exceeds p. Among finitely many weighted edge types, some (x,p,y) occurs infinitely often. Between two sufficiently late occurrences there is a threshold return from y to x; the original prefix reaches x. This gives the second clause of (23). Again no bound on original excursion lengths is used.

A losing closed history contributes $\bot_{\mathrm{play}}$, the least outcome. If none exists, every closed history wins. Fixing a complete legal σ preserves totality, so at least one infinite history exists from every start. Hence the closed minimum is never an empty row, and the stated formula recovers it. This proves every field of ρ_C. The raw closed flags A_C are distinct from the boundary-sensitive no-return flags N_C. □

6.6 Original witnesses and record size

Lemma 6.8 (Short existence witnesses)**.** *Every true spectral bit has an original positive first-return witness of at most $2n$ edges. Every true no-return bit has an original lasso with a stem of at most $n-1$ edges and a positive period of at most n edges. These are existence witnesses; arbitrary histories are not required to be ultimately periodic.*

Proof. For a spectral weight p, mark a visited vertex of priority p. If it is internal, erase cycles separately before and after it. Each resulting path has at most $n-1$ edges, retains the marked maximum, and introduces neither an interior boundary visit nor a nonpolicy edge. If p occurs at an endpoint, take a shortest admissible path when the endpoints differ, or a shortest positive return cycle when they coincide. These have at most n edges. The loose common bound $2n$ follows.

For a no-return history of recurrent maximum p, choose a recurrent vertex w of priority p. It lies outside C. After all priorities greater than p have disappeared, consecutive returns to w give a positive closed walk outside C of maximum p. Erasing cycles away from w yields a positive simple cycle through it, of at most n edges. A shortest path from the start u to w avoiding positive visits to C has at most $n-1$ edges; larger transient priorities on it do not affect parity. Both pieces obey the same original policy and give the required lasso. □

Choose short witnesses from Lemma 6.8 for the current input record of each construction. The following loose bounds count original edge-ID occurrences:

- Formula (20) uses at most three spectral paths, hence at most $6n \leq 72$ edges.
- Formula (21) uses an old lasso, or an initial spectral path followed by an old no-return lasso or an endlessly repeated spectral path. In the former new case the stem has at most $3n - 1 \leq 35$ edges and the period at most $n \leq 12$; in the latter, stem and period have at most $2n \leq 24$ each.
- Formula (22) uses at most $2q - 1$ boundary edges, giving at most $2n(2q - 1) \leq 120$ original edges.
- The first clause of (23) gives a stem of at most $2n(q - 1) + n - 1 \leq 59$ edges and a period of at most $n \leq 12$. Its second clause gives a stem of at most $2n(q - 1) \leq 48$ and a period of at most $2nq \leq 72$.

Expanded periods may revisit original vertices; they are still positive legal closed walks. These bounds are per construction with short witnesses for its current input record. When applying reconstruction after a forget, short witnesses for the newly attained record may be chosen by Lemma 6.8; no sharper bound is claimed for naive composition of previously expanded witness lists.

Size of a spectral record. The behavioral fields of H_C use

$$5q^2 + 2q \leq 51$$

incidence bits, plus at most q literal selected edge IDs. Reconstruction involves at most forty-five weighted edges on at most three vertices and five threshold-reachability relations. The original graph, complete σ, and optional witness lists are additional data. One short witness per true spectral or no-return bit uses at most

$$(5q^2)(2n) + (2q)(2n - 1) \leq 1218$$

original edge-ID occurrences, each charged at its actual encoded length. These witnesses establish the asserted positive bits; by themselves they do not verify completeness or false bits of an arbitrary bit string. The theorems take attained records as inputs. The bound is per record in the stated caps, not an all-policy relation size or running-time bound, and H_C does not recover σ.

6.7 The spectral repair on the equal-key pair

In the eight-vertex arena under σ_j, $j = 4, 1$, the only nonempty old first-return spectra are

$$S_B(a, z) = \{0\}, \qquad S_B(z, z) = \{3, j\}, \qquad S_B(c, c) = \{2\}.$$

Their original witnesses are a, z, z, y, p_3, z, z, y, x, p_j, z, and c, c. All old no-return flags are false. The positive repeated-return spectra are

	σ_4	σ_1
(a, z)	$\{0, 3, 4\}$	$\{0, 1, 3\}$
(z, z)	$\{3, 4\}$	$\{1, 3\}$
(c, c)	$\{2\}$	$\{2\}$

with all other entries empty. Their minima are respectively three, three and two. Raw closed parities at a, z are $\{0, 1\}$ for σ_4 and $\{1\}$ for σ_1; at c they are $\{0\}$. Thus raw closed parity distinguishes the policies although their closed minima agree.

There is no old excursion from z to a or c, so (20) leaves only $S_{B'}(c,c) = \{2\}$. Formula (21) gives $N_{B'}(a,1)$ for both policies through the weight-three loop, and $N_{B'}(a,0)$ exactly for σ_4 through the weight-four loop. At c both flags stay false. The edge $a \to z$ is retained, the only repeated-return minimum is two at (c,c), and the new closed minima are $(\bot_{\mathrm{play}}, \top_{\mathrm{play}})$. The full spectra therefore distinguish the policies before forgetting and preserve precisely the information lost by their common old minimum key. Positive and negative flag assertions follow from the complete episode partition in Theorem 6.2.

Fearnley and Schewe use fixed-strategy profiles and ask for an algorithm of running time $f(k)N^C$, with C independent of treewidth k and no separate priority bound [1, Sections 3 and 9, especially p. 30]. The result here concerns one fixed-policy boundary operation under explicit caps. It gives no join or introduce operation, theorem for successive forgets, all-policy representation, policy synthesis or treewidth-parameterized winner algorithm. The distinctions between raw observations and minima are compatible with Watanabe's ordered outcome framework [8]; the present reconstruction rests on the finite-walk and infinite-block arguments above.

7 Completed pivots and exact evaluation on paths

We specialize the strategy-iteration algorithm of Koh and Loho to bidirected paths with priorities one and two [5, Section 3, Algorithm 1]. Two features of this setting are decisive. At a completed evaluation, a finite label records a distance to one of at most two fixed anchors. Moreover, each pivot strictly increases the label of its source. These facts bound the number of completed pivots and permit an exact evaluator with constant-size coordinate domains.

The algorithmic state is an Odd policy together with a tree labeling. Its order is the full order of the tree labels, not the outcome or row order of the open-game semantics. We write its greatest label as $\mathsf{TOP}_{\mathrm{KL}}$. At the terminal output, a vertex is Even-winning precisely when its label is finite [5, Theorems 2.3 and 3.1]; thus $\mathsf{TOP}_{\mathrm{KL}}$ has a different meaning from the Even-winning play outcome $\top_{\mathrm{play}}$.

7.1 The path, its labels, and initialized iteration

Definition 7.1 (Two-priority path)**.** Let $n_G \geq 1$ and $V = \{0, \dots, n_G - 1\}$. The arc set contains both $i \to i+1$ and $i+1 \to i$ for $0 \leq i < n_G - 1$, together with the endpoint loops $0 \to 0$ and $n_G - 1 \to n_G - 1$, and contains no other arcs. If $n_G = 1$, these specify one loop. Fix any ownership partition $V = V_E \sqcup V_O$ and priority map $\pi : V \to \{1, 2\}$. The parity condition uses the numerical maximum recurring priority. Set $O_p = \{v \in V_O : \pi(v) = p\}$ for $p = 1, 2$. The source priority bound is $d = 2$, so its universal tree has height $d/2 = 1$.

Definition 7.2 (Ordered tree labels)**.** An ordered tree is a finite prefix-closed set of tuples over a linearly ordered alphabet, with lexicographic order and proper prefixes before their extensions. All leaves have the same depth. An embedding is an injective order-preserving map on vertices that preserves immediate edges and sends leaves to leaves. A height-h tree is (n,h)-universal if every height-h ordered tree with at most n leaves embeds into it [5, Definitions 2.1–2.2].

Fix a height-one $(n_G, 1)$-universal tree T, with ordered leaves $a_0 < \cdots < a_{\ell_T - 1}$, where $\ell_T = |L(T)|$. Adjoin $\mathsf{TOP}_{\mathrm{KL}}$ above all finite tuples, including the empty tuple. Define the order isomorphism

$$\rho : L(T) \cup \{\mathsf{TOP}_{\mathrm{KL}}\} \longrightarrow \{0, \dots, \ell_T\}, \qquad \rho(a_j) = j, \quad \rho(\mathsf{TOP}_{\mathrm{KL}}) = \ell_T.$$

A labeling is $\lambda_{\mathrm{KL}} : V \to L(T) \cup \{\mathsf{TOP}_{\mathrm{KL}}\}$, ordered pointwise; write $r(v) = \rho(\lambda_{\mathrm{KL}}(v))$. Leaf names themselves need not be integers.

For a height-h leaf $(\xi_{2h-1}, \xi_{2h-3}, \ldots, \xi_1)$, the source p-truncation keeps components whose indices are at least p, and $\mathsf{TOP}_{\mathrm{KL}}|_p = \mathsf{TOP}_{\mathrm{KL}}$. At height one this is exactly

$$a_j|_1 = a_j, \qquad a_j|_2 = (), \qquad \mathsf{TOP}_{\mathrm{KL}}|_1 = \mathsf{TOP}_{\mathrm{KL}}|_2 = \mathsf{TOP}_{\mathrm{KL}}, \qquad () < \mathsf{TOP}_{\mathrm{KL}}.$$

All embeddings here preserve edges; they do not contract them or change leaf depth.

Definition 7.3 (Arc constraints and lifts)**.** An arc $v \to w$ is *satisfied* by λ_{KL} when

$$\begin{cases} \lambda_{\mathrm{KL}}(v)|_2 \geq \lambda_{\mathrm{KL}}(w)|_2, & \pi(v) = 2, \\ \lambda_{\mathrm{KL}}(v)|_1 > \lambda_{\mathrm{KL}}(w)|_1 \text{ or } \lambda_{\mathrm{KL}}(v) = \lambda_{\mathrm{KL}}(w) = \mathsf{TOP}_{\mathrm{KL}}, & \pi(v) = 1. \end{cases}$$

Otherwise it is *violated*. A satisfied arc is *tight* if its source has the least full label that satisfies the arc upon replacement of that coordinate; otherwise it is *loose*. Replacement at a self-loop changes both occurrences of the same coordinate.

For a rank vector $x \in \{0, \ldots, \ell_T\}^V$, the least satisfying replacement rank is

$$\theta_x(v,w) = \begin{cases} \min\{x(w)+1, \ell_T\}, & v \neq w,\ \pi(v) = 1, \\ 0, & v \neq w,\ \pi(v) = 2,\ x(w) < \ell_T, \\ \ell_T, & v \neq w,\ \pi(v) = 2,\ x(w) = \ell_T, \\ \ell_T, & v = w,\ \pi(v) = 1, \\ 0, & v = w,\ \pi(v) = 2. \end{cases} \tag{24}$$

The inflationary lift $\mathrm{lift}(\lambda_{\mathrm{KL}}, vw)$ is the least label at least $\lambda_{\mathrm{KL}}(v)$ that satisfies the arc after replacement; its rank is $\max\{r(v), \theta_r(v,w)\}$.

These formulas specialize the source's *Tighten* and lift operations [5, Section 2.2, pp. 24:7–24:8]. A finite tight priority-one nonself arc decreases rank by exactly one. A priority-one loop is violated at every finite label and becomes tight only at $\mathsf{TOP}_{\mathrm{KL}}$; its lift therefore jumps directly to $\mathsf{TOP}_{\mathrm{KL}}$. A priority-two loop is always satisfied and has identity lift, although its unconstrained tight label is a_0. Whether a loop can be pivoted depends separately on its source being Odd-owned.

Definition 7.4 (Policy evaluation)**.** A legal positional Odd policy is a map $\tau_O : V_O \to V$ with $v \to \tau_O(v) \in E$. Its graph G_{τ_O} keeps all Even arcs and just the selected arc at each Odd vertex. The original coordinate operators replace $\lambda_{\mathrm{KL}}(v)$ by

$$\begin{cases} \min\limits_{vw \in E(G_{\tau_O})} \mathrm{lift}(\lambda_{\mathrm{KL}}, vw), & v \in V_E, \\ \mathrm{lift}(\lambda_{\mathrm{KL}}, v\tau_O(v)), & v \in V_O, \end{cases}$$

leaving every other coordinate unchanged. A labeling is *feasible* in G_{τ_O} if every selected Odd arc is satisfied and each Even vertex has a satisfied available arc. For a floor labeling f, denote by $\mathrm{Eval}(\tau_O, f)$ the least simultaneous fixed point of these operators that is pointwise at least f.

The operators are monotone and inflationary, as follows directly from (24). Their common fixed points are exactly the feasible labelings. This condition imposes no constraint on ignored Odd arcs and does not require every Even arc to be satisfied. The finite labeling lattice is nonempty, and the all-$\mathsf{TOP}_{\mathrm{KL}}$ labeling is feasible. Starting at any floor f, a maximal sequence of strict updates must terminate at a common fixed point. If $z \geq f$ is any common fixed point, monotonicity bounds every iterate by z, inductively. The terminal iterate is therefore the least such fixed point, independent of exhaustive update order. This also proves that Eval is defined for every floor, without yet giving a time bound [5, Proposition 2.5].

Definition 7.5 (Iteration and reachable floors)**.** Choose any legal $\tau_O^{(0)}$, let $f_{\min}(v) = a_0$, and set $\lambda_{\mathrm{KL}}^{(0)} = \mathrm{Eval}(\tau_O^{(0)}, f_{\min})$. At a completed state $(\tau_O^{(k)}, \lambda_{\mathrm{KL}}^{(k)})$, an eligible arc is a violated original arc $v \to w$ with $v \in V_O$. Replace only $\tau_O^{(k)}(v)$ by w, obtaining $\tau_O^{(k+1)}$, and set

$$\lambda_{\mathrm{KL}}^{(k+1)} = \mathrm{Eval}(\tau_O^{(k+1)}, \lambda_{\mathrm{KL}}^{(k)}). \tag{25}$$

Stop when there is no eligible Odd arc. Let I count the completed single-edge pivots, excluding initialization.

A *reachable-floor call* is the initial evaluation or an evaluation in (25). The floor is the immediately preceding completed labeling, including when a policy occurs more than once. Two particular selection rules are $R1$, the least eligible identity pair (v, w), and $R2$, the eligible arc with greatest full target label, with ties broken by the least identity pair. In $R2$, $\mathsf{TOP}_{\mathrm{KL}}$ is the greatest possible target label.

This is single-admissible-edge strategy iteration in [5, Algorithm 1]. Reachability of the floor is an operative hypothesis of the bounds below. In particular, evaluating a repeated policy from zero would define a different iteration.

Theorem 7.6 (Linear number of completed pivots)**.** *For every path, ownership partition, priority map and height-one universal tree in Definitions 7.1–7.2, every legal initial Odd policy and every eligible single-edge order in Definition 7.5 satisfy*

$$I \le 2|O_1| + |O_2| \le 2n_G.$$

The count concerns completed pivots, not internal lifts or evaluation time.

The proof will identify the finite labels at completed states. A priority-two vertex has only rank zero or $\mathsf{TOP}_{\mathrm{KL}}$. A finite priority-one vertex has the distance to the nearest original priority-two vertex on one of its two sides. Since labels increase, each source has only a constant number of remaining strict increases.

7.2 Completed labels and fixed anchors

Lemma 7.7 (Completed-state invariants)**.** *At every completed state the labeling is feasible in G_{τ_O}, and that graph has no loose arcs. Every selected Odd arc is tight and every Even vertex has an available tight arc. Labels never decrease through successive evaluations, and $\mathsf{TOP}_{\mathrm{KL}}$ is absorbing. Other Even arcs may be violated; ignored Odd arcs need not be tight.*

Proof. Feasibility is the fixed-point condition, while inflationarity and the carried floor give monotonicity and absorption of the greatest label. It remains to prove absence of loose arcs. At the all-minimum labeling, every priority-one arc is violated and every priority-two arc is tight. An eligible policy replacement introduces a violated arc and leaves the other arcs of the policy graph unchanged.

Suppose the graph has no loose arc before an original coordinate update. A changing Odd source is assigned its selected arc's least satisfying label. A changing Even source must have every outgoing arc violated: otherwise a satisfied arc would supply the identity lift and prevent any change. Its new rank is the minimum required rank of its outgoing arcs. An attaining arc is tight, and each other arc is tight or still violated. For an incoming nonself arc, increasing the target cannot decrease $\theta_x(v, w)$; an unchanged source that was tight or below its threshold remains tight or below it. The loop cases follow directly from (24): a priority-one loop remains violated until top; an available

priority-two loop at an Even vertex, or a selected one at an Odd vertex, prevents a source increase. Thus no update creates a loose arc. Induction through every evaluation and eligible replacement proves the assertion. At completion, feasibility turns the available satisfied arcs into tight arcs. □

Lemma 7.8 (Strict source progress)**.** *Completing an eligible pivot $v \to w$ strictly raises $\lambda_{\mathrm{KL}}(v)$.*

Proof. A top source satisfies every arc, including the both-top exception at priority one, so an eligible source is finite. For a nonself arc, keeping the source fixed while its target increases cannot repair a violation: the threshold in (24) is nondecreasing. The new selected arc must be satisfied at completion, so its source must increase. A priority-two loop is never eligible, and an eligible priority-one loop can become satisfied only at top. Finally, a source that has reached top remains there and has no further eligible arc. □

Lemma 7.9 (Priority-two labels)**.** *At every internal update of the initialized original iteration, a priority-two vertex has rank zero or ℓ_T. In particular, a finite priority-two vertex at a completed state has rank zero.*

Proof. These vertices start at zero. At priority two a nonself finite target gives identity lift on a finite source, while a top target requires top. A loop gives identity lift. A selected Odd arc therefore preserves zero or changes it to top. At an Even vertex, one finite target or its own loop preserves zero; only the absence of such an identity choice permits a change to top. Induction over the updates, including the carried floors, proves the claim. An arbitrary externally inflated floor would not have this property. □

Anchors. For each vertex v, let $a_{\mathrm{left}}(v)$ and $a_{\mathrm{right}}(v)$ be the nearest original priority-two vertices strictly to its left and right, when they exist. These are graph positions, not tree leaves. For a priority-one vertex define

$$D(v) = \{v - a_{\mathrm{left}}(v) : a_{\mathrm{left}}(v) \text{ exists}\} \cup \{a_{\mathrm{right}}(v) - v : a_{\mathrm{right}}(v) \text{ exists}\}.$$

Thus $D(v)$ is a fixed set of at most two positive integers; equal distances count once. It depends only on the original priority map.

Lemma 7.10 (Finite labels are anchor distances)**.** *At every completed state, each finite priority-one vertex v has rank in $D(v)$. A tight walk from v ends at the nearest original priority-two anchor on one side, and its length is that rank. A top vertex, a top nearest anchor or a missing anchor excludes that side; it never exposes a farther priority-two vertex as a new candidate.*

Proof. Lemma 7.7 supplies a tight selected successor at an Odd vertex and a tight available successor at an Even vertex. From a finite priority-one vertex, such an arc is nonself, has finite target, and decreases rank by one. Following tight arcs while priority remains one therefore produces no repeated vertex. It cannot stop at a rank-zero priority-one vertex, since no outgoing arc there is satisfied. It must reach a priority-two vertex, whose rank is zero by Lemma 7.9. Consequently the walk length equals the starting rank.

Every step is to an original path neighbor. A change of direction would immediately revisit the preceding vertex, contrary to strict descent. The walk is therefore a monotone interval ending at its first priority-two vertex, exactly the nearest original anchor on that side. A finite anchor ends descent at zero; a top anchor cannot be the finite target of a tight priority-one arc. The same excludes any intermediate top vertex. At an endpoint without an anchor, the priority-one loop is violated and reversing direction would revisit a vertex. This proves every claimed boundary case. In particular, if the path has no priority-two vertex, all completed labels are top. □

Proof of Theorem 7.6. After the first completed evaluation, a priority-one vertex has at most two possible finite ranks, the members of $D(v)$, followed possibly by top. By Lemmas 7.7 and 7.8, every pivot at that source strictly advances along this nondecreasing list. There can be at most one finite-to-finite and one finite-to-top source pivot. Coincident distances, absent anchors, a larger initial rank or initial top reduce this budget. Increases caused by another source's pivot consume, rather than replenish, the same budget.

A priority-two source has only zero and top by Lemma 7.9, and hence at most one strict source pivot. Each pivot has one Odd-owned source, so summing gives $I \le 2|O_1| + |O_2| \le 2n_G$. No particular eligible order was used. Every evaluation terminates by Definition 7.4, and the finite source budgets exclude infinitely many completed pivots. For $n_G = 1$, initialization gives top at priority one and zero at priority two; the sole loop provides no alternative policy, so $I = 0$. □

Lemma 7.11 (Height-one leaf bound)**.** *Every $(n_G, 1)$-universal ordered tree has $\ell_T \ge n_G$.*

Proof. The height-one star with n_G leaves is an allowed source. Its embedding into T sends these leaves injectively to leaves of T. The bound uses only this injection, with no numerical assumption on leaf names. □

Corollary 7.12 (An additive pivot bound)**.** *Under the hypotheses of Theorem 7.6, $I \le 2|O_1|+|O_2| \le 2n_G \le n_G + \ell_T$. Thus the completed-pivot inequality $I \le n_G + C\ell_T$ holds with $C = 1$ in this setting.*

Proof. Combine Theorem 7.6 and Lemma 7.11. □

7.3 Exact evaluation by rounded coordinate updates

For the remainder of this section put $n = n_G$, $\ell = \ell_T$. Use ρ coordinatewise to identify a labeling with its rank vector; $\mathrm{Eval}(\tau_O, f)$ in rank formulas denotes the image of the original evaluation under this order isomorphism. The next lemma concerns that independently defined evaluation, before any accelerated algorithm is introduced.

Lemma 7.13 (Candidate domains for reachable calls)**.** *Set*

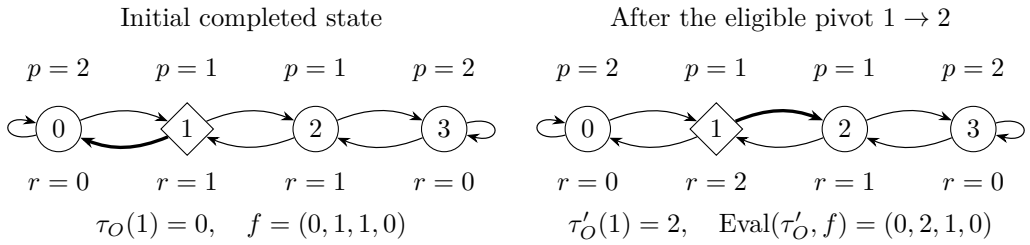
$$C(v) = \begin{cases} \{0\} \cup D(v) \cup \{\ell\}, & \pi(v) = 1, \\ \{0, \ell\}, & \pi(v) = 2. \end{cases}$$

Each domain has at most four distinct members. All domains can be constructed in $O(n)$ rank/index operations. For every reachable-floor call, its floor f and its original answer $z = \mathrm{Eval}(\tau_O, f)$ both belong to $\mathcal{C} = \prod_{v \in V} C(v)$.

Proof. Two sweeps find the nearest original priority-two vertex on each side. Every resulting distance lies in $\{1, \dots, n-1\}$, hence is below ℓ by Lemma 7.11. Each domain can be sorted and deduplicated in constant work. Initialization has $f = 0$. A later floor is a completed labeling, so Lemmas 7.9 and 7.10 give its membership in $\mathcal{C}$. They also apply to the completed result z of the current original evaluation. This existence and endpoint statement does not require knowing which candidate is $z(v)$. Inaccessible or top anchors may make candidates unusable, but do not enlarge the domains. □

Original and rounded coordinate maps. For $x \in \{0, \dots, \ell\}^V$ define

$$F_v(x) = \max\left\{ x(v), \min_{vw \in E(G_{\tau_O})} \theta_x(v, w) \right\}. \tag{26}$$



Circles: Even; diamond: Odd. Bold arrow: selected Odd arc.
Every original arc is drawn; $\ell \geq 4$, and $\mathsf{TOP}_{\mathrm{KL}}$ has rank ℓ.

Figure 7: An exact carried-floor pivot on the four-vertex bidirected path. Priorities are $(2, 1, 1, 2)$, and vertex one is the sole Odd vertex. The old floor $f = (0, 1, 1, 0)$ is retained when $1 \to 2$ is selected; its threshold is $\theta_f(1, 2) = 2$. Both panels show completed evaluations. The figure depicts the full graph and selected Odd policy, not intermediate rounded updates.

At Odd vertices the minimum has the selected arc only; at Even vertices it has all original outgoing arcs. The identity $\min_w \max\{x(v), t_w\} = \max\{x(v), \min_w t_w\}$ shows that $x[v \leftarrow F_v(x)]$ is the original coordinate operator. For $0 \leq t \leq \ell$ put

$$\lceil t \rceil_{C(v)} = \min\{c \in C(v) : c \geq t\}, \qquad H_v(x) = \lceil F_v(x) \rceil_{C(v)}. \tag{27}$$

The ceiling is total, monotone, extensive, and fixes every member of $C(v)$; it scans at most four candidates. Thus $x[v \leftarrow H_v(x)]$ is monotone and inflationary on $\mathcal{C}$.

Figure 7 illustrates an exact evaluation from a nonzero carried floor. Its two panels are completed states.

Definition 7.14 (Rounded worklist evaluation)**.** Precompute the domains $C(v)$. For a reachable-floor call, initialize $x = f$, a FIFO queue containing all vertices, and membership flags that prevent duplicate entries. Repeatedly remove a vertex v, clear its flag, and compute $t = H_v(x)$. If $t > x(v)$, set $x(v) = t$ and enqueue v and each of its original nonself path neighbors whose flag is false. Set the flag of each newly enqueued vertex. When the queue is empty, return x. A circular queue needs at most n slots.

Theorem 7.15 (Exact evaluation above a reachable floor)**.** *Under the hypotheses of Theorem 7.6, the rounded worklist returns exactly* $\mathrm{Eval}(\tau_O, f)$ *on every reachable-floor call. It makes at most* $3n$ *strict coordinate changes and* $O(n)$ *queue removals. Replacing every original evaluation by this procedure preserves the completed policy/labeling trajectory for every prescribed eligible single-edge order, including repeated policies.*

Proof. Let $z = \mathrm{Eval}(\tau_O, f)$. The reason rounding is safe is that the original least answer already lies in the candidate product. More precisely, maintain

$$f \leq x \leq z, \qquad x \in \mathcal{C}.$$

The invariant holds initially by Lemma 7.13. If it holds before a coordinate update, monotonicity of the original operator gives $F_v(x) \leq F_v(z) = z(v)$. Since $z(v) \in C(v)$, rounding to the least candidate above $F_v(x)$ cannot exceed $z(v)$. Inflationarity preserves $x \geq f$, and the ceiling preserves membership in $\mathcal{C}$. This proves the invariant without any assumption about arbitrary feasible labelings.

A priority-one coordinate has at most three strict changes and a priority-two coordinate at most one. Thus there are at most $3n$ changes, each causing at most three enqueue attempts. Including

the initial queue, there are at most $n + 9n$ successful insertions and hence removals. The procedure terminates.

At queue exhaustion every rounded coordinate is stable. Each vertex was processed initially. Subsequently every change of a coordinate on which H_v depends—namely v or an outgoing path neighbor—enqueues v, unless it is already pending. Thus no such change after its last processing can remain unprocessed at exhaustion. For every v,

$$x(v) \le F_v(x) \le H_v(x) = x(v).$$

Consequently x is an original feasible labeling above f. Leastness gives $z \le x$, and domination gives $x \le z$, proving equality. Decoding ranks therefore returns the exact original full labels.

The comparison point z is never an input to the worklist. Its candidate membership came from the original completed-state lemmas. Rounded internal states need not be states of a literal original lift schedule and need not have the no-loose property. For the trajectory claim, initialize with the equality just proved. Equal completed states give the same prescribed eligible arc and the same next floor; applying the equality again gives equal next completed states. Induction proves the assertion. A repeated policy is evaluated from that occurrence's floor, rather than from a cached policy-only answer or a zero restart. □

7.4 Rank work, representations, and certificates

Corollary 7.16 (Rank and bit costs)**.** *Assume indexed arrays of path priorities, owners, legal Odd choices and floor ranks, together with a constructed interface for the actual leaf order. One reachable-floor call uses $O(n)$ rank/index operations and $O(n)$ working words. Let $B = \max\{1, \lceil \log_2(\ell + 1) \rceil\}$, and use word and address width $W = O(B)$, large enough for $O(n)$ working entries and constant tags. On a RAM charging $O(W)$ for an indexed word access, copy, comparison, addition or subtraction, the call uses $O(nB)$ time and $O(nB)$ working bits. Under $R1$ or $R2$, the entire ranked trajectory, including initialization and the final stopping scan, uses $O(n^2)$ word operations, or $O(n^2B)$ time in this bit-cost RAM.*

Proof. The domain construction takes $O(n)$ operations once. Each queue removal examines at most two outgoing arcs and four candidates and performs a constant number of queue operations. Theorem 7.15 therefore gives the per-call bound. The required primitives are min/max, comparison, saturated increment, subtraction of path indices, bounded list scans, flags and circular-queue indexing. To implement saturation in (24), test whether the target is top before incrementing. Because $\ell \ge n$, all ranks, path indices and distances fit in B bits; a constant-factor enlargement handles addresses and tags. Each listed primitive has $O(B)$ cost in the stated model.

There are $I + 1 \le 2n + 1$ evaluations. A scan of the $O(n)$ original Odd arcs tests eligibility by $x(v) < \theta_x(v, w)$. Rule $R1$ retains the least eligible identity pair; $R2$ retains the greatest full target rank, including top, with the prescribed tie-breaking. There are I successful selection scans and one final unsuccessful scan. The total is therefore $O(n(I + 1)) = O(n^2)$, with the stated bit-cost multiplier. For any other eligible order, add its actual selection cost instead. □

Representation and output costs. The corollary bounds computation on ranks. Let $P(T)$ be the one-time cost of reading and validating the supplied height-one ordered tree, determining ℓ, and constructing the rank-to-leaf interface. An explicit ordered list can be scanned and indexed, charging all bits read or stored and the actual pointer widths. If order is not explicit, charge the needed comparisons. An implicit representation must specify and pay for its own preprocessing and

access operations. Once the height-one representation is valid, universality is exactly $\ell \geq n$. There is no bound in n alone on tree reading, tree storage or long leaf-name comparisons.

Let I_G charge conversion from an external graph encoding to original path positions, including the verification of neighbor arcs, endpoint loops, ownership, priorities and legal policy. Any sorting of external names belongs here. Reading the indexed arrays costs $O(n)$ word operations and their actual encoded length in bits. A floor supplied in arbitrary leaf names incurs an additional inverse-conversion cost I_f; a rank-to-leaf table does not make inverse name lookup free. Internally carried floors already have ranks. These representation checks do not certify reachability of an arbitrary externally supplied floor.

Each requested B-bit rank vector entails $O(nB)$ bits of reading or writing. Decoding to actual leaf names costs its actual interface accesses and at least the total emitted name length, denoted $D_T(\text{output})$. One may instead return ranks with the retained order interface and explicitly identify that output representation. Thus full costs add $P(T) + I_G$, any external floor conversions, and all requested intermediate/final output and decoding charges, measured in the same model. Name pointers wider than B occur in these separate interface operations, not in the rank computation. Keeping all states or certificates instead of streaming them requires their total output storage. The RAM bit bound does not assert a same-order multitape Turing-machine simulation.

Proposition 7.17 (A replay certificate for exact evaluation)**.** *Fix the original graph, ordered-tree interface, policy and floor of a reachable-floor call. A certificate consists of a supplied final rank vector $\widehat{x}$ and a list $(v_i, t_i)_{i=1}^{k}$, with $0 \leq k \leq 3n$. The verifier checks valid vertex/rank encodings, sets $y_0 = f$, and for $i = 1, \ldots, k$ requires*

$$t_i = H_{v_i}(y_{i-1}) > y_{i-1}(v_i), \qquad y_i = y_{i-1}[v_i \leftarrow t_i].$$

After replay it requires both

$$\widehat{x} = y_k \quad \textit{and} \quad F_v(y_k) = y_k(v) \quad \textit{for every } v \in V. \tag{28}$$

Every accepted certificate has $\widehat{x} = \mathrm{Eval}(\tau_O, f)$. Such a certificate exists for every reachable-floor call, and verification takes $O(n)$ rank operations, with the representation charges above.

Proof. Let $z = \mathrm{Eval}(\tau_O, f)$. Each verified update is exactly a strict rounded coordinate update. The domination argument of Theorem 7.15 applies to this list in any order, so $f \leq y_i \leq z$ and $y_i \in \mathcal{C}$ for every i. The final feasibility test is on this same replay endpoint y_k; leastness of z gives $z \leq y_k$. Hence $y_k = z$, and the explicit equality test gives $\widehat{x} = z$.

For existence, take the list of strict updates from the rounded worklist and its actual final vector. Theorem 7.15 bounds the list by $3n$ and proves final original feasibility. Replay and the final coordinatewise equality/feasibility scans use $O(n)$ rank operations. The verifier needs no queue transcript. It certifies the least answer under the reachable-floor hypothesis; it does not certify that hypothesis. □

The endpoint equality in (28) is essential. Take $n = \ell = 1$, priority two, the sole loop, and $f = 0$, with either owner. Then $\mathrm{Eval}(\tau_O, f) = 0$, but the top vector is also feasible. An empty update list and a supplied top vector would satisfy the update tests vacuously and a separate feasibility test, yet they fail $\widehat{x} = y_0 = 0$. The verifier rejects them. The empty list with supplied vector zero is accepted. This also shows why feasibility alone is not a minimality certificate.

Boundary examples. For every $n \geq 2$, make vertex zero priority two with its loop selected, and all other vertices priority one, Odd-owned and selecting left. With zero Odd-owned too, the least vector is $(0, 1, \ldots, n-1)$: selected predecessor inequalities force these lower bounds and the displayed vector is feasible. Making zero Even-owned preserves the result. In Figure 7, the least initial vector is $(0, 1, 1, 0)$; the eligible replacement $1 \to 2$ has threshold two and gives exactly $(0, 2, 1, 0)$ above that floor. Here $C(1) = \{0, 1, 2, \ell\}$.

For priorities $(2, 1, 2)$, Even endpoints and Odd middle, either policy has least vector $(0, 1, 0)$. The two distances deduplicate and the other middle arc is satisfied, hence not eligible. For priorities $(2, 1, 1)$, all vertices Odd-owned and policy $(0, 2, 2)$, the vector is $(0, \mathsf{TOP}_{\mathrm{KL}}, \mathsf{TOP}_{\mathrm{KL}})$: the right loop forces top despite the remaining left-anchor candidate. With no priority-two vertex all completed labels are top by Lemma 7.10. At $n = 1$, the only loop yields top at priority one or zero at priority two, and there is no pivot.

Comparison with the general algorithm. Koh and Loho's naive-lifting analysis already gives total $O(m\gamma(T)\ell_T)$ work [5, Section 3, p. 24:10]. For the canonical n_G-leaf height-one star, $m = O(n_G)$, $\gamma(T) = O(1)$, and $\ell_T = n_G$, so quadratic total work is already covered. Their perfect-tree Algorithm 3 also computes exact evaluation above a supplied floor when the policy graph has no loose arcs [5, Theorem 4.5, pp. 24:15–24:16]. The height-one star is the base case of the construction presented by Fijalkow, with credit there to Jurdziński and Lazić [3, Section 5.1, Theorem 4].

The results here give an explicit constant-domain implementation for every height-one input tree and exact reachable-floor correspondence. Surplus leaves cause no rank-by-rank iteration, although their reading cost can dominate $P(T)$. These restricted bounds do not answer the general pivot-rule question in [5, Section 1, p. 24:3], prove optimality, or extend to arbitrary graphs, priorities, tree heights or starting floors. No state or valuation simulation with the open-game sections is used, and no higher-height extremal-tree result is a premise.

8 Further questions

The joint compiler shows that exact realizability can be retained in a succinct representation when the arena grammar bounds the interface and admits common-policy transport. Extending this method requires a replacement for that transport argument and a bound on the variables retained across interfaces. Its legal-neighbor corollary decides one finite neighborhood; an iteration theorem would additionally have to control the number of strict updates or exploit reuse between successive calls.

The feedback example and the acyclic-context theorem identify two sides of the strictness problem. A monotone context can identify strictly ordered local rows. The exact context transformer makes this loss decidable in the stated syntax. More general feedback, shared physical choices between subexpressions and a correspondence with the full equational calculus would require additional arguments preserving original policies and all opponent histories. The marked feedback collision of Proposition 4.11 shows that the forward compiler's minimum-profile output alone cannot be used as a feedback evaluator. It does not preclude a compiler that retains the arena, common policy templates or richer excursion data.

For boundary operations, positive first-return spectra retain distinctions discarded by minimum antichains. The one-forget theorem is an exact factorization on attained records for one policy. To obtain a tree-decomposition algorithm, one would need compatible introduce and join operations, a representation for policy choice, and closure under successive operations with explicit parameter dependence. The collision theorem alone does not obstruct other summaries or winner-solving

algorithms.

For path iteration, reachability of the carried floor is the structural promise supporting the candidate sets. Extending the evaluator to wider arenas or more priorities requires an analogous finite description of the original least answer. A bound on completed pivots must still be combined with the work needed to select and evaluate them. The universal-tree leaf minimum in the appendix concerns a separate embedding problem; any algorithmic use would require an explicit relation between its tree domain and the iteration labels.

A Universal trees for seven-leaf sources of height three

The size of an ordered universal tree is measured here by its leaves. We prove the fixed minimum $M(7,3) = 31$, using the published recursive construction for the upper bound and an order obstruction for the lower bound. The obstruction refines threshold deletion: a target with at most thirty leaves has almost no freedom in its degree counts, and two ordered source patterns force its remaining large branches onto incompatible sides.

A.1 Trees, embeddings, and the extremal statement

Definition A.1 (Ordered universality measured in leaves)**.** Fix positive integers n, h. A tree is finite, rooted and ordered, and every leaf has depth exactly h, the root having depth zero. An internal node has positive outdegree; unary nodes are allowed. An embedding is an injective map on all nodes that sends root to root, preserves immediate parent–child edges and sibling order, and sends leaves to leaves. A tree is (n, h)-universal if it contains an embedded copy of every such height-h tree with at most n leaves. Let $M(n, h)$ be the minimum number of leaves in a universal tree.

The class over which the minimum is taken is nonempty: the complete n-ary tree of height h contains every source in the definition, since each source node has at most n children. These conventions are the fixed-height ordered-tree conventions of [3, Section 4.1, pp. 8–9]. In particular, the embedding does not contract edges, and leaf count differs from total node count.

Theorem A.2 (The fixed leaf minimum)**.** *For rooted ordered trees with full-node, immediate-edge embeddings, unary nodes allowed and all leaves at depth three,*

$$M(7,3) = 31.$$

We first describe the upper construction and the representation of embeddings. We then establish the small height-two minima needed to exclude every target with at most thirty leaves. All orders in this section are the given sibling and leaf orders.

Lemma A.3 (Padding)**.** *If $1 \le m \le n$ and $h \ge 1$, every m-leaf tree of height h embeds in one with exactly n leaves. Hence universality for exactly n leaves implies universality for at most n leaves.*

Proof. Choose a node at depth $h-1$ and append $n-m$ leaf children after its existing children. The identity on the original nodes preserves every immediate edge, root, order and leaf. Compose it with an embedding of the padded tree into the proposed universal tree. □

A.2 The recursive upper construction

For a height-three tree, write $((a_1, \ldots, a_k), \ldots)$ for its nested degree description: outer entries represent root children in order, and each inner sequence gives the positive degrees of that child's children. This description retains a separate vertex at every level, including every unary vertex.

Lemma A.4 (The published upper construction)**.** *There is a* $(7, 3)$*-universal tree with* 31 *leaves, namely*

$$((1), (1, 3, 1), (1), (1, 3, 1, 7, 1, 3, 1), (1), (1, 3, 1), (1)). \tag{29}$$

The same recursive construction gives a $(7, 2)$*-universal tree with* 17 *leaves.*

Proof. We use the construction presented by Fijalkow, who credits its streamlined Jurdziński–Lazić lineage [3, Section 5.1, Theorem 4, p. 12]. Let $U(1, h)$ be the unary chain of height h, and let $U(n, 1)$ be the star with n leaves. For $n \in \{3, 7\}$, put $k = (n-1)/2$. For $h > 1$, form $U(n, h)$ under a new root by concatenating three groups of subtrees: the root-child subtrees of $U(k, h)$, the single middle subtree $U(n, h-1)$, and a second copy of the root-child subtrees of $U(k, h)$. Their roots are children of the new root, so all leaves have depth h. Each recursive call decreases $n + h$ until a chain or star is reached.

We prove universality by induction on $n + h$. The chain and star cases follow directly from the definition. By Lemma A.3, it suffices to embed a source with exactly $n = 2k + 1$ leaves. Its $(k+1)$-st leaf lies under some root child c. At most k leaves occur under root children before c, and at most k under root children after c. Each nonempty side forest, with a root added, is a height-h tree with at most k leaves. Induction embeds it into the corresponding $U(k, h)$. Removing only the added root and its image places the source root-child subtrees into the corresponding side group. The subtree rooted at c has height $h - 1$ and at most n leaves, so induction embeds it into the middle $U(n, h-1)$. An empty side forest needs no images.

Map the source root to the new root. The three maps have disjoint images in correctly ordered groups. Their union is injective on all nodes and preserves siblings and leaves. Within a group every immediate edge is preserved by induction; edges from the source root go to edges from the new root. Thus the resulting map is precisely the required embedding.

Let $f(n, h)$ denote the number of leaves of this construction. Then

$$f(1, h) = 1, \qquad f(n, 1) = n, \qquad f(n, h) = 2f((n-1)/2, h) + f(n, h-1).$$

In particular,

$$f(3, 2) = 5, \quad f(3, 3) = 7, \quad f(7, 2) = 17, \quad f(7, 3) = 7 + 17 + 7 = 31.$$

The degree sequence of $U(7, 2)$ is $(1, 3, 1, 7, 1, 3, 1)$, while the root-child descriptions of $U(3, 3)$ are $(1), (1, 3, 1), (1)$. Substituting these into the recursion gives (29). □

Figure 8 draws the entire tree in (29). Its levels contain $1, 7, 17, 31$ vertices respectively. In particular, the number 31 counts its leaves, whereas its total node count is 56.

A.3 Adjacent-ancestor words and literal embeddings

Let $x_1, \ldots, x_m$ be the leaves of a height-h tree in order. Its adjacent-ancestor word is

$$a = (a_1, \ldots, a_{m-1}), \qquad a_i = \mathrm{depth}(\mathrm{LCA}(x_i, x_{i+1})),$$

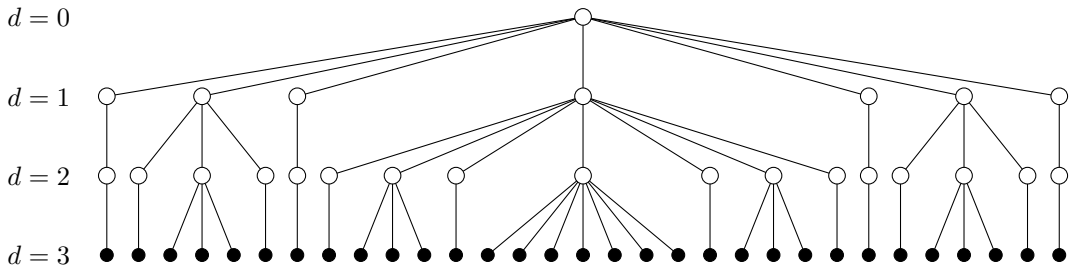


Figure 8: The complete ordered tree in (29), with root at the top and leaf order from left to right. Every node and immediate edge is drawn: all 16 unary vertices remain visible, and every leaf has depth three. The seven root-child subtrees have $1, 5, 1, 17, 1, 5, 1$ leaves. This is the published recursive upper construction presented by Fijalkow, with credit to Jurdziński and Lazić [3, Section 5.1, Theorem 4].

with alphabet $\{0, \ldots, h-1\}$. For $i < j$,

$$\operatorname{depth}(\mathrm{LCA}(x_i, x_j)) = \min_{i \le k < j} a_k. \tag{30}$$

To see this, fix a depth d. The leaves below any depth-d node form an interval. Two leaves have the same depth-d ancestor if and only if every adjacent gap between them has ancestor depth at least d. Applying this for all d proves the identity.

Lemma A.5 (Words and full-node embeddings)**.** *Every word of length $m-1$ over $\{0, \ldots, h-1\}$ determines a unique ordered height-h tree with m leaves, up to rooted ordered isomorphism. Let height-h source and target trees have words a and b, the source having m leaves. An embedding exists if and only if target leaf indices $j_1 < \cdots < j_m$ satisfy*

$$a_i = \min_{j_i \le k < j_{i+1}} b_k \qquad (1 \le i < m). \tag{31}$$

Proof. At each depth $d = 0, \ldots, h$, partition $\{1, \ldots, m\}$ into maximal intervals all of whose interior gaps have $a_i \ge d$. Make one node for each interval at that depth and join every depth-$(d+1)$ interval to its containing depth-d interval, in interval order. At depth zero there is one interval; at depth h the intervals are singletons. Every node has a leaf descendant. If an interval is unchanged at consecutive depths, retain both nodes and their connecting edge. Thus the construction includes all unary chains, including the case $m = 1$ of an empty word. It has adjacent ancestor word a. In any original tree these intervals are exactly the descendant-leaf sets at each depth, so the construction recovers all its nodes, edges and orders and proves uniqueness.

An embedding preserves depths because it preserves the root and immediate edges. Distinct children of a source node have distinct image children by injectivity. Therefore the lowest common ancestor of two source leaves maps to the lowest common ancestor of their images, with the same depth. Sibling order makes the target leaf indices increasing, and (30) implies (31).

Conversely, assume the latter equations. Map a source node at depth d to the depth-d ancestor of the selected image of any one of its descendant leaves. This is well-defined: every adjacent source gap within the node's interval has depth at least d, so the equations ensure that all selected images have the same depth-d ancestor. Distinct source nodes at depth d have intervals separated by a gap of depth less than d; the corresponding selected target leaves cannot have a common

depth-d ancestor. Thus their images are distinct. Images at different depths are also distinct, proving full-node injection. For a child and its parent use the same descendant leaf: their images are consecutive ancestors and hence adjacent. Interval order preserves sibling order. The root and leaf images are the prescribed ones, and a unary chain has a separate image at every depth. This proves sufficiency. □

In particular, exactly-seven-leaf height-three sources correspond to the $3^6 = 729$ words. This identity is a representation fact; the lower bound below excludes arbitrary targets directly.

A.4 Threshold deletion and height-two constraints

Threshold trees. For a height-three tree T and $\delta \geq 1$, let c_δ be the number of depth-two nodes of outdegree at least δ. Retain these nodes and their ancestors, delete all depth-three leaves, and delete any other depth-two node or now-empty depth-one node. If $c_\delta > 0$, this gives an ordered height-two tree T_δ with exactly c_δ leaves. Its remaining immediate edges are unchanged; in particular no shallow leaf or contracted edge is introduced.

Lemma A.6 (Threshold deletion)**.** *If T is $(7,3)$-universal, then T_δ is $(\lfloor 7/\delta \rfloor, 2)$-universal for $1 \leq \delta \leq 7$. Writing $L = |L(T)|$ for the number of leaves of the original height-three tree T, one has*

$$L = \sum_{\delta \geq 1} c_\delta, \qquad c_1 \geq c_2 \geq \cdots . \tag{32}$$

Proof. Take any height-two source with at most $\lfloor 7/\delta \rfloor$ leaves. Attach δ new children to each leaf. The resulting height-three source has at most seven leaves and embeds into T. Each former leaf has at least δ distinct image children, so its depth-two image survives in T_δ, as do all its ancestors. Restricting the embedding to the original source nodes gives an injective, root-, immediate-edge-, order- and leaf-preserving map into T_δ. This also proves that $c_\delta > 0$ for these values. It is the threshold-deletion argument of Fijalkow's Theorem 5, with the pruning specified explicitly [3, Section 5.2, Theorem 5, pp. 13–14].

A depth-two node of degree d contributes one to each of $c_1, \ldots, c_d$. Summing these contributions counts every original depth-three leaf exactly once and proves the identity. The sets counted by c_δ are nested, proving monotonicity. Degrees above seven are allowed, and their contributions for $\delta \geq 8$ remain in the sum. □

Height-two sequences. A height-two tree is specified by the positive degree sequence $(b_1, \ldots, b_r)$ of its root children. A source sequence $(a_1, \ldots, a_k)$ embeds exactly when there are indices $j_1 < \cdots < j_k$ with $a_i \leq b_{j_i}$. Necessity follows from node injection and sibling order. For sufficiency map each source root child to its matched target child and select the required number of its leaf children in order.

Lemma A.7 (Height-two minima and their equality case)**.** *One has $M(7,2) = 17$, $M(3,2) = 5$, and $M(2,2) = 3$. The unique five-leaf $(3,2)$-universal degree sequence is $(1,3,1)$.*

Proof. For an $(n,2)$-universal degree sequence put $d_\delta = |\{i : b_i \geq \delta\}|$. The source consisting of $\lfloor n/\delta \rfloor$ root children of degree δ forces $d_\delta \geq \lfloor n/\delta \rfloor$. At $n = 7$,

$$d_1 \geq 7, \quad d_2 \geq 3, \quad d_3 \geq 2, \quad d_4, \ldots, d_7 \geq 1.$$

These lower bounds sum to sixteen. If the tree had at most sixteen leaves, equality would hold throughout, with no contribution at $\delta \geq 8$. There would be exactly one child q of degree at least

four and exactly one other child of degree at least three. The source $(3,4)$ requires the other child before q, whereas $(4,3)$ requires it after q, a contradiction. Hence at least seventeen leaves are needed, and Lemma A.4 attains the bound.

For $n=3$, the inequalities $d_1 \geq 3, d_2 \geq 1, d_3 \geq 1$ give at least five leaves. At equality the sources $(1,1,1)$ and (3) force exactly three children with degrees $3,1,1$ in some order. Sources $(1,2)$ and $(2,1)$ require a child on each side of the unique degree-at-least-two child, placing degree three in the middle. Conversely, $(1,3,1)$ embeds $(3),(1,2),(2,1),(1,1,1)$, the four compositions of three; Lemma A.3 handles smaller sources. For $n=2$, $d_1 \geq 2, d_2 \geq 1$ force three leaves, attained by $(1,2)$, which embeds both (2) and $(1,1)$. □

A.5 The order obstruction below thirty-one leaves

Proof of Theorem A.2. Lemma A.4 gives the upper bound. For the lower bound suppose that T is $(7,3)$-universal and that its original leaf count L is at most thirty. Lemmas A.6 and A.7 give

$$c_1 \geq 17, \qquad c_2 \geq 5, \qquad c_3 \geq 3, \qquad c_4, \ldots, c_7 \geq 1. \tag{33}$$

The last four bounds use the one-leaf height-two source. The displayed minimum contributions sum to twenty-nine. By (32), there is at most one extra unit across all counts, including the tail $\delta \geq 8$. In particular $1 \leq c_4 \leq 2$.

We shall repeatedly use the consequence of $c_2 = 5$. In that case the equality classification in Lemma A.7 applied to T_2 shows that exactly three original depth-one parents have children of degree at least two, with respective counts $1,3,1$ in order. Call the middle parent C. Other original depth-one parents may exist, but all their depth-two children have degree one. Thus C is the only parent that can host two or more grandchildren of degree at least two in a source root-child subtree. This is a statement about the original target nodes, not contracted or merged parents.

Case 1: $c_4 = 1$. Let q be the unique depth-two node of degree at least four and P its parent. The source $((3,4))$ must map its degree-four grandchild to q, and therefore requires another child of P of degree at least three strictly before q. The source $((4,3))$ requires one strictly after q. Thus P has at least three children of degree at least three. The source $((3),(3))$ requires such children under two distinct depth-one parents, so at least one lies outside P. Consequently $c_3 \geq 4$.

This uses the sole extra unit in (33), forcing

$$c_1 = 17, \quad c_2 = 5, \quad c_3 = 4, \quad c_4 = \cdots = c_7 = 1, \quad c_\delta = 0 \quad (\delta \geq 8).$$

The preceding equality classification applies, and $P = C$ because P has three children of degree at least two. In the source $((2,2),(3))$, the first root child must map to C; its second root child therefore requires a degree-at-least-three child under a parent strictly after C. The source $((3),(2,2))$ requires one under a parent strictly before C. These are two distinct children outside C, in addition to the three already inside it. This gives $c_3 \geq 5$, contradicting $c_3 = 4$.

Case 2: $c_4 \geq 2$. The available extra unit now forces

$$c_1 = 17, \quad c_2 = 5, \quad c_3 = 3, \quad c_4 = 2, \quad c_5 = c_6 = c_7 = 1, \quad c_\delta = 0 \quad (\delta \geq 8).$$

Again C is the unique parent with at least two children of degree at least two. Both $((3,4))$ and $((4,3))$ must use C, so at least two of the three children counted by c_3 lie in C. The source $((3),(3))$ requires distinct parents, so exactly two lie in C and one outside it. List the two inside as u, v in order. With only these two available, $((3,4))$ forces $\deg(v) \geq 4$, and $((4,3))$ forces $\deg(u) \geq 4$. Both children counted by c_4 therefore belong to C.

Now $((3),(4))$ must send its degree-four grandchild into C, so it requires a degree-at-least-three child under a parent strictly before C. Conversely $((4),(3))$ requires one strictly after C. The sole such child outside C cannot be on both sides, which is a contradiction.

Every source pattern used in these cases has six or seven leaves, so universality requires each embedding. The two cases exhaust $1 \le c_4 \le 2$. They exclude all targets with at most thirty leaves without imposing an arity bound, eliminating unary nodes, limiting the number of root children or discarding high-degree tail contributions. Hence $M(7,3) \ge 31$, completing the proof. □

Relation to the general minimum problem. Fijalkow's author-hosted discussion asks for the minimum leaf size of an (n,h)-universal tree [2, "Between lower and upper bounds?"]. The cited migrated page is undated. The fixed equality above concerns that full-node ordered domain and does not settle the general minimum problem. The upper construction is published, and the height-two bounds are auxiliary exposition; the external novelty of this exact fixed-case lower argument has not been established. No claim about current openness or priority follows from the cited discussion. Finally, this leaf minimum alone supplies no pivot or evaluation bound for strategy iteration.

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